

(12) **United States Patent**  
**Kim**

(10) **Patent No.:** **US 12,399,533 B2**  
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **PORTABLE DISPLAY DEVICE**

USPC ..... 361/679.01  
See application file for complete search history.

(71) Applicant: **Folding All Inc.**, Seoul (KR)

(72) Inventor: **Si-Han Kim**, Gwangju-si (KR)

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(73) Assignee: **Folding All Inc.**, Seoul (KR)

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(\*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **18/775,213**

(22) Filed: **Jul. 17, 2024**

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(65) **Prior Publication Data**

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**Related U.S. Application Data**

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(63) Continuation of application No. 18/121,531, filed on Mar. 14, 2023, now Pat. No. 12,072,737, which is a continuation of application No. 17/362,847, filed on Jun. 29, 2021, now Pat. No. 11,640,186, which is a continuation of application No. 14/557,338, filed on Dec. 1, 2014, now abandoned, which is a continuation-in-part of application No. 13/809,355, filed as application No. PCT/KR2011/002980 on Apr. 25, 2011, now abandoned.

International Search Report dated Dec. 19, 2011, issued to corresponding International Application No. PCT/KR2011/002980.

*Primary Examiner* — Calvin C Ma

(74) *Attorney, Agent, or Firm* — Stein IP LLC

(30) **Foreign Application Priority Data**

Jul. 9, 2010 (KR) ..... 10-2010-0066332

(51) **Int. Cl.**  
**G06F 1/16** (2006.01)

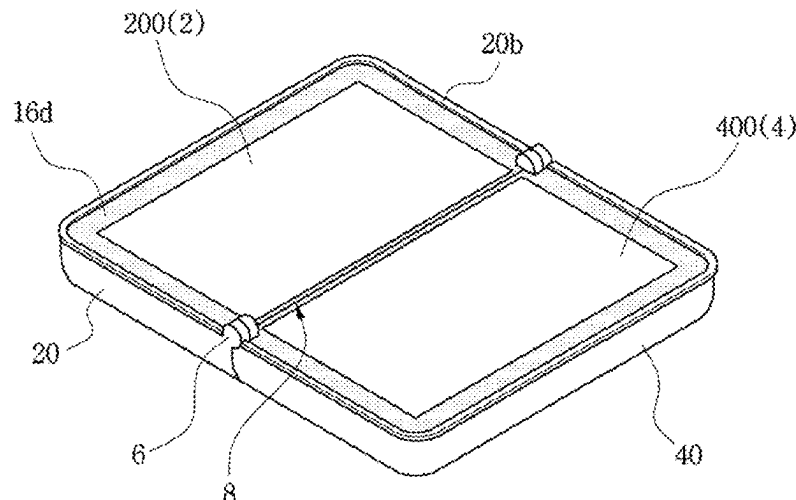
(52) **U.S. Cl.**  
CPC ..... **G06F 1/1641** (2013.01); **G06F 1/1618** (2013.01); **G06F 1/1624** (2013.01); **G06F 1/1643** (2013.01); **G06F 1/1698** (2013.01)

(58) **Field of Classification Search**  
CPC .... G06F 1/1641; G06F 1/1618; G06F 1/1624; G06F 1/1643

(57) **ABSTRACT**

A portable display device is provided in which two display panels are joined to each other to form a single screen when two panel housings are unfolded from a folded state. The portable display device includes at least two panel housings and display panels that are mounted on the panel housings, respectively. When the panel housings are unfolded, the display panels are joined to each other to form a single screen. The distance between the display panels when the display panels are contiguous to each other is less than 3 mm and the distance between input units or between input sensors is less than 3 mm.

**20 Claims, 76 Drawing Sheets**



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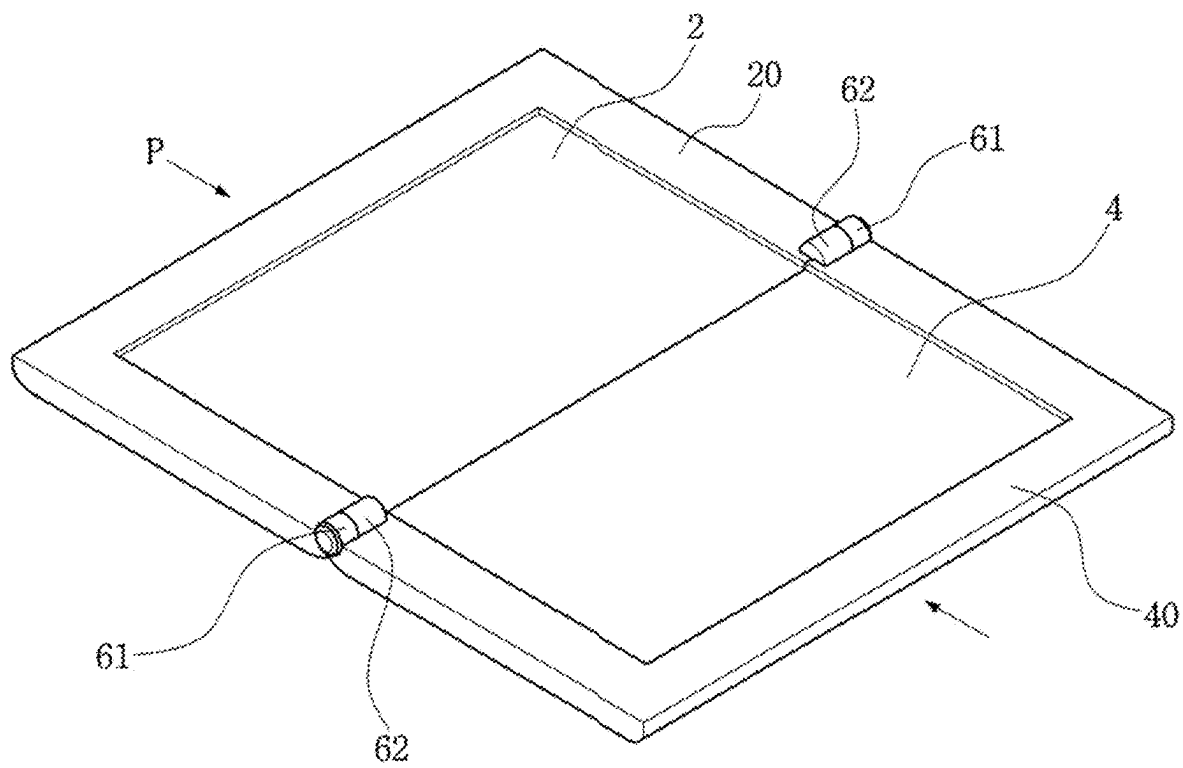


FIG. 1

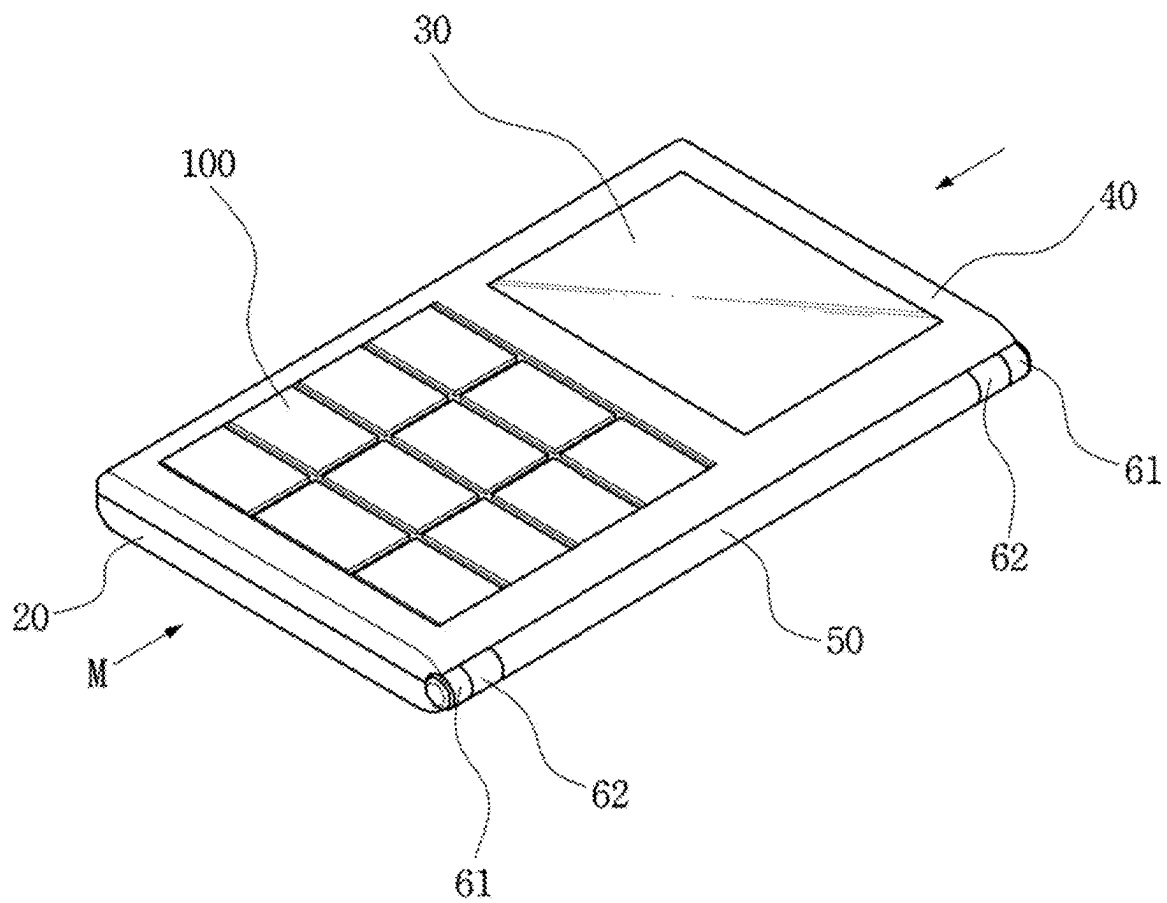


FIG. 2

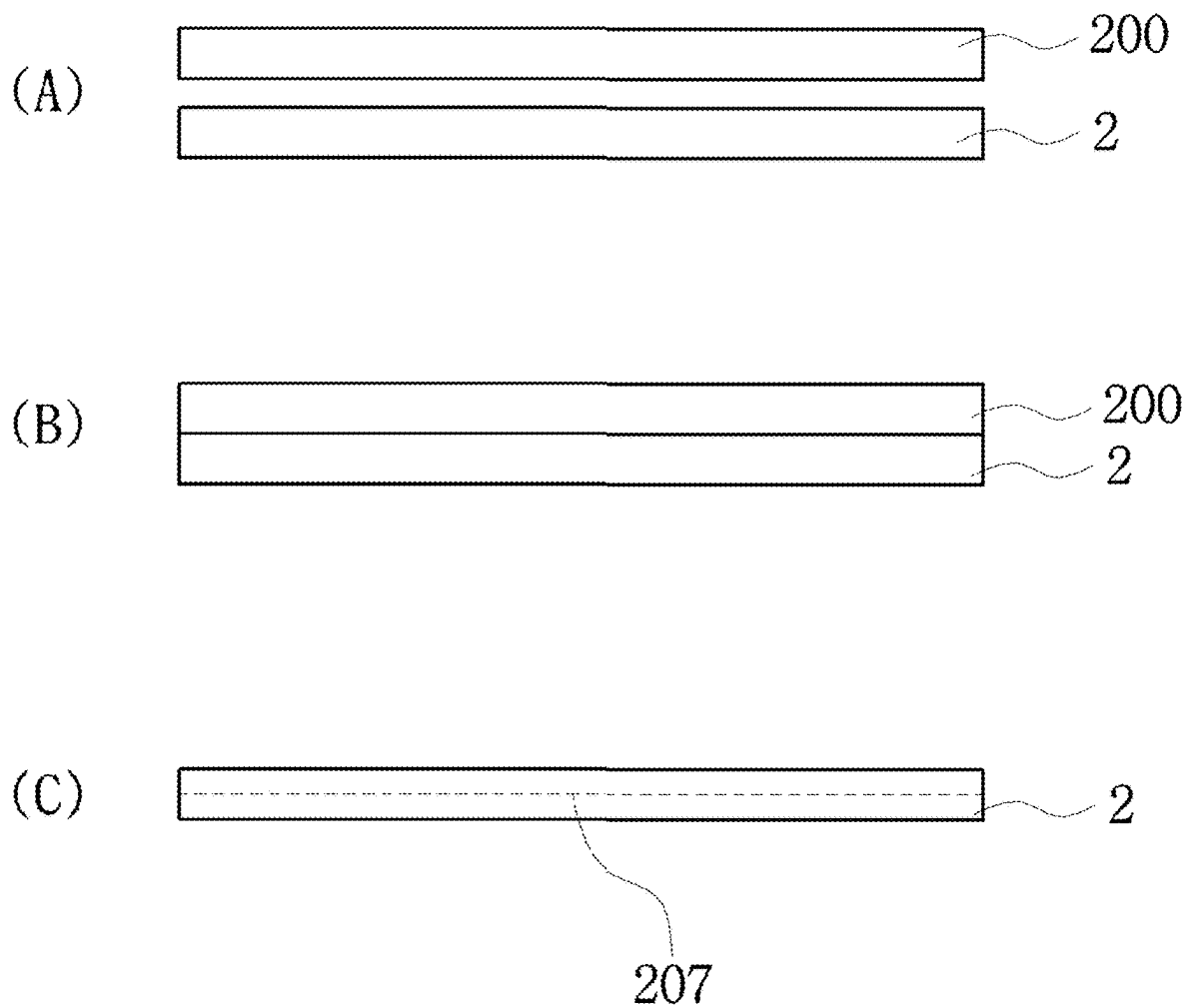
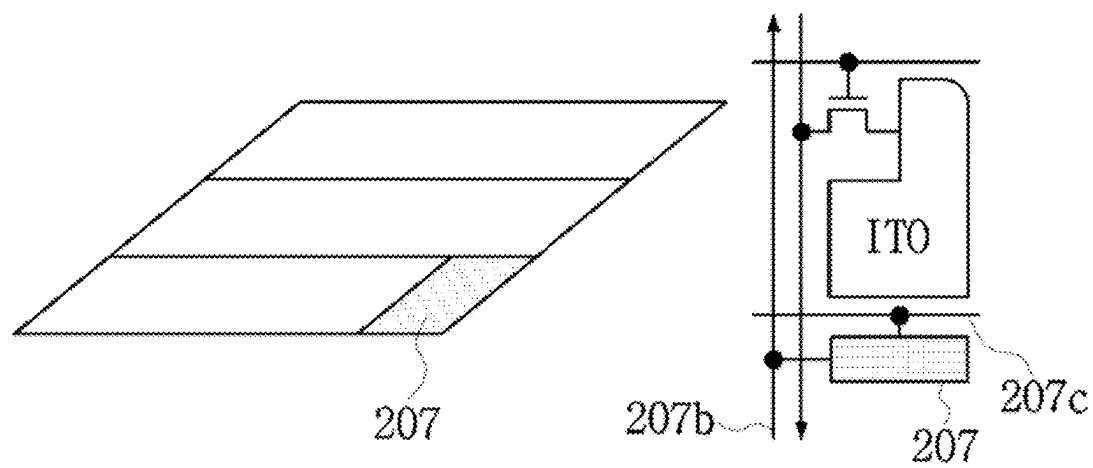
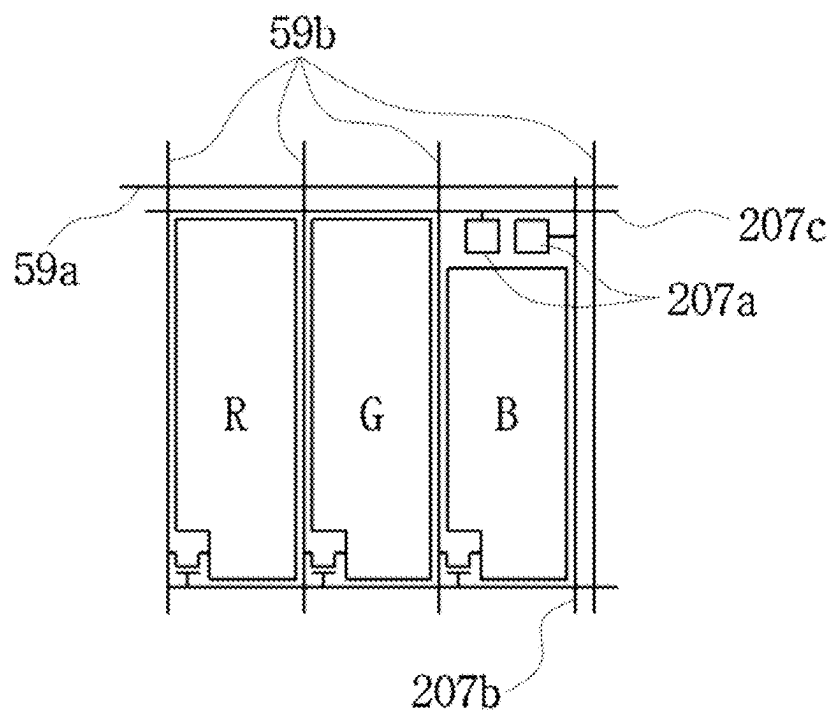


FIG. 3



(A)



(B)

FIG. 4

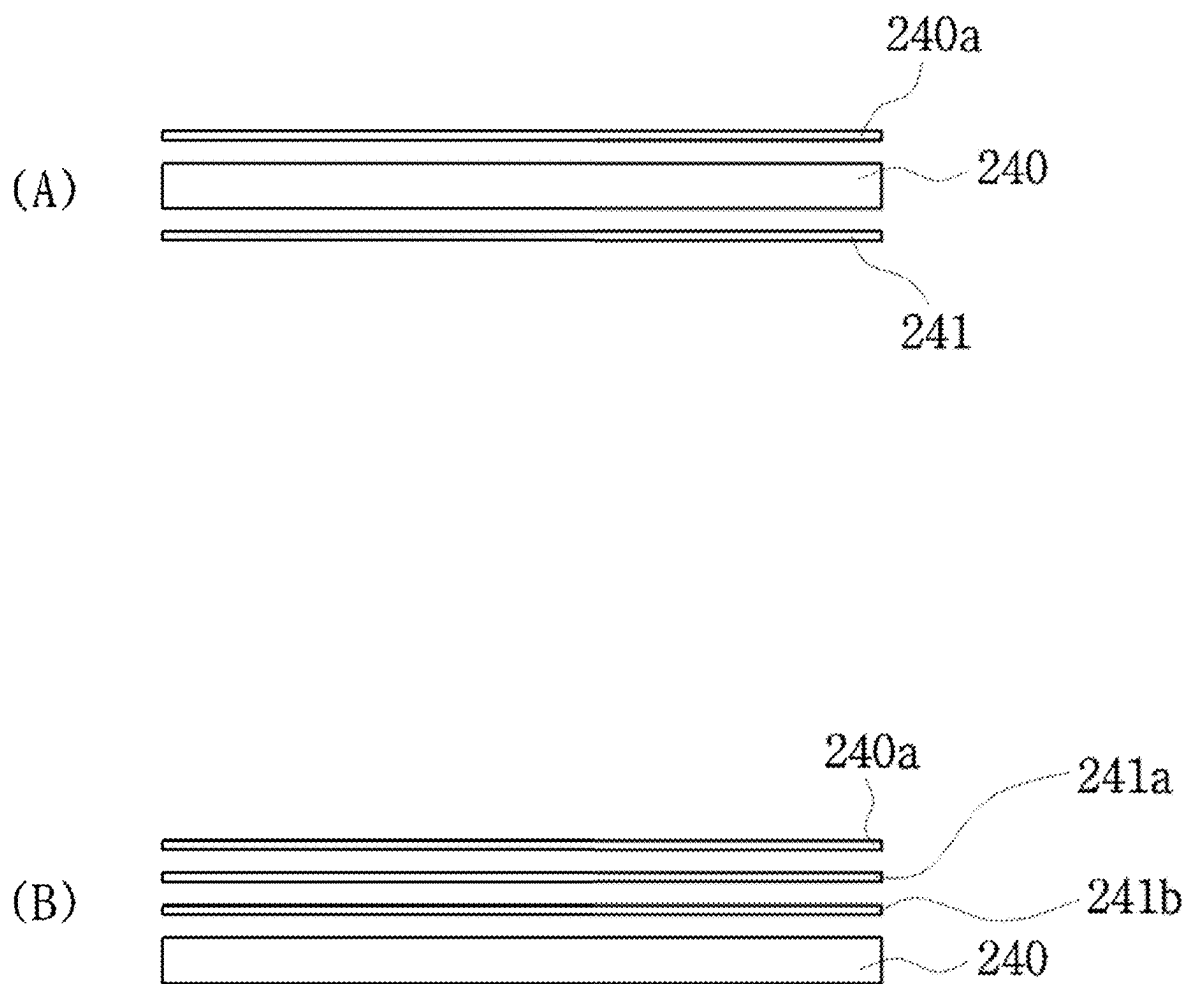


FIG. 5

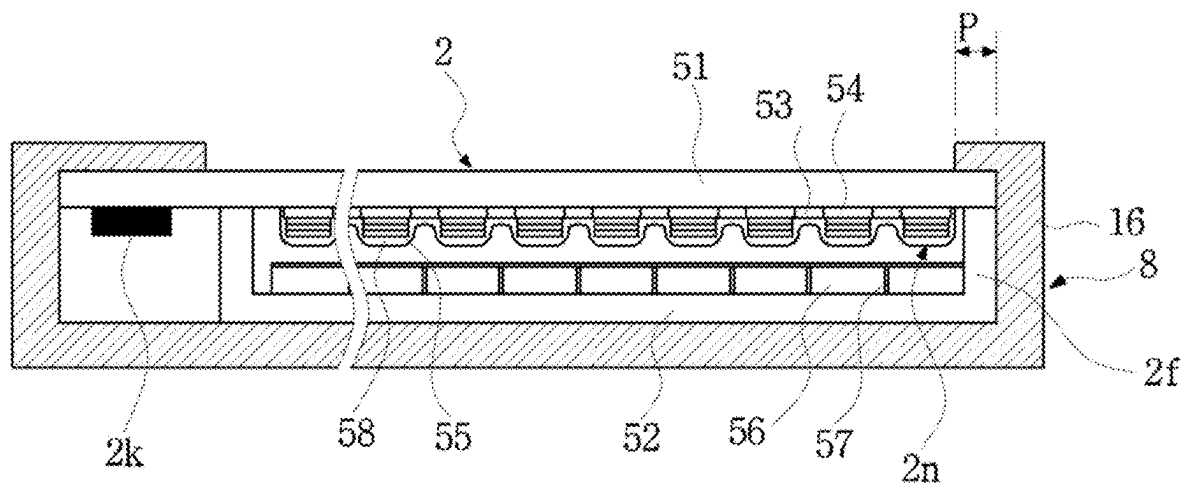


FIG. 6

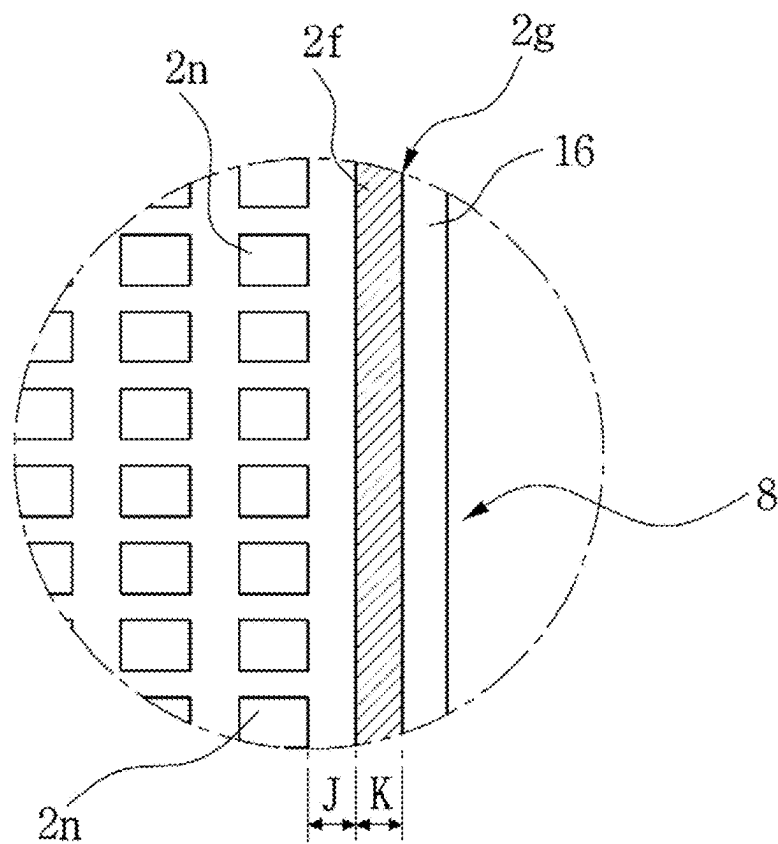


FIG. 7



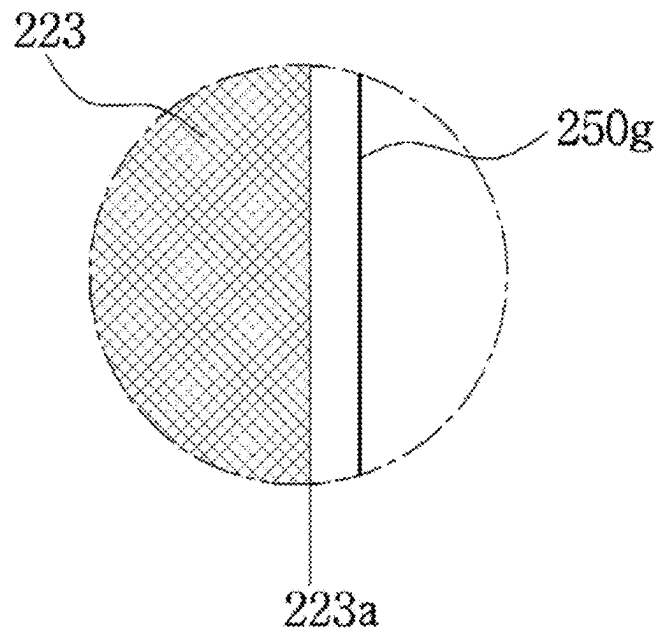


FIG. 8

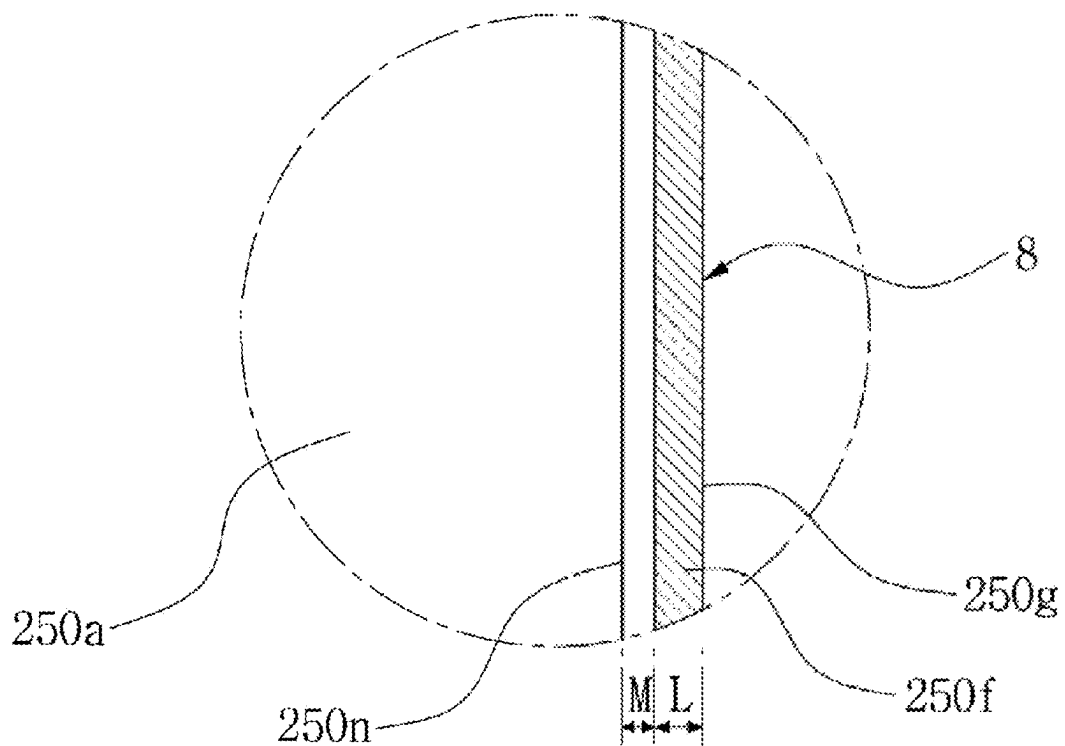


FIG. 9

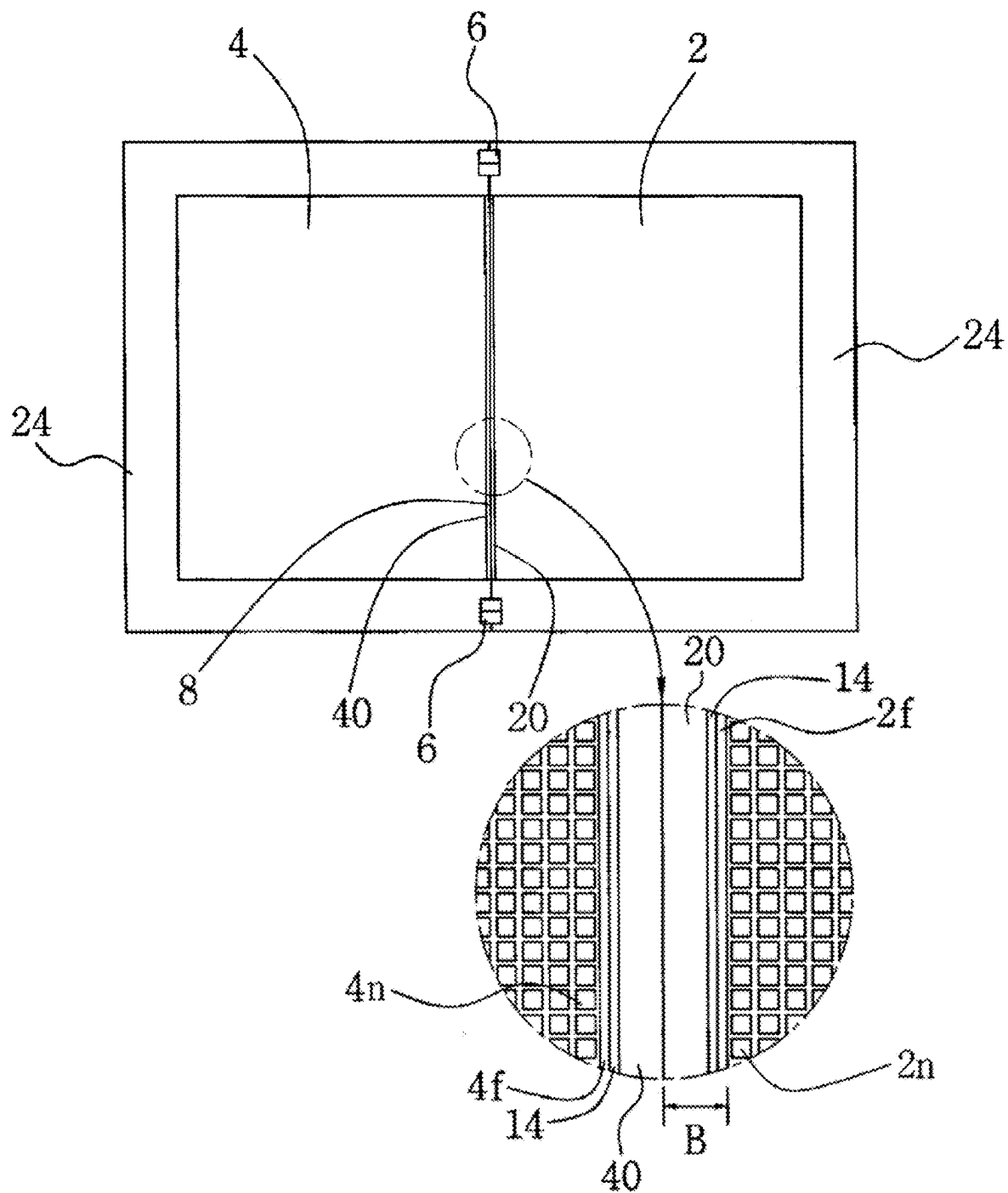


FIG. 10

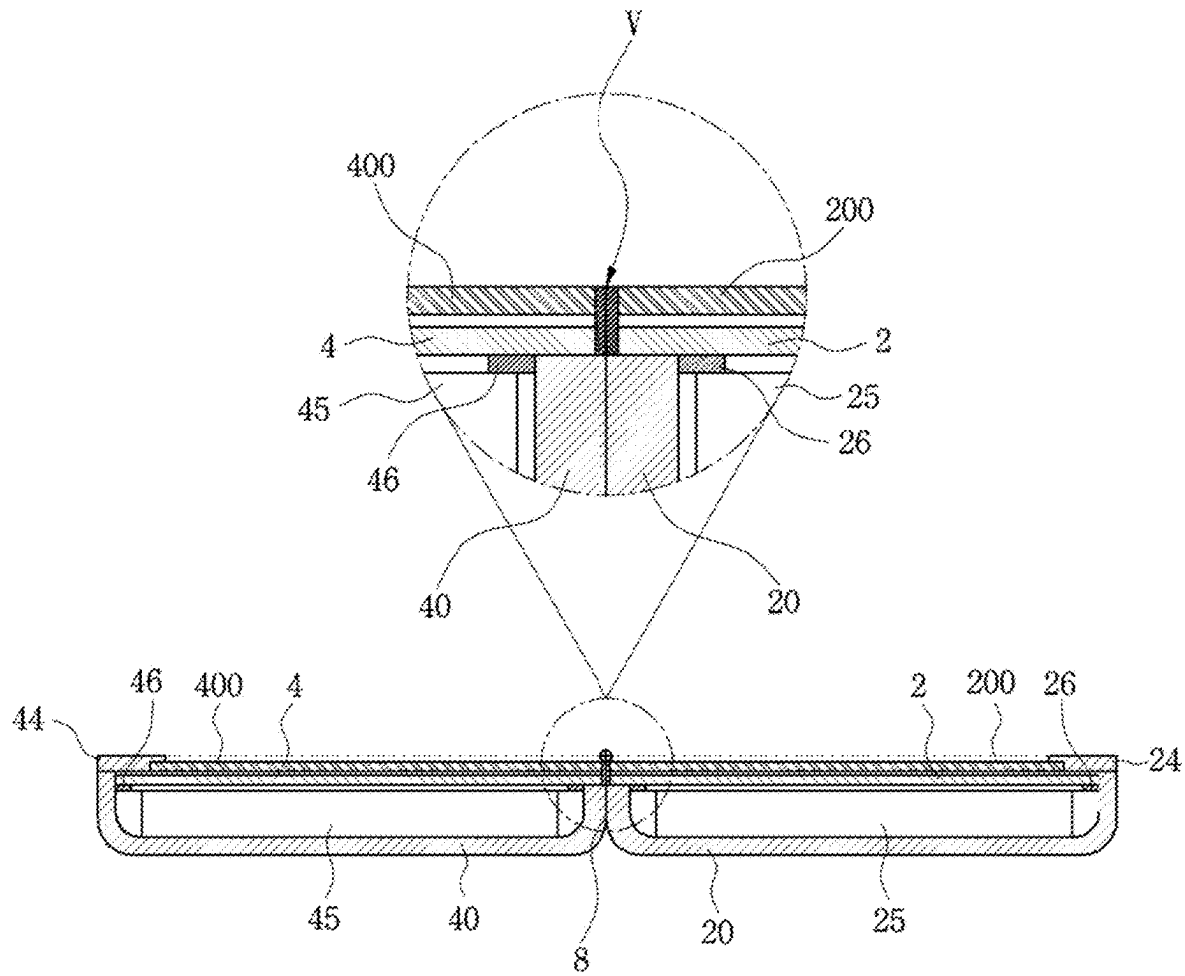


FIG. 11

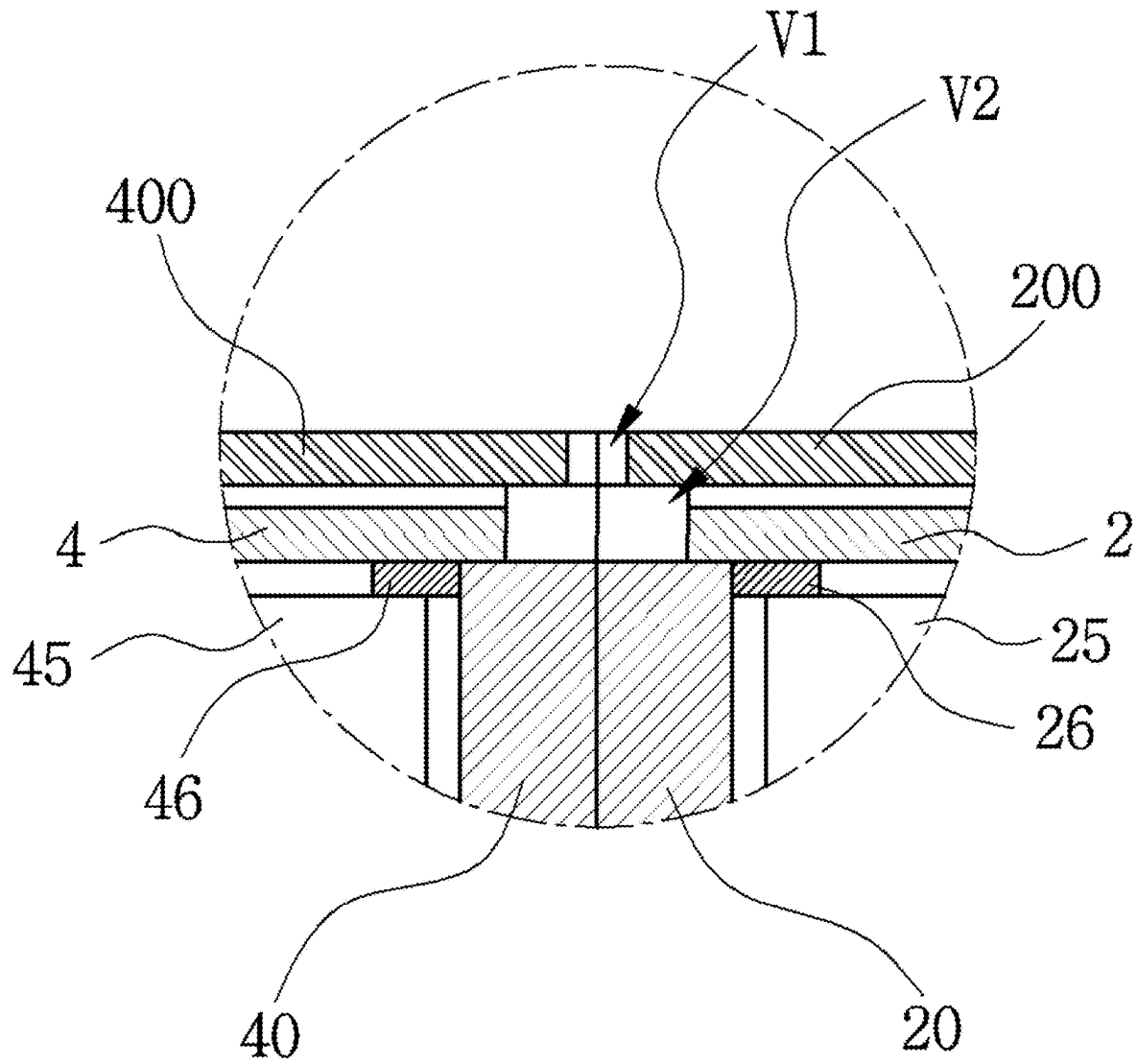


FIG. 12

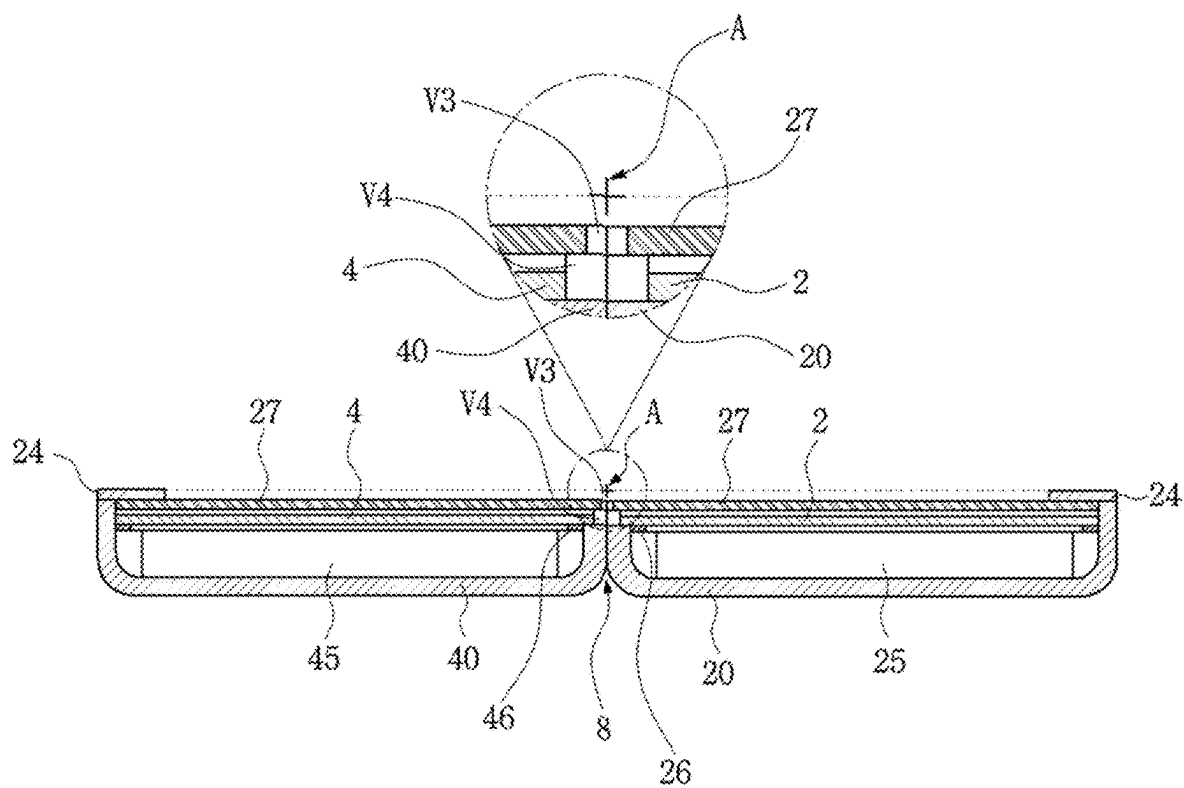


FIG. 13

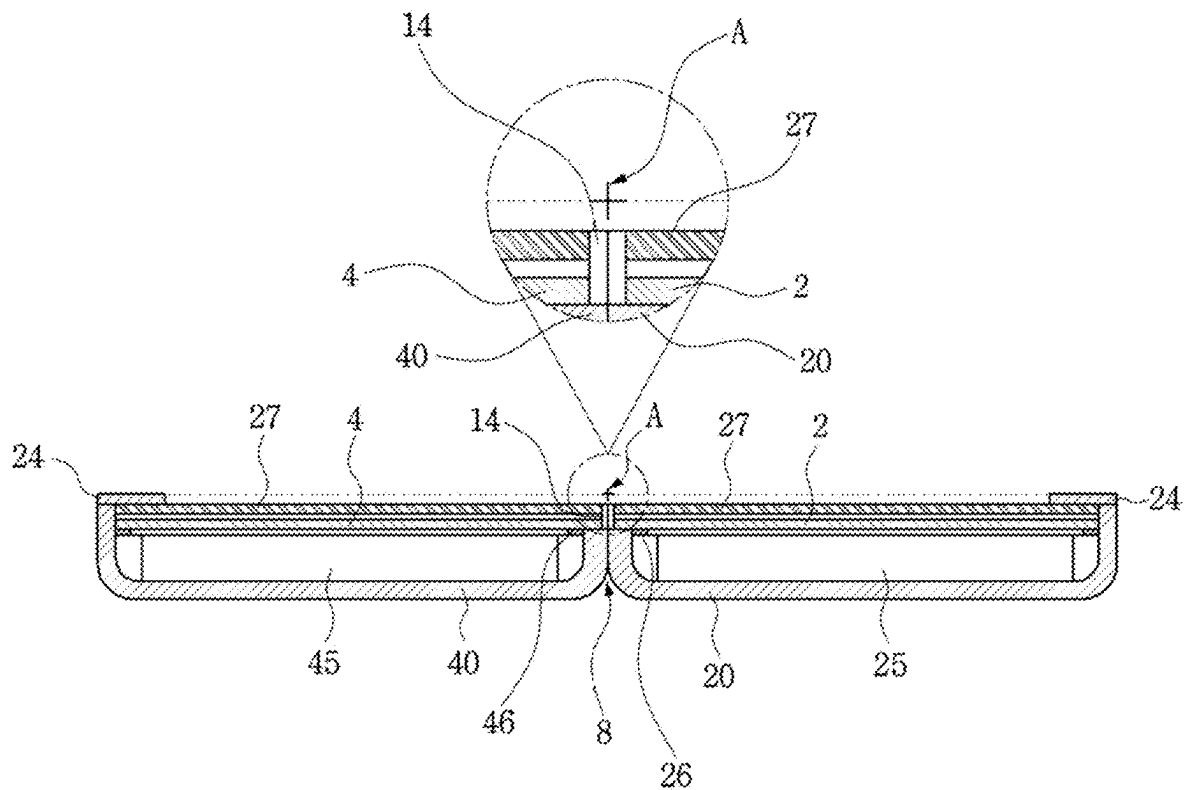


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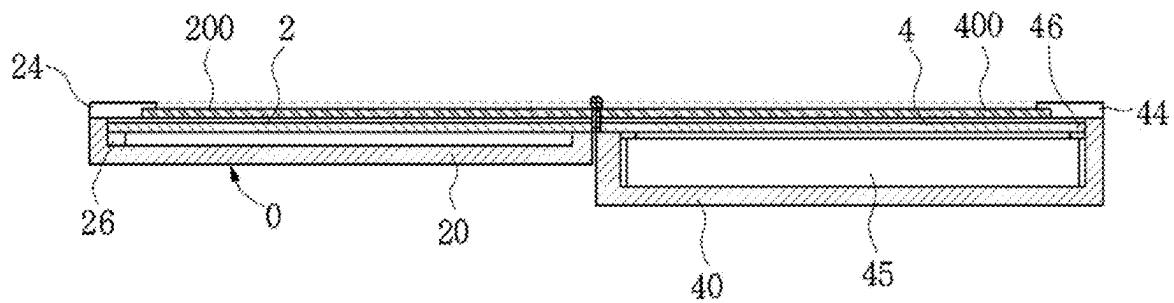


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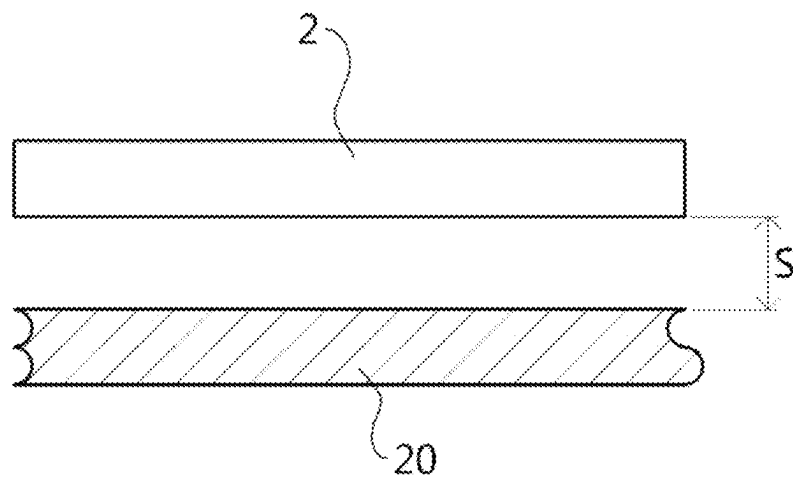


FIG. 16

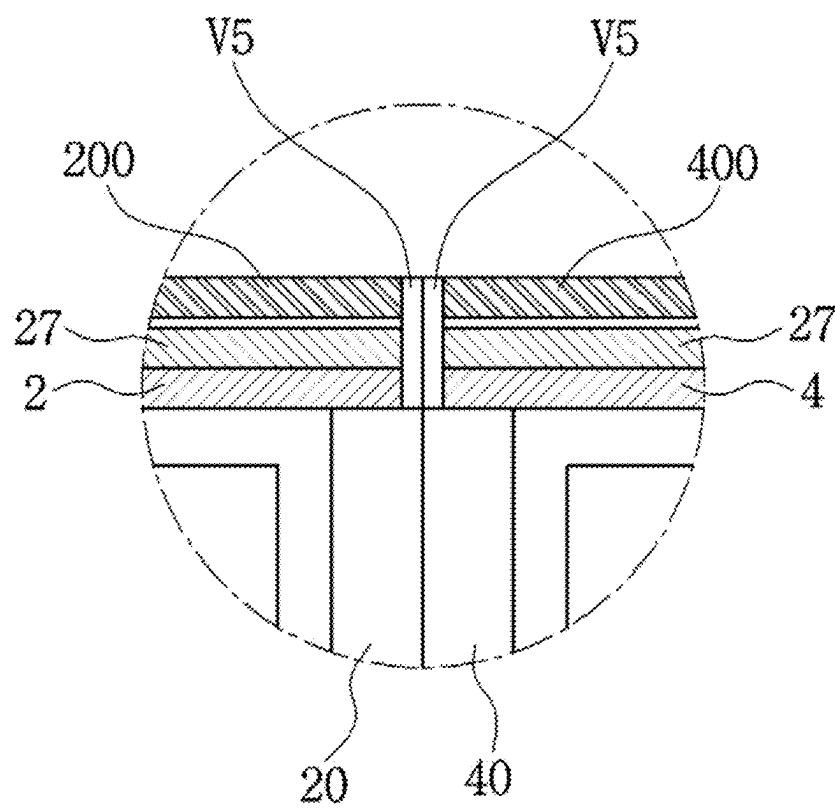


FIG. 17

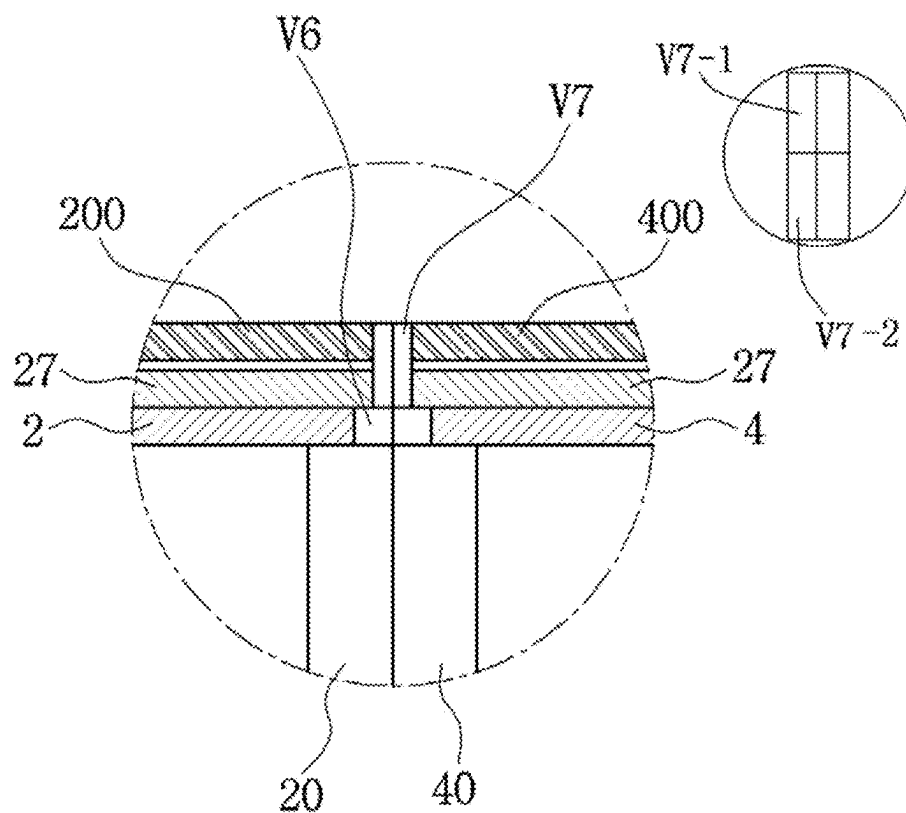


FIG. 18

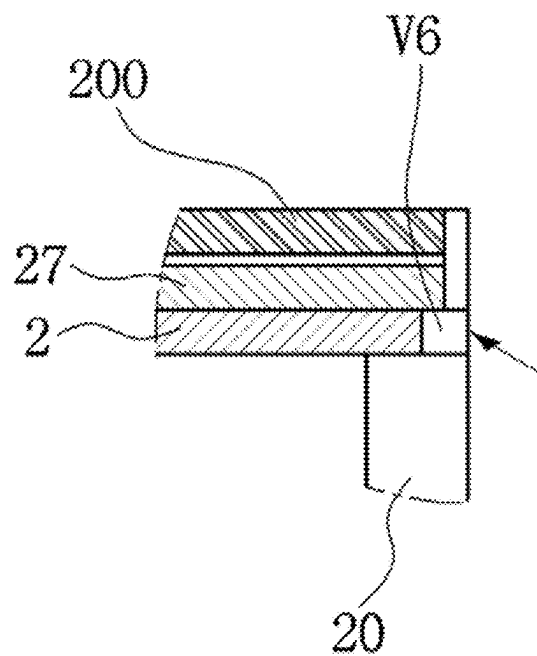


FIG. 19



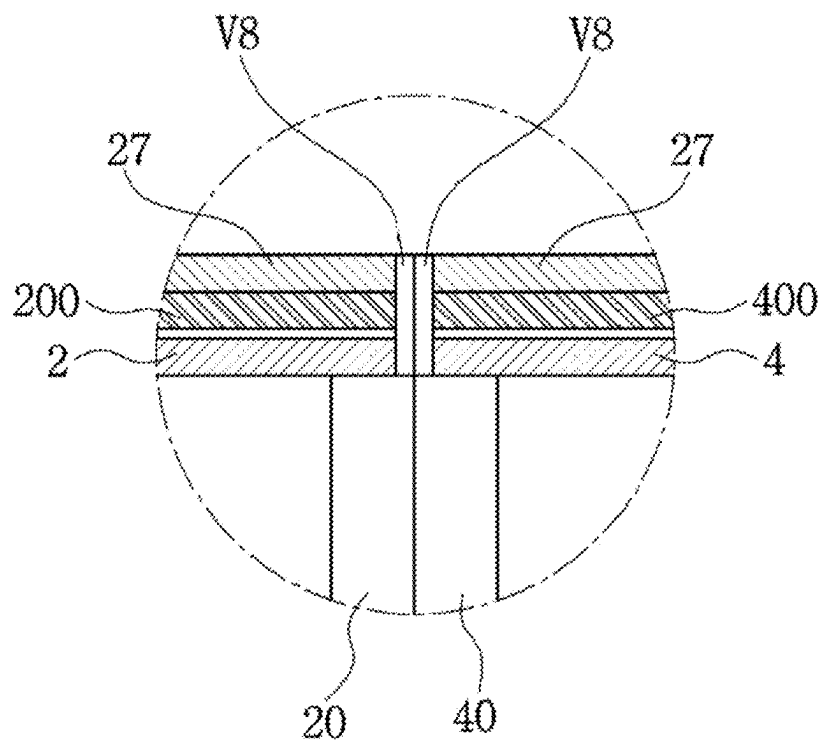


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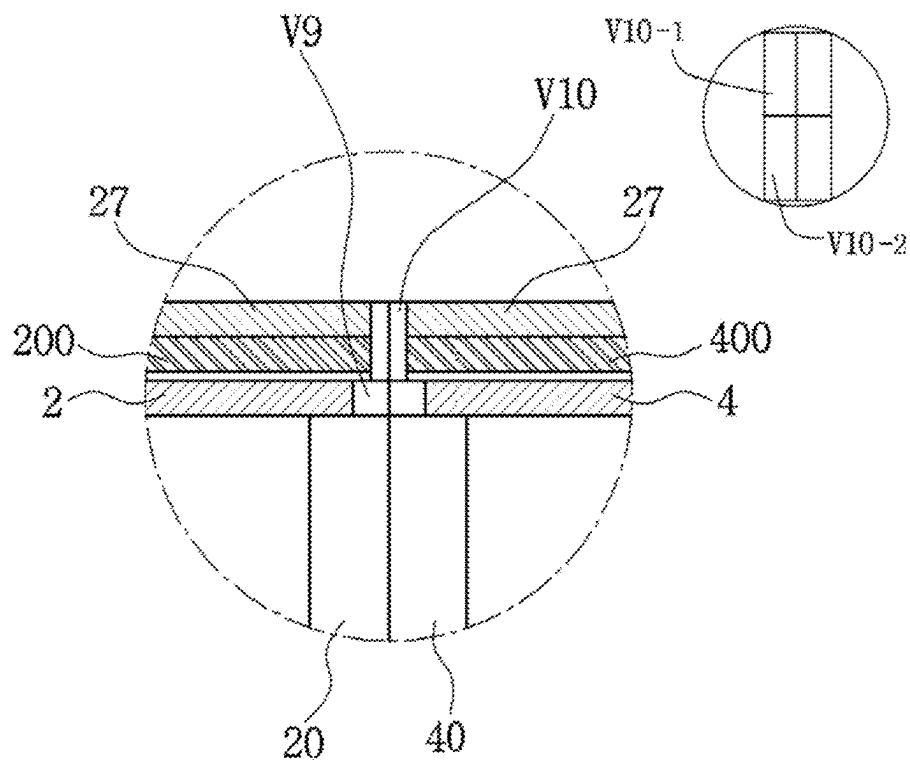


FIG. 21

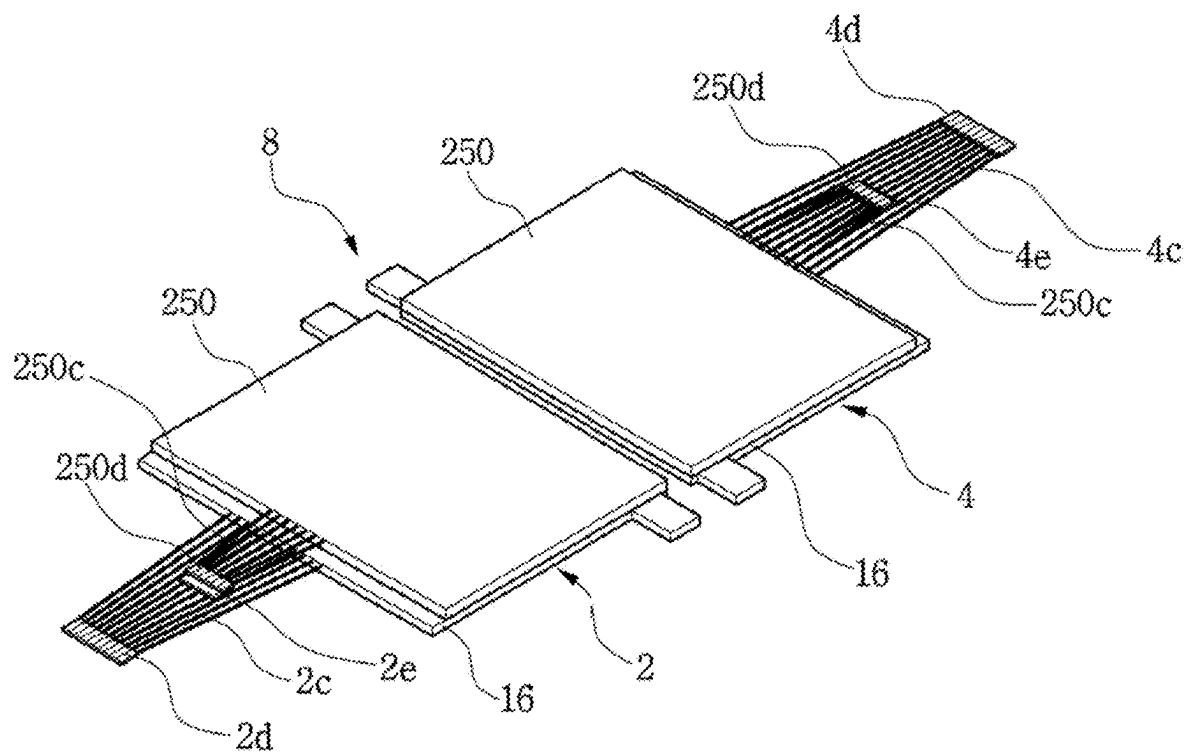


FIG. 22

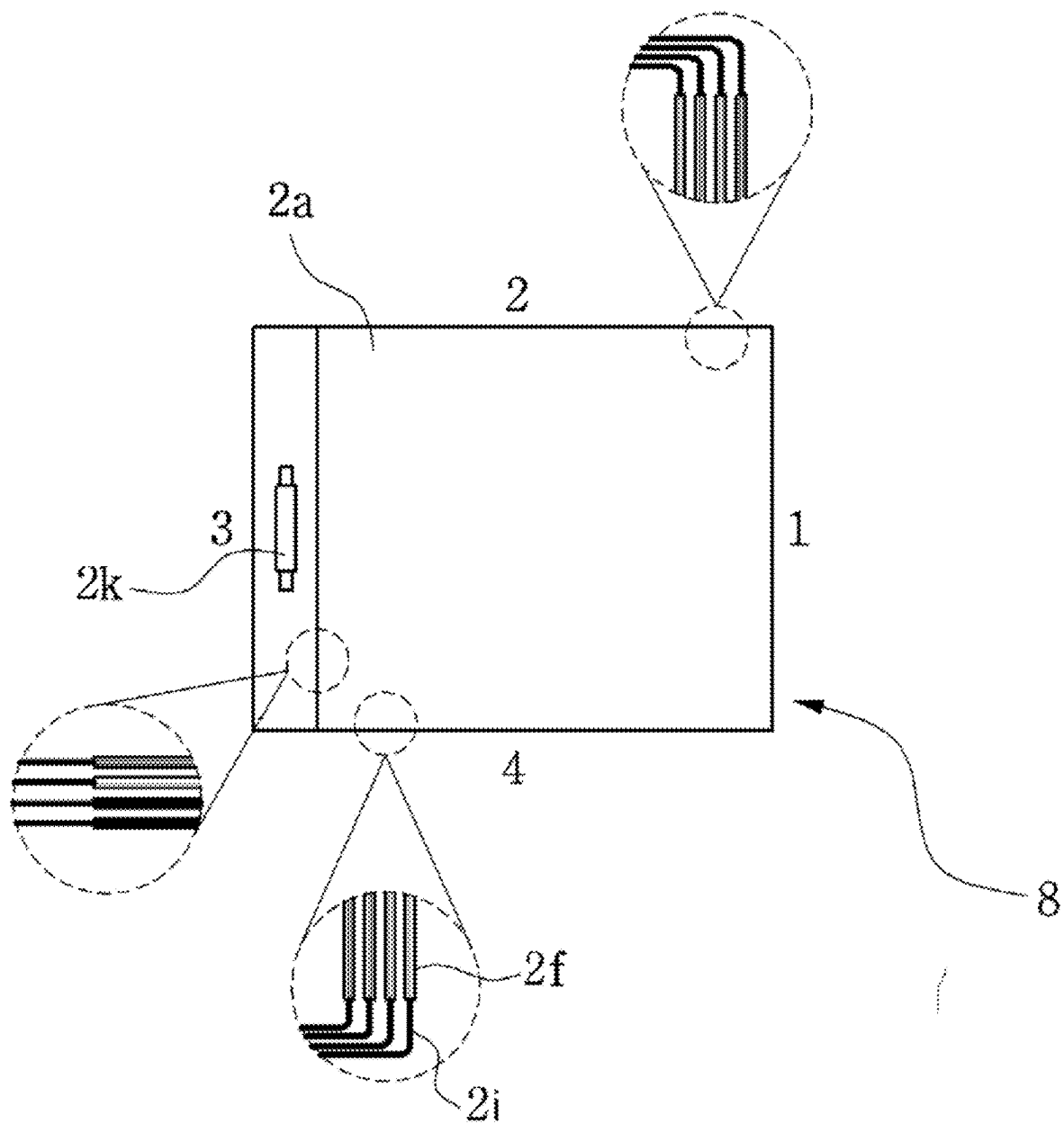


FIG. 23

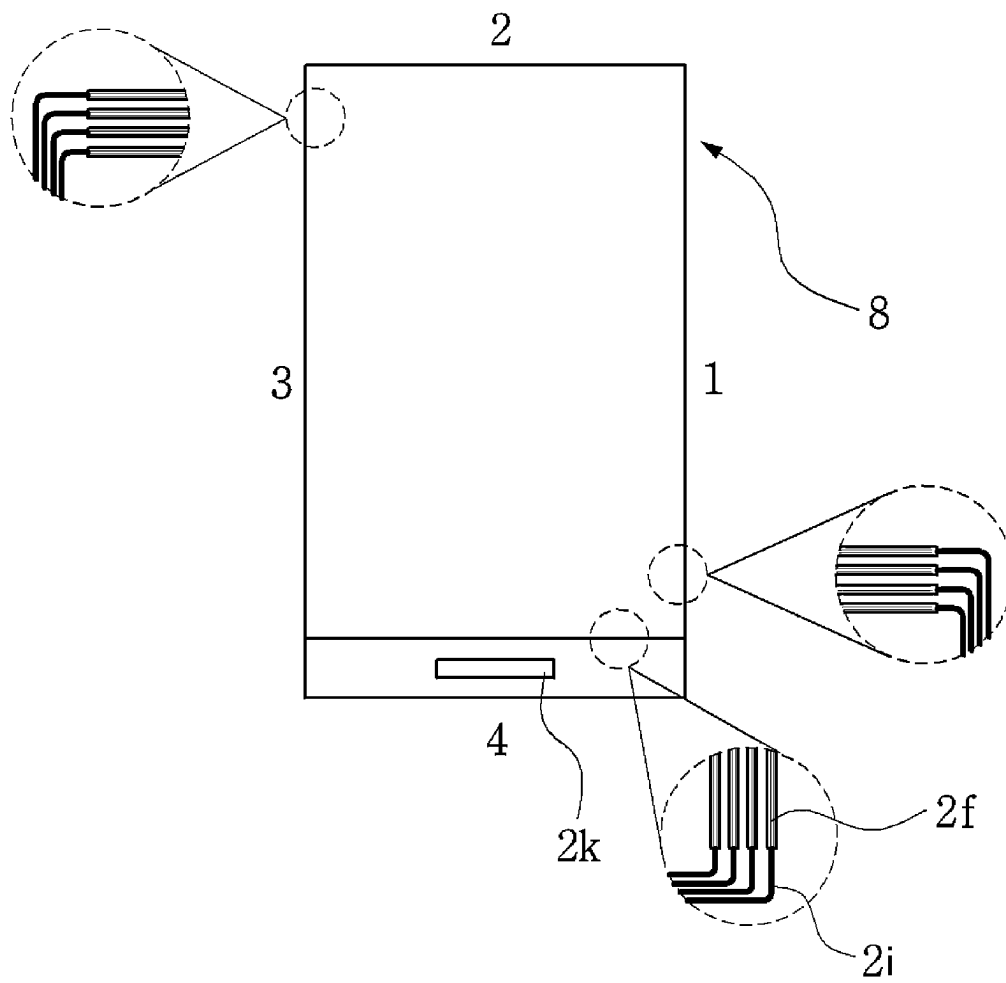


FIG. 24

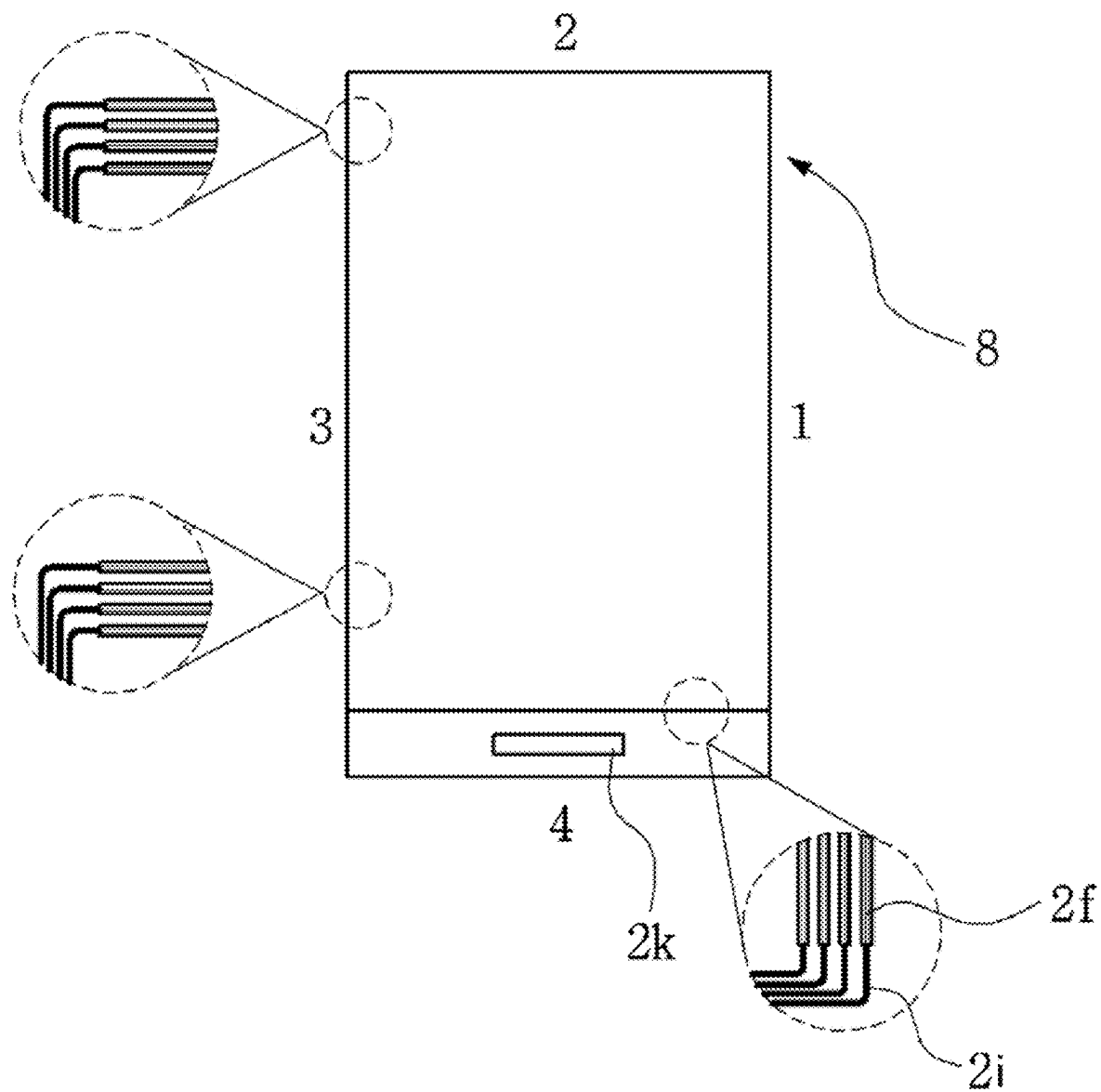


FIG. 25

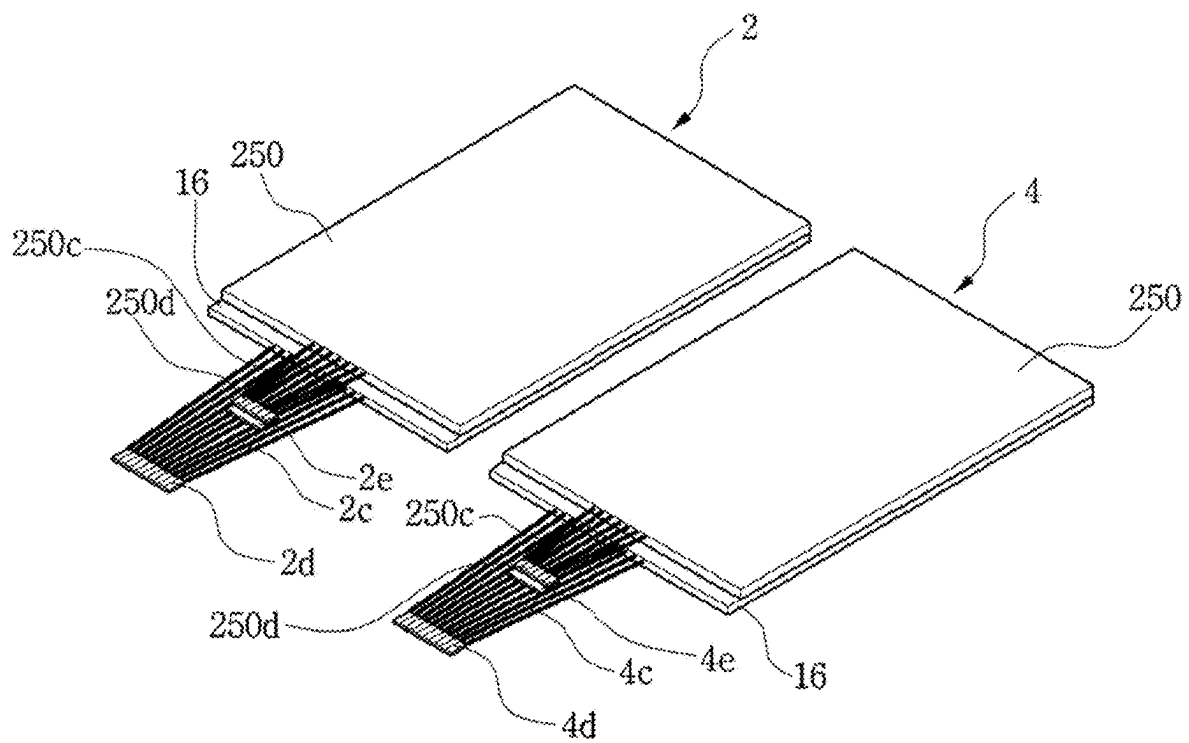


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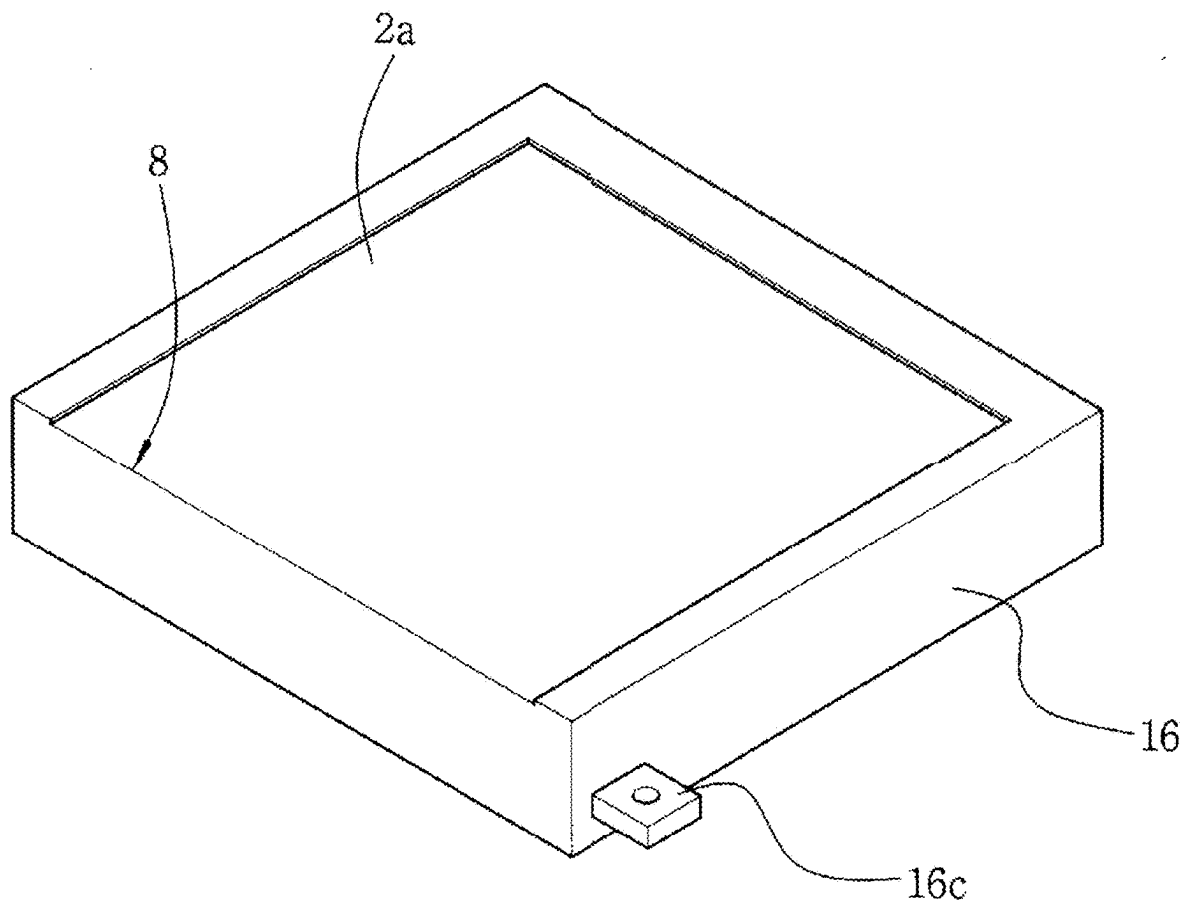


FIG. 27

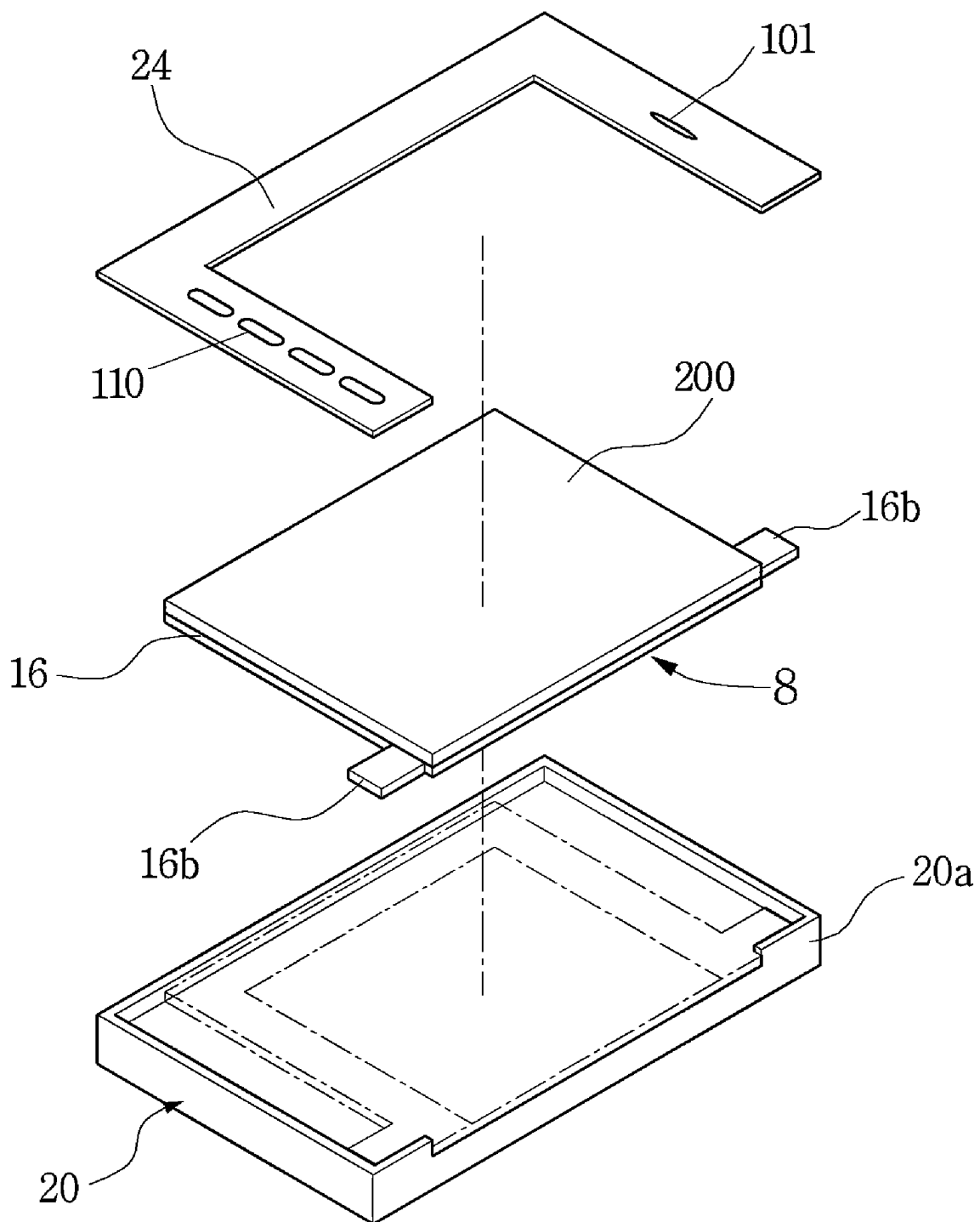


FIG. 28



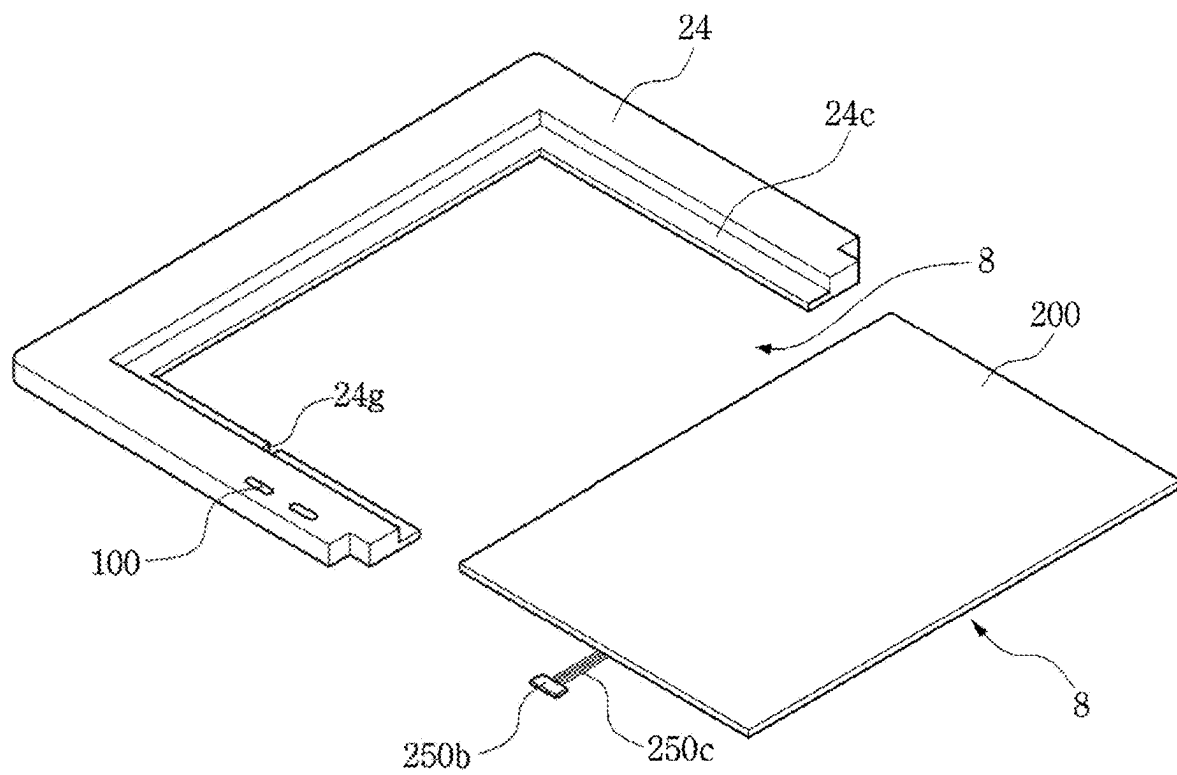


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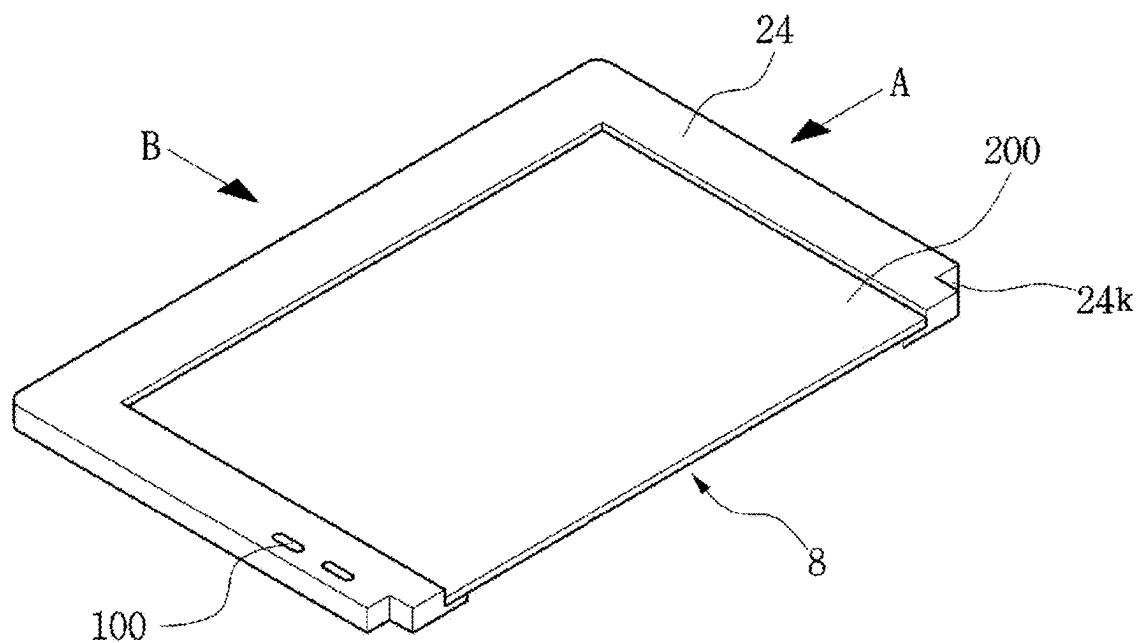


FIG. 30

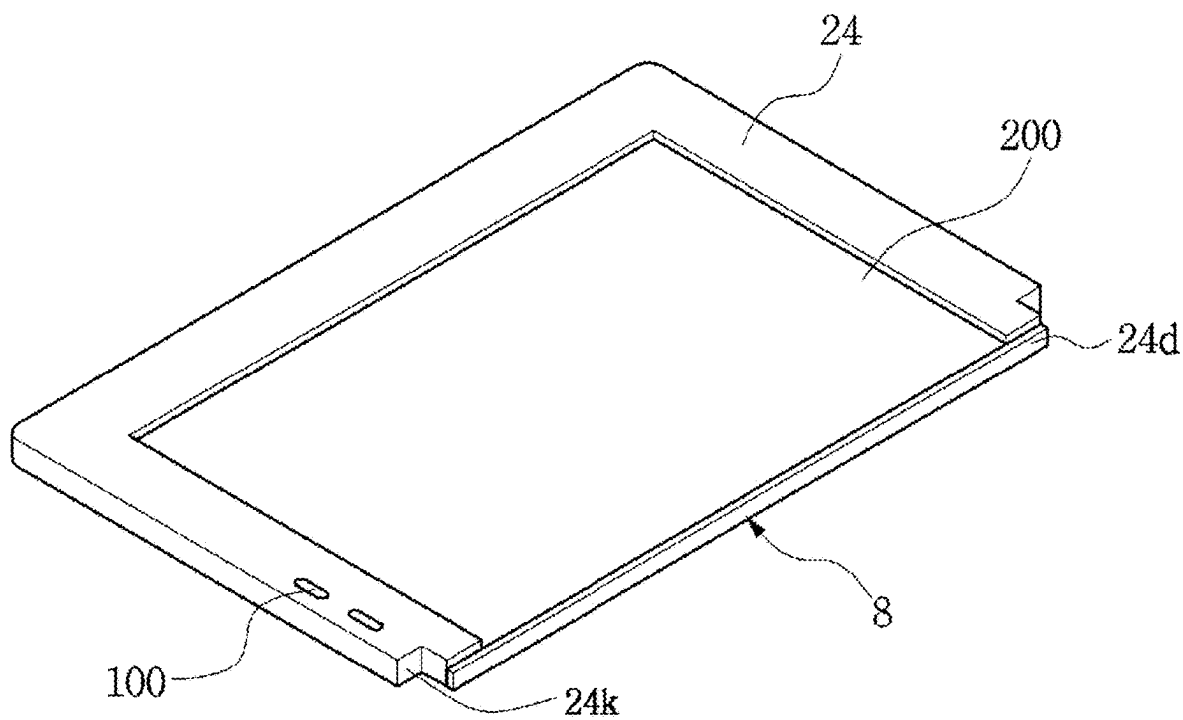


FIG. 31

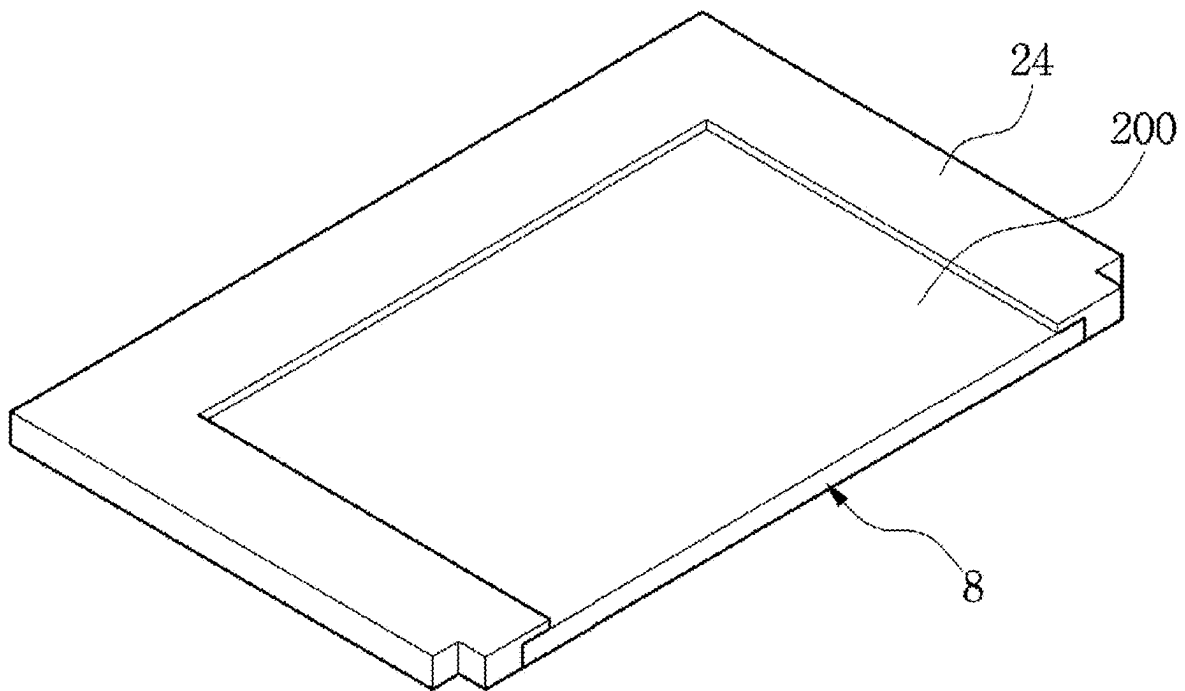


FIG. 32

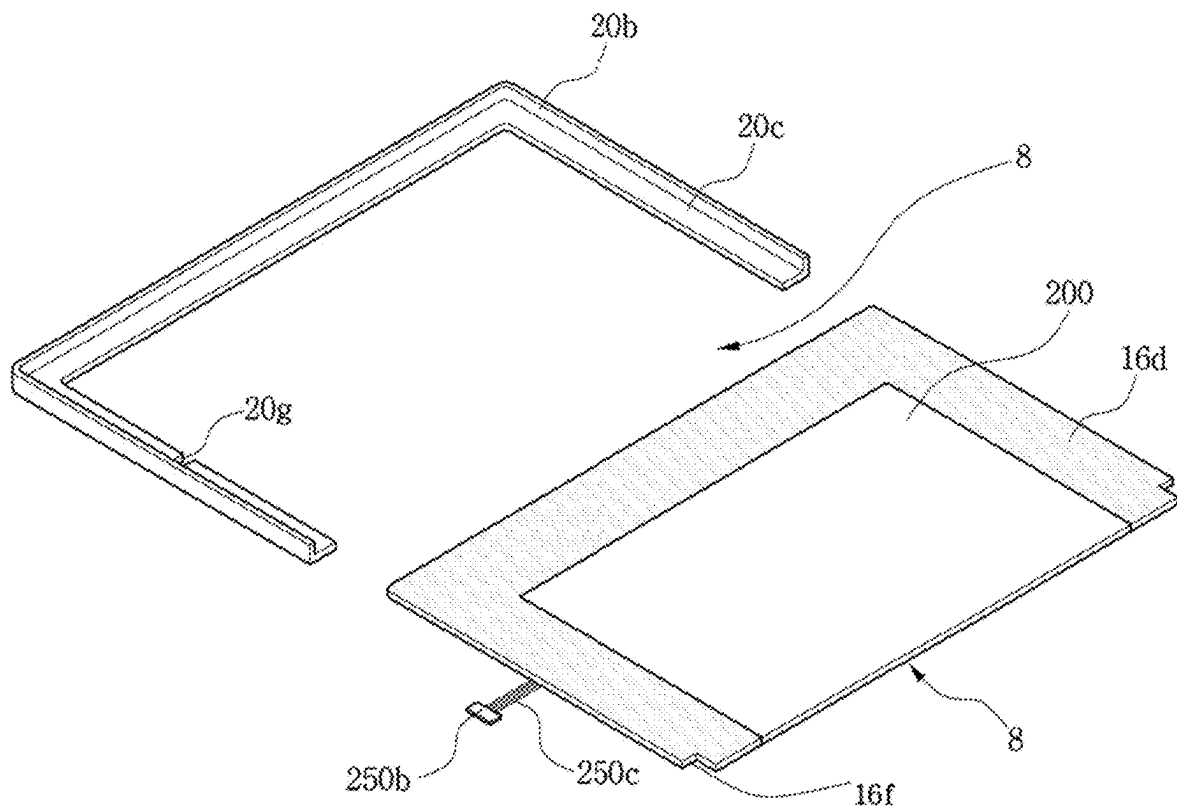


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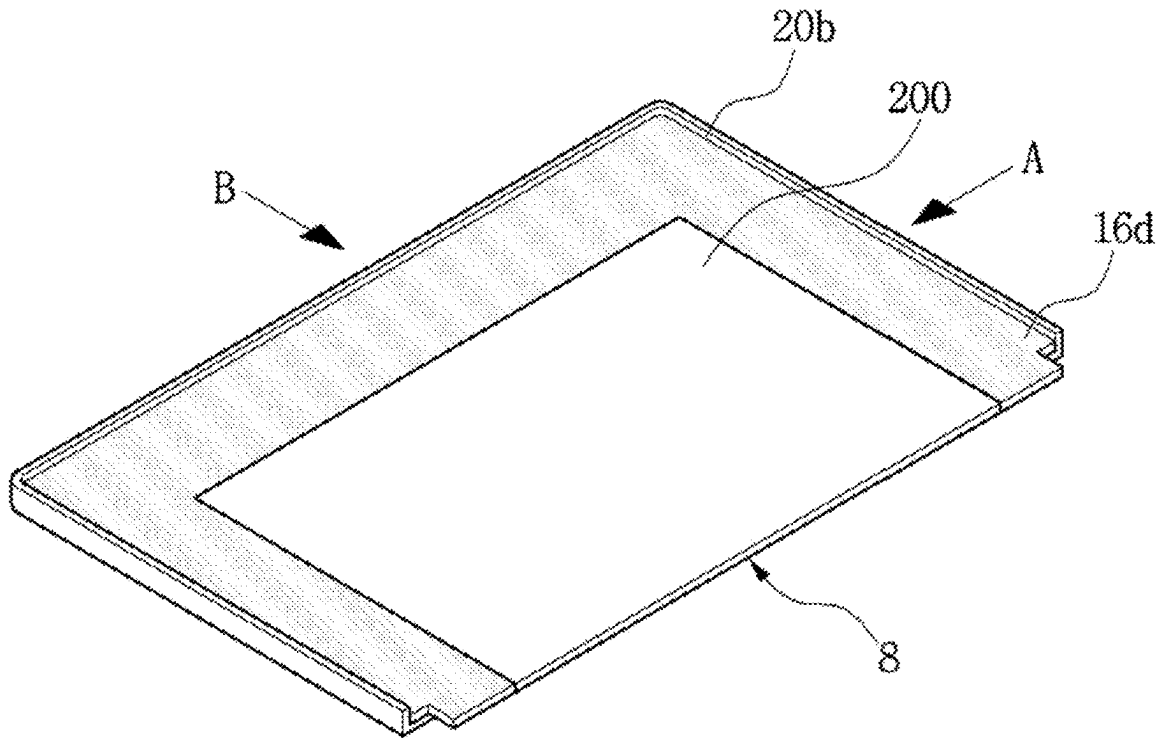


FIG. 34

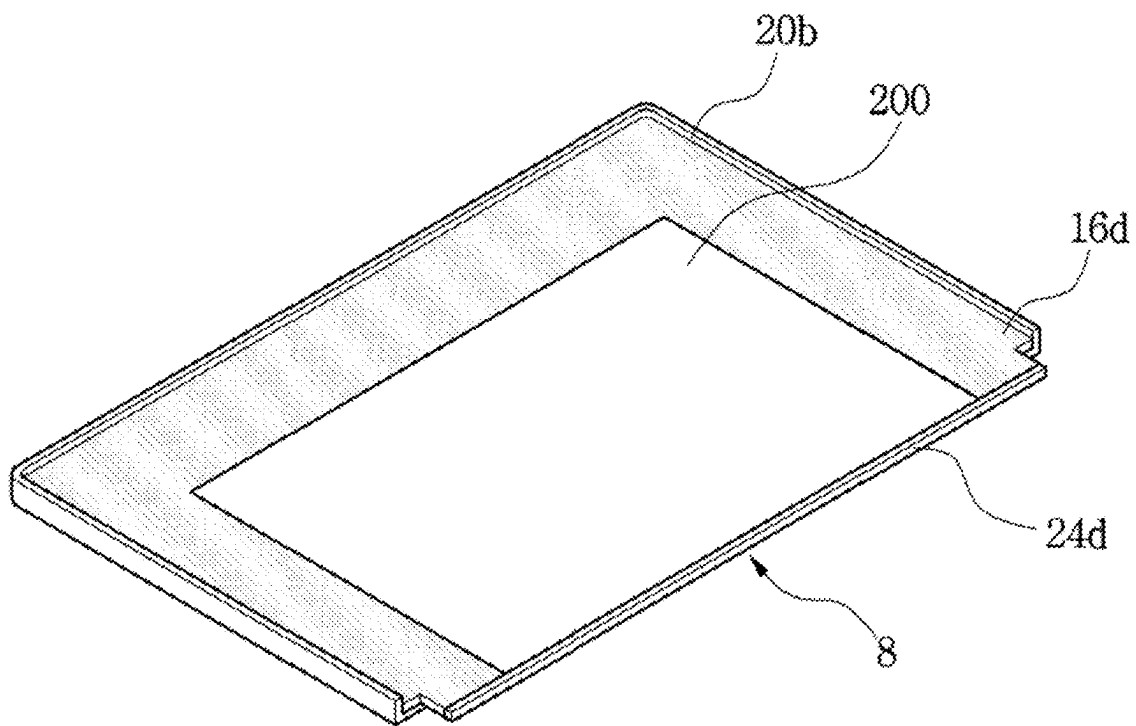


FIG. 35

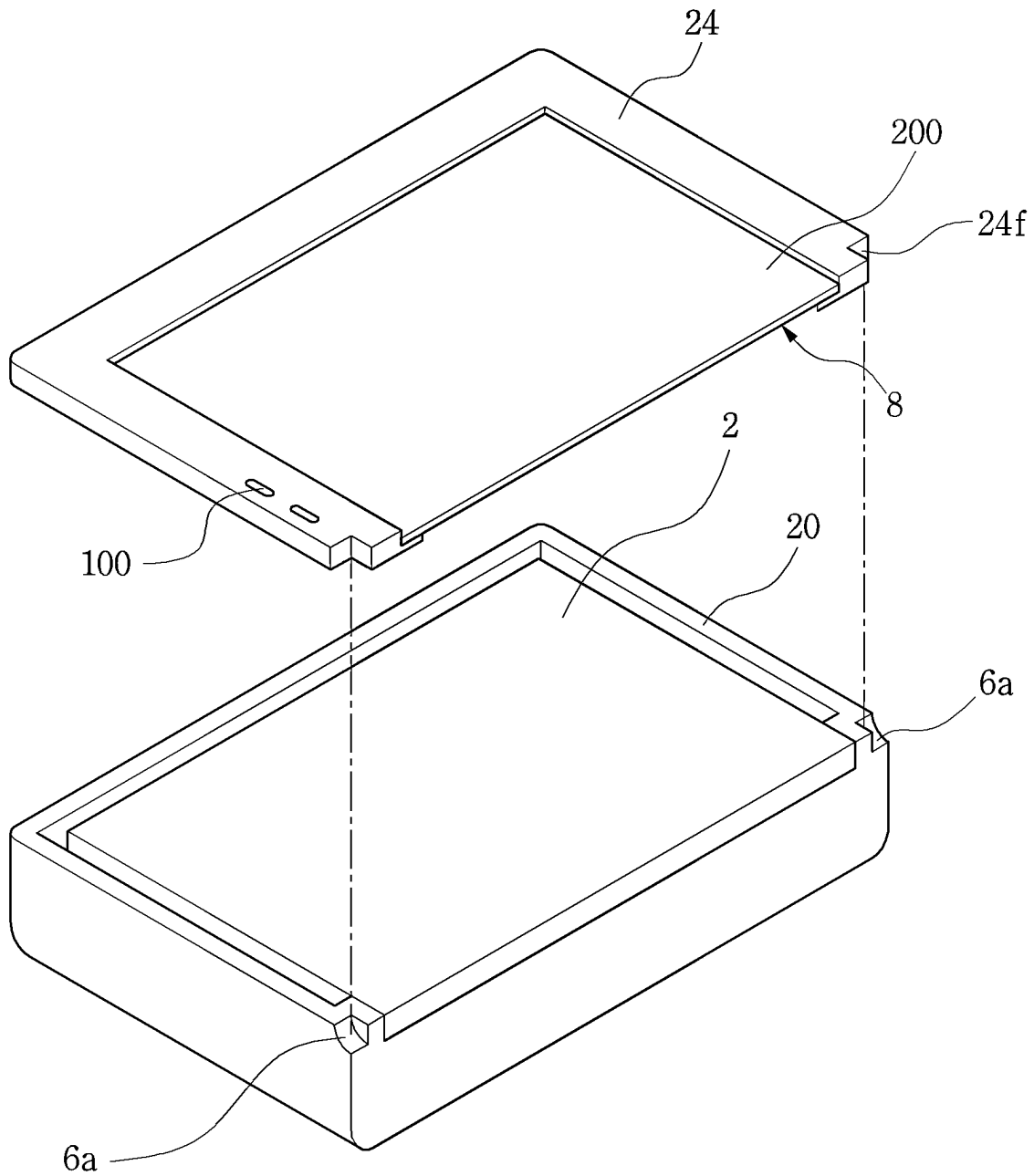


FIG. 36

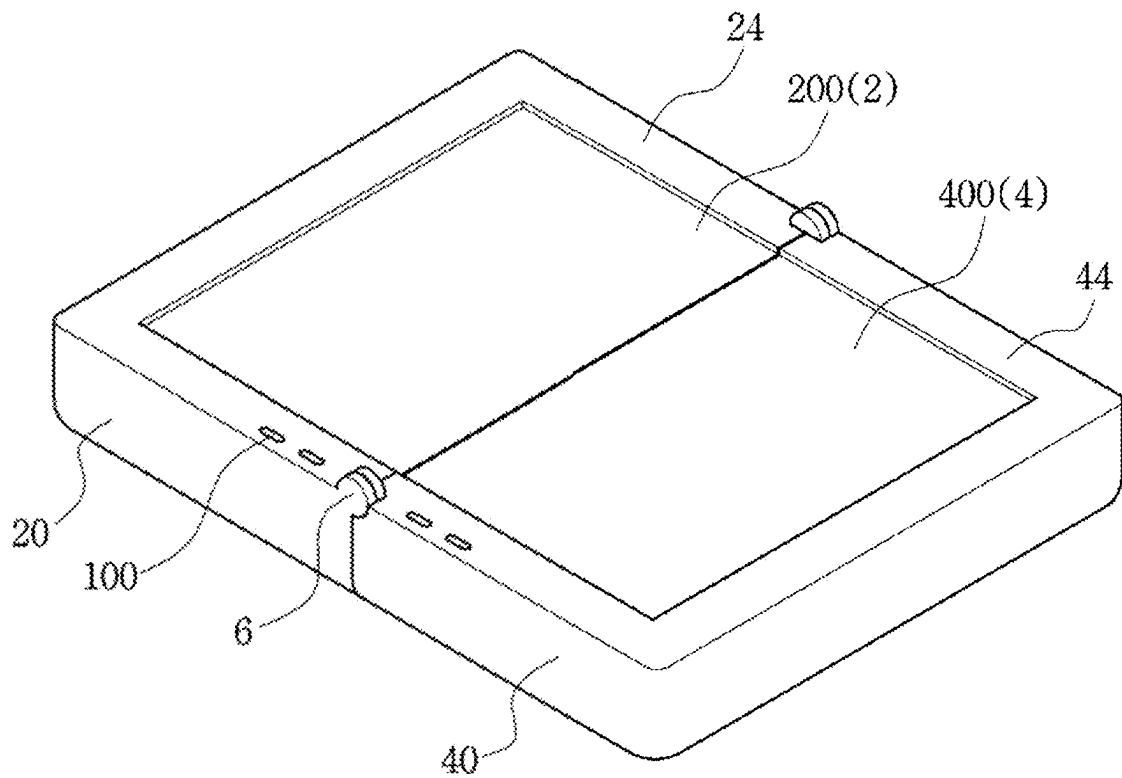


FIG. 37

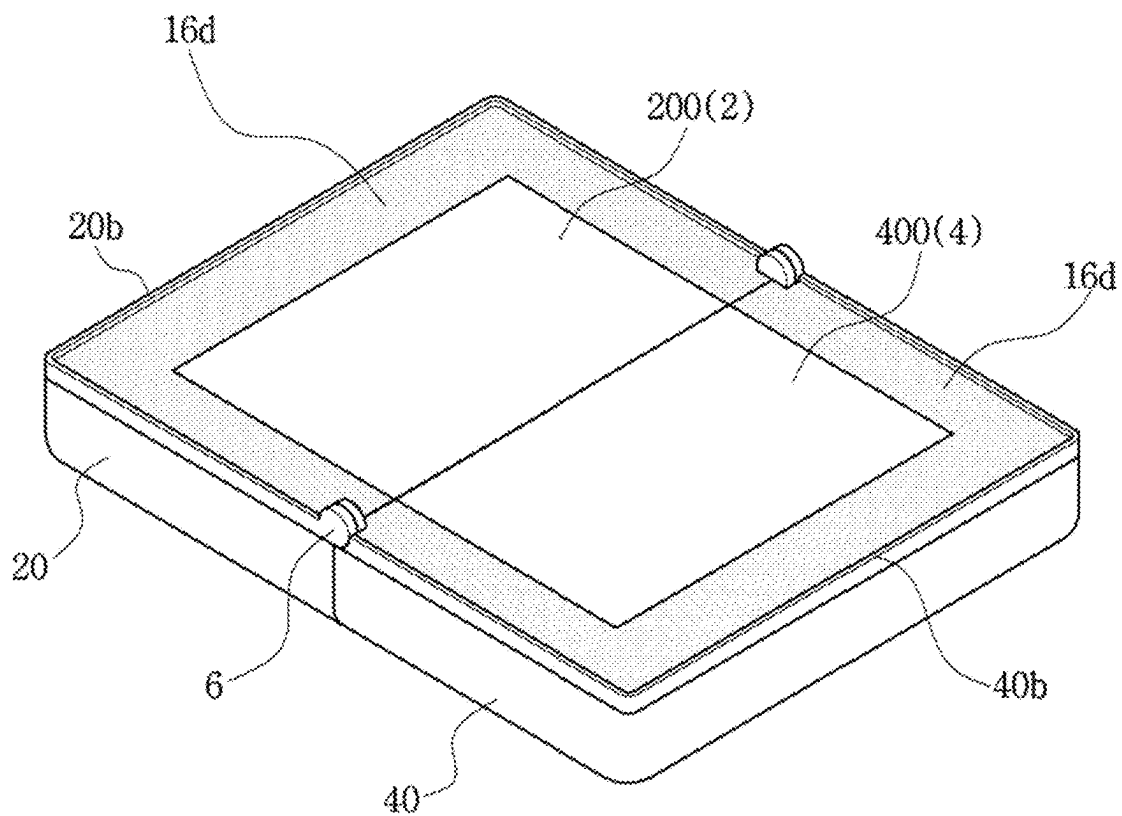


FIG. 38

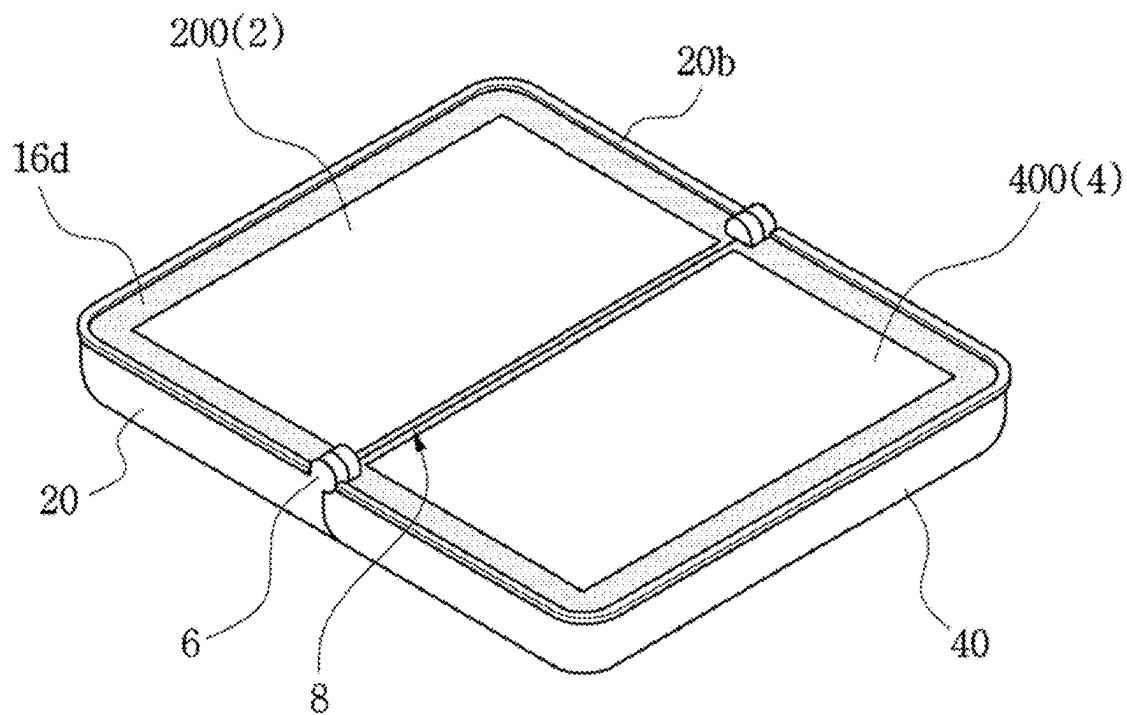


FIG. 39

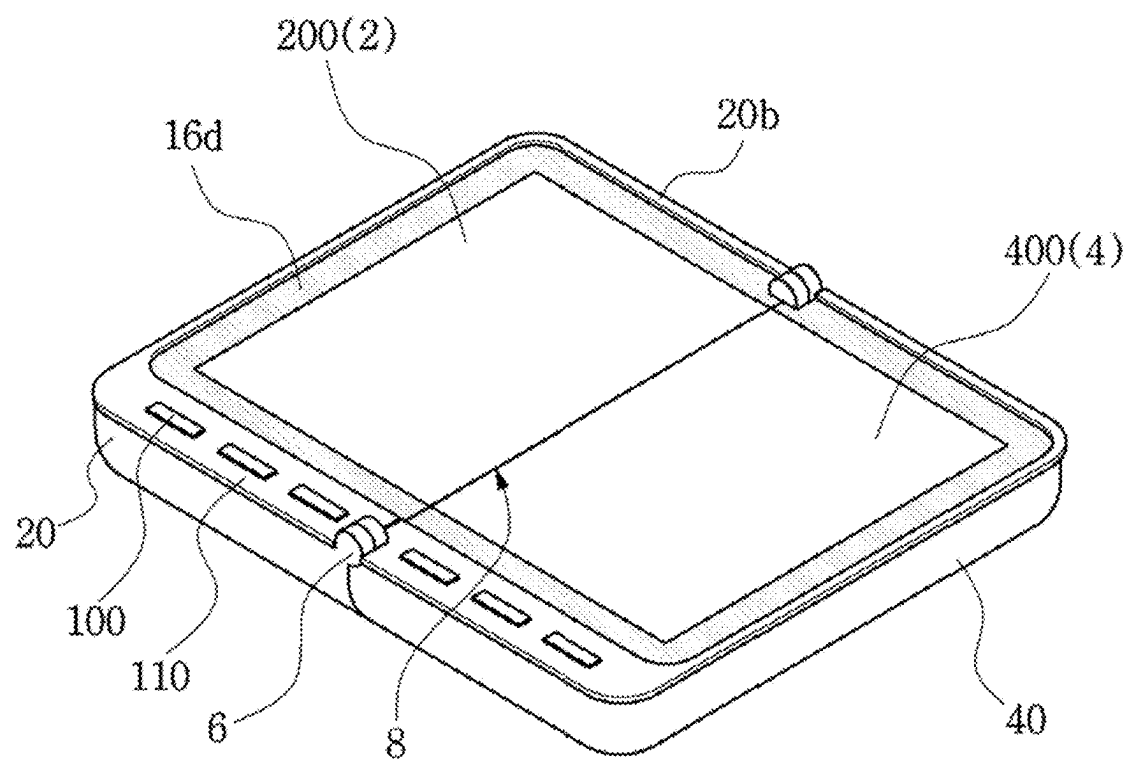


FIG. 40



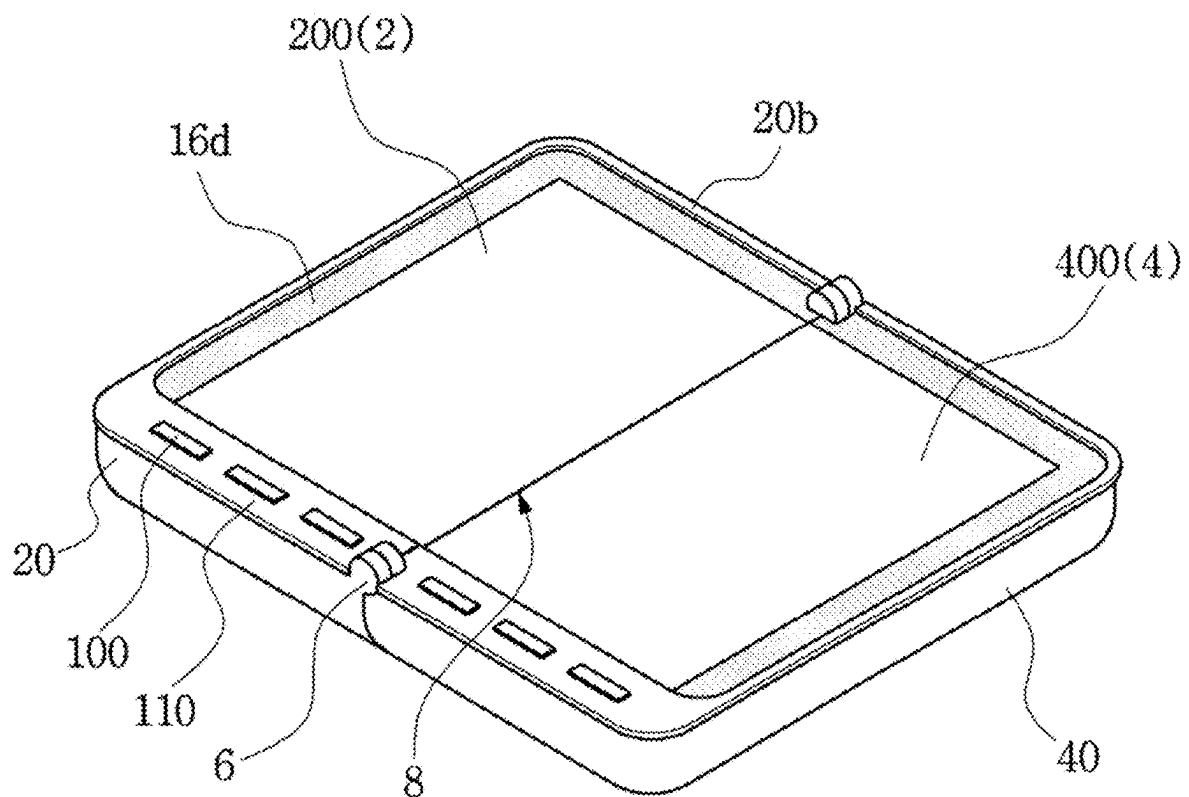


FIG. 41

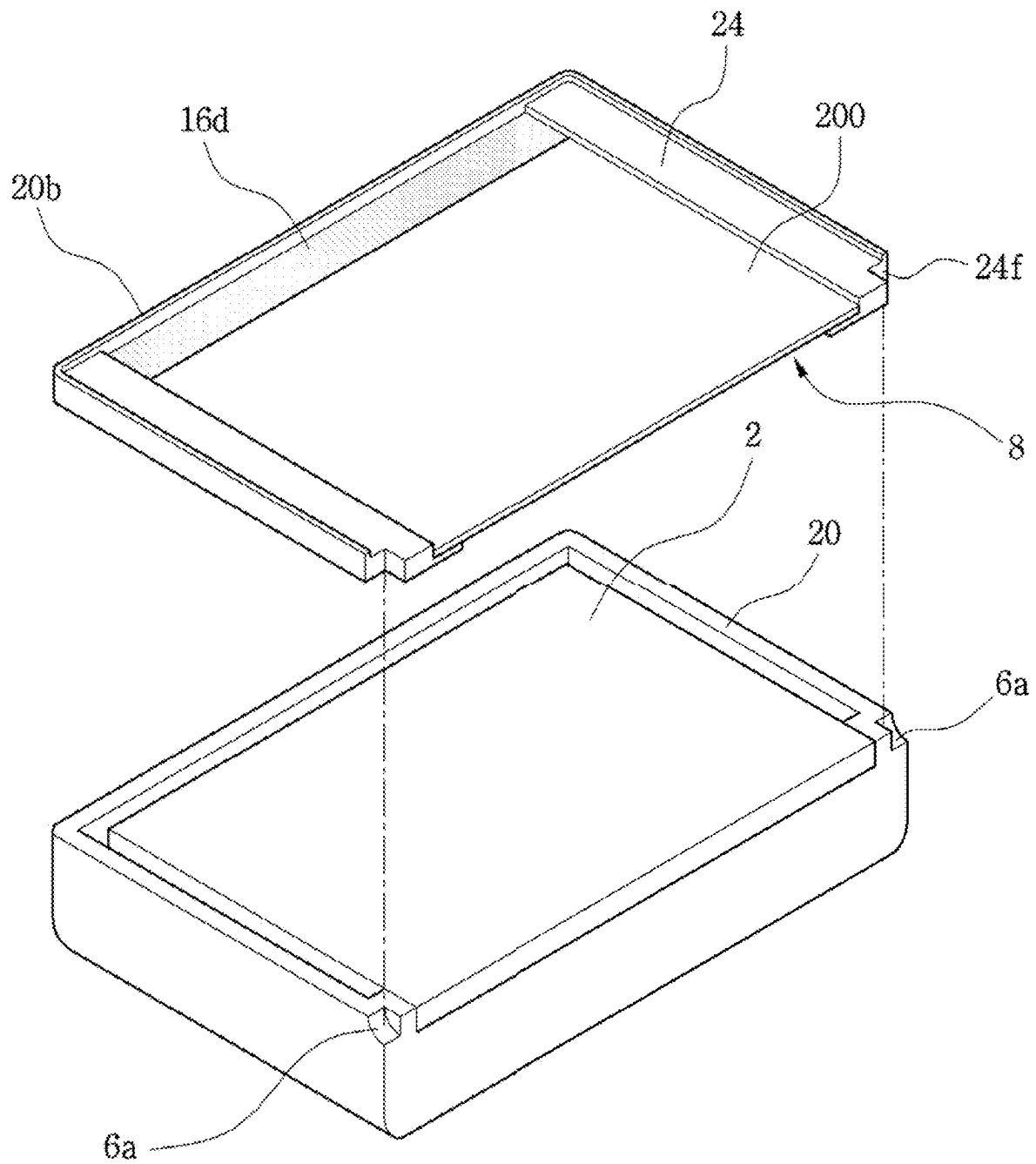


FIG. 42

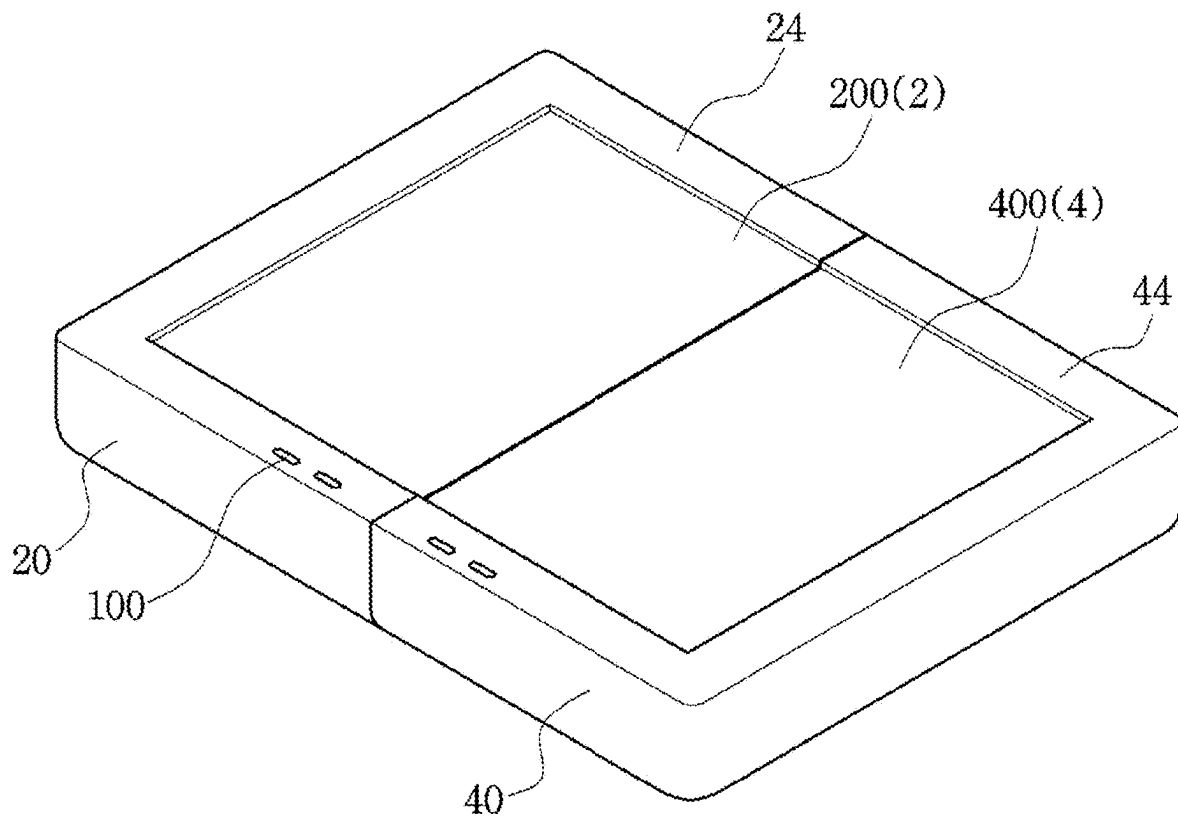


FIG. 43

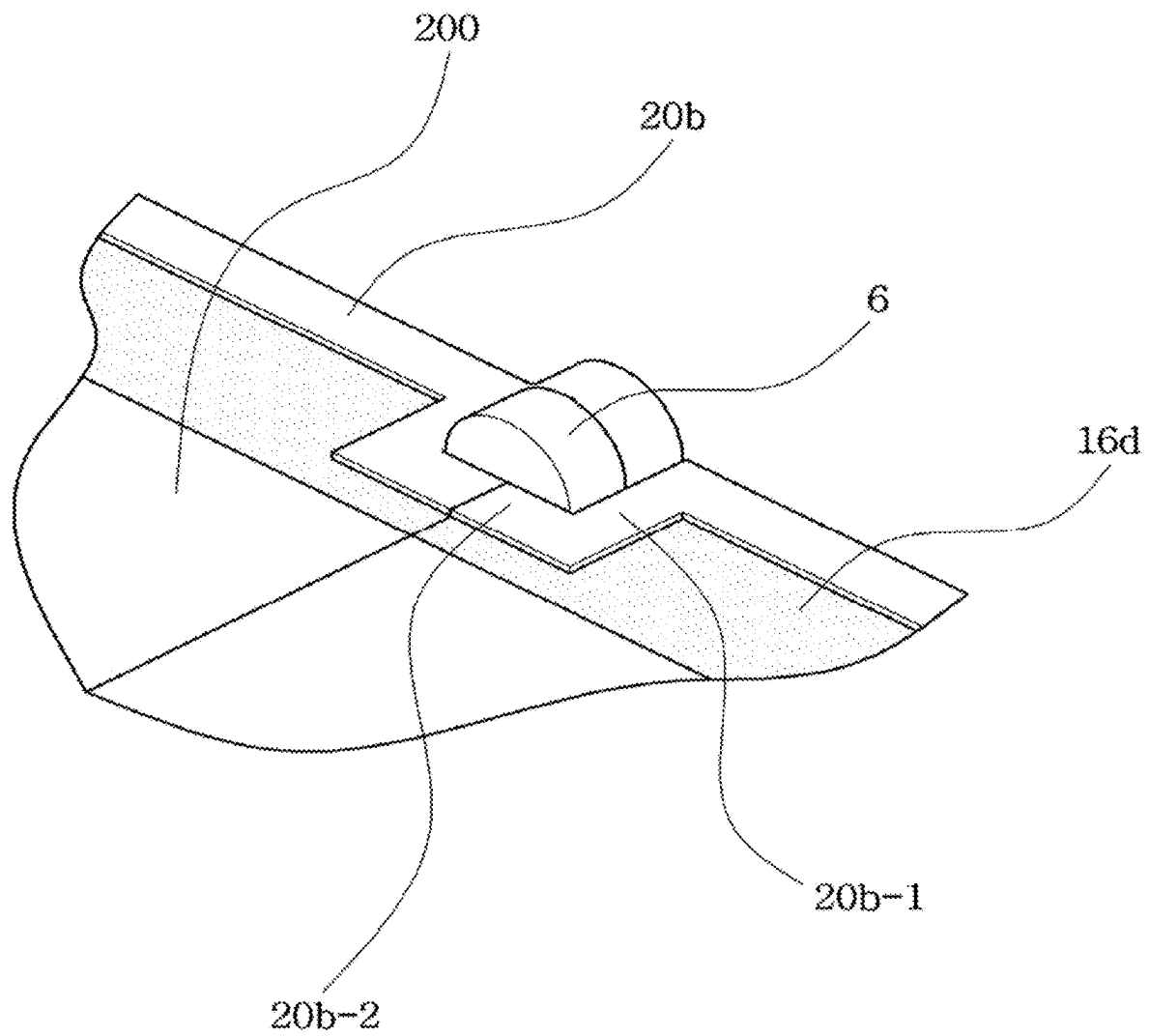


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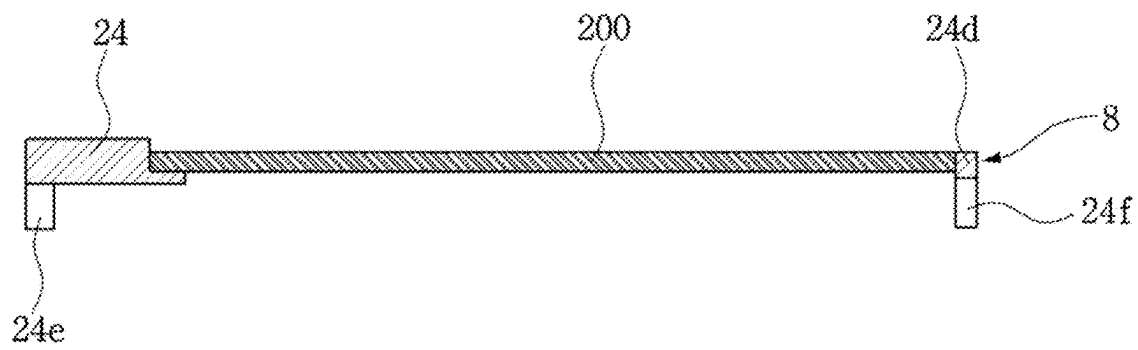


FIG. 45

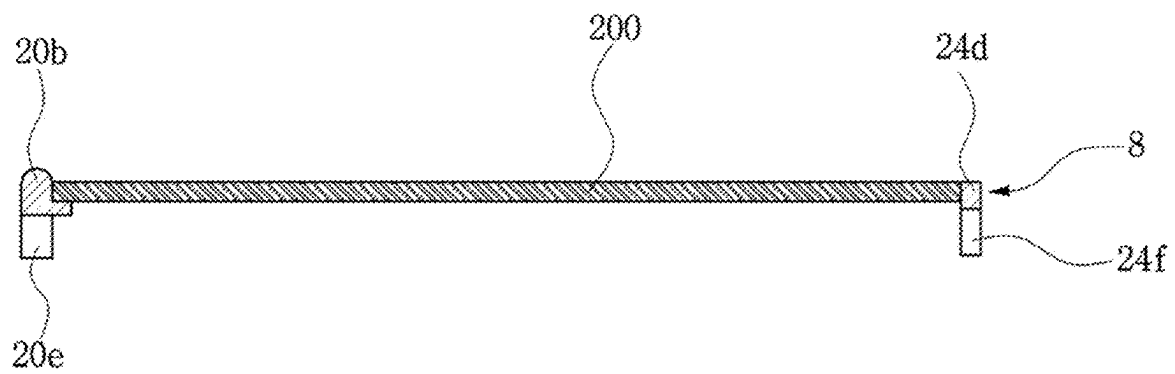


FIG. 46

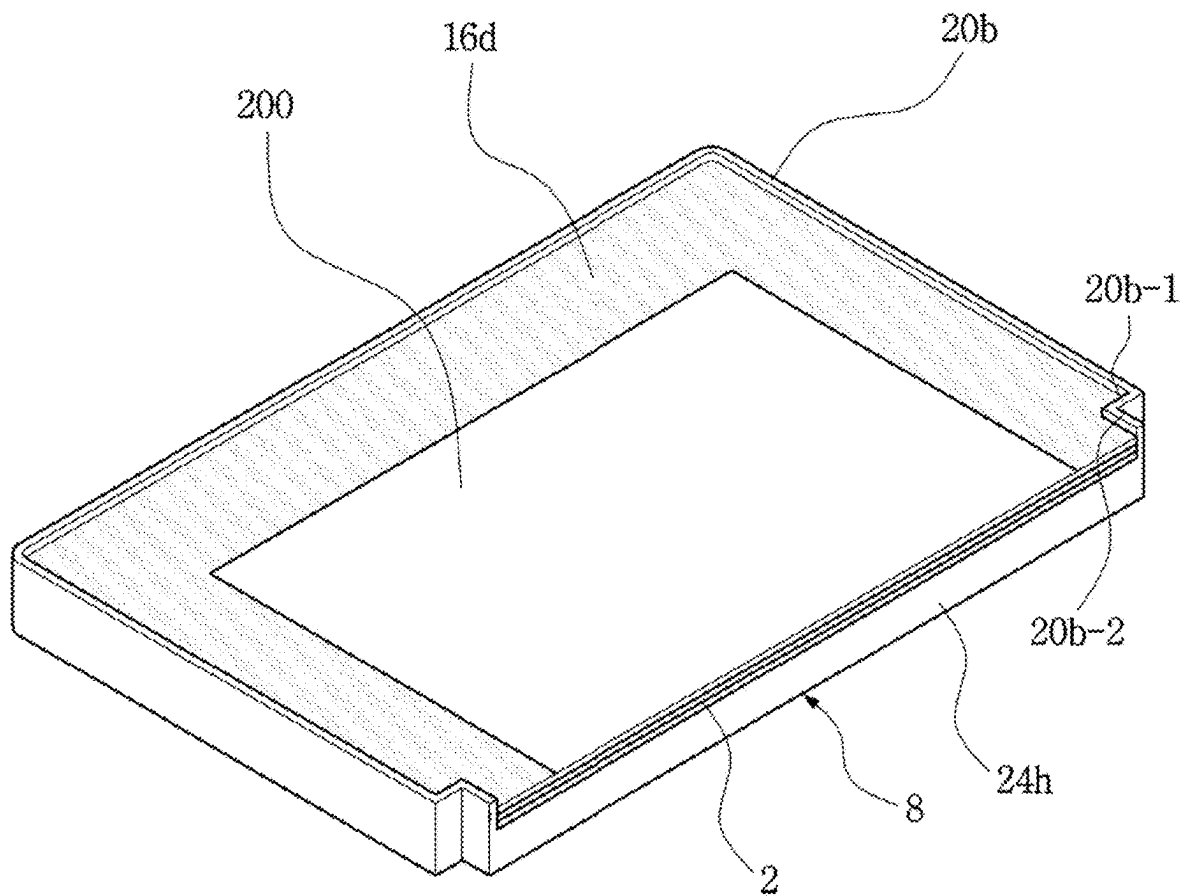


FIG. 47

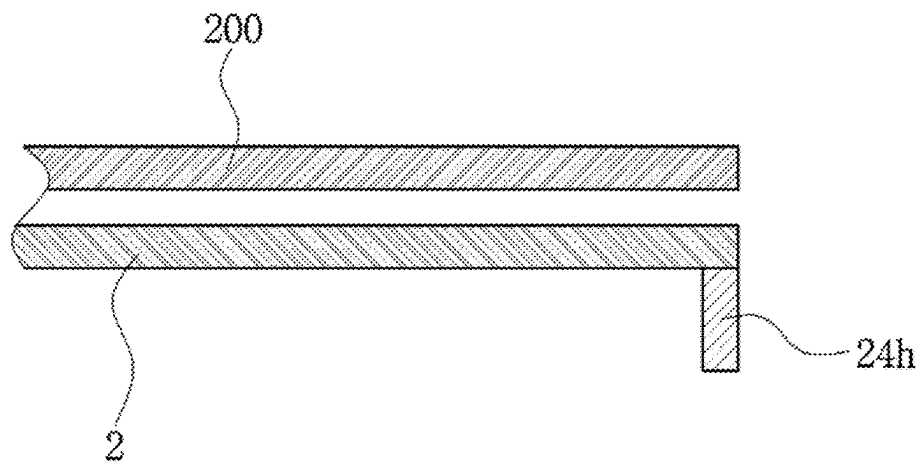


FIG. 48

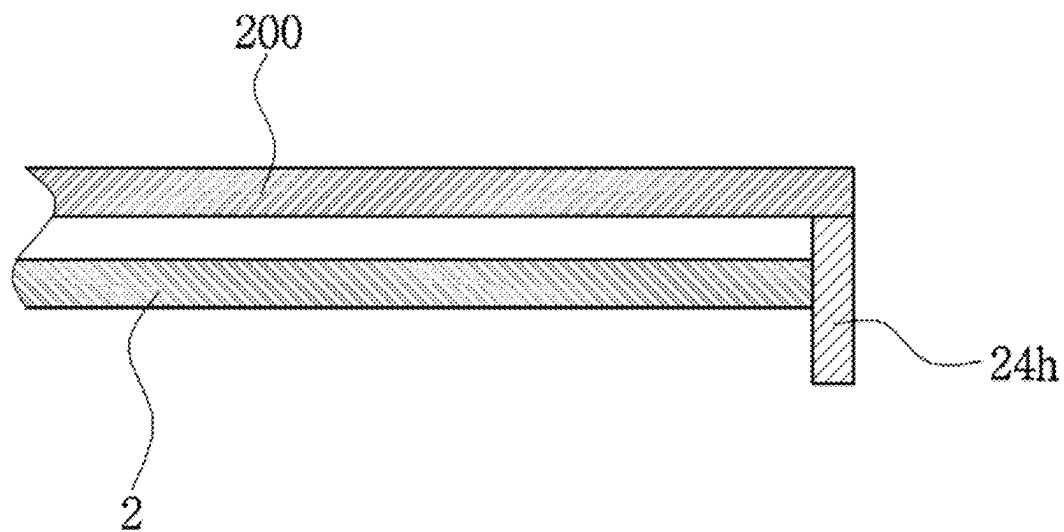


FIG. 49

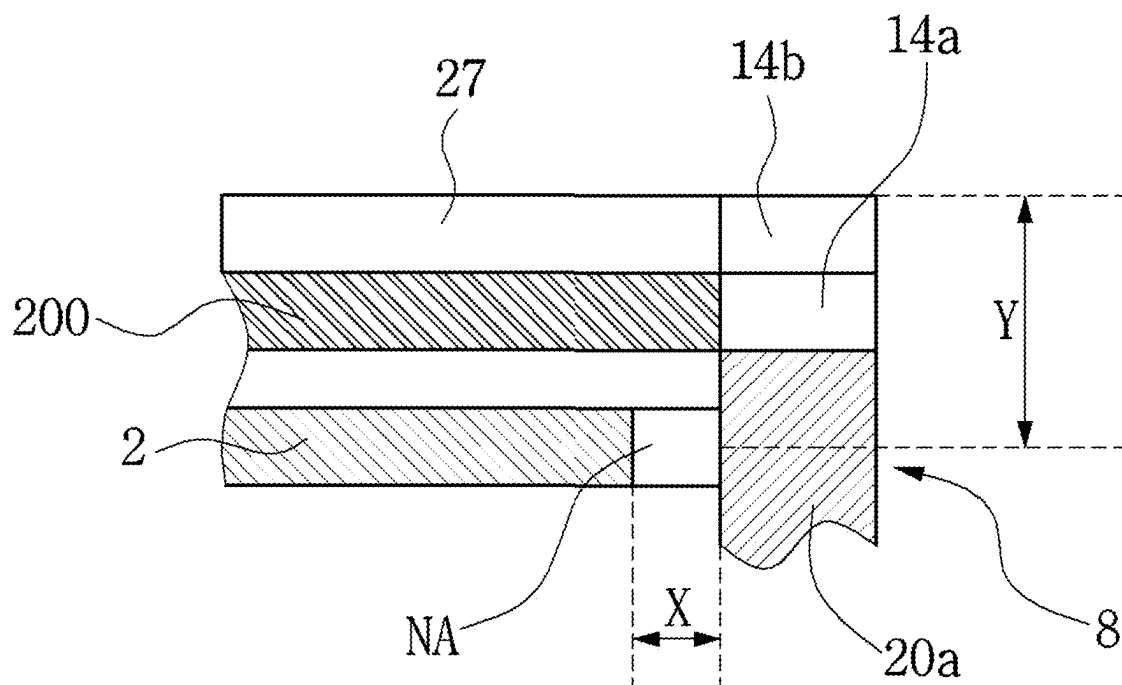


FIG. 50

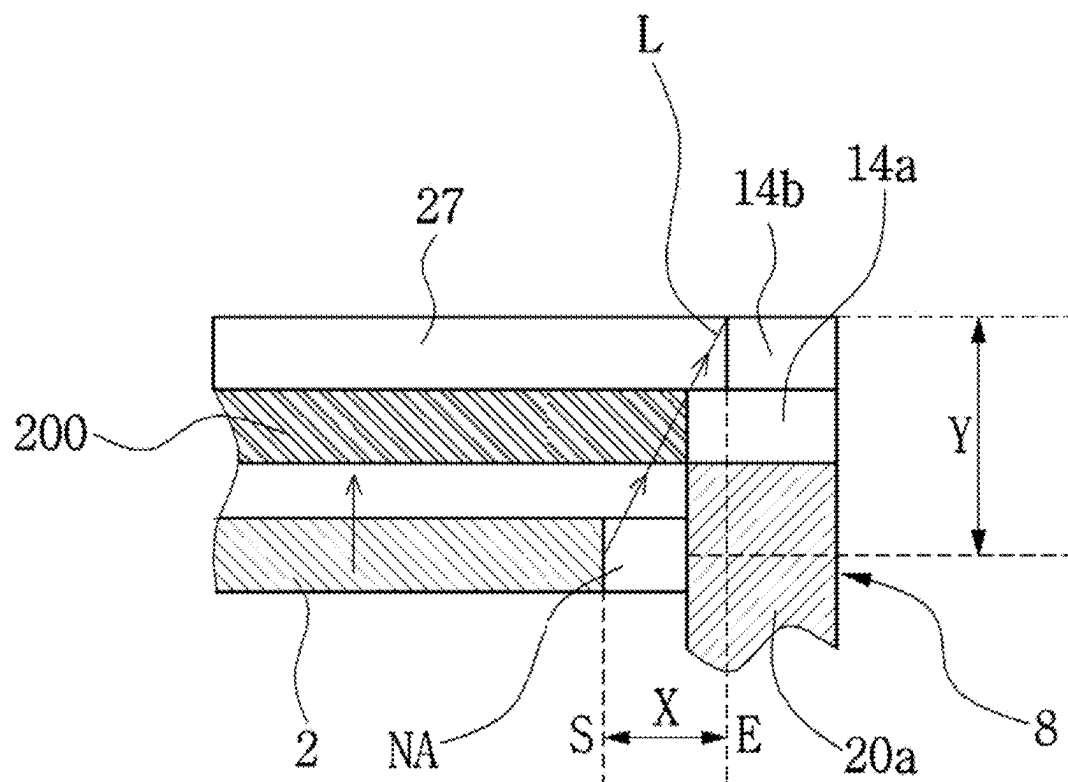


FIG. 51

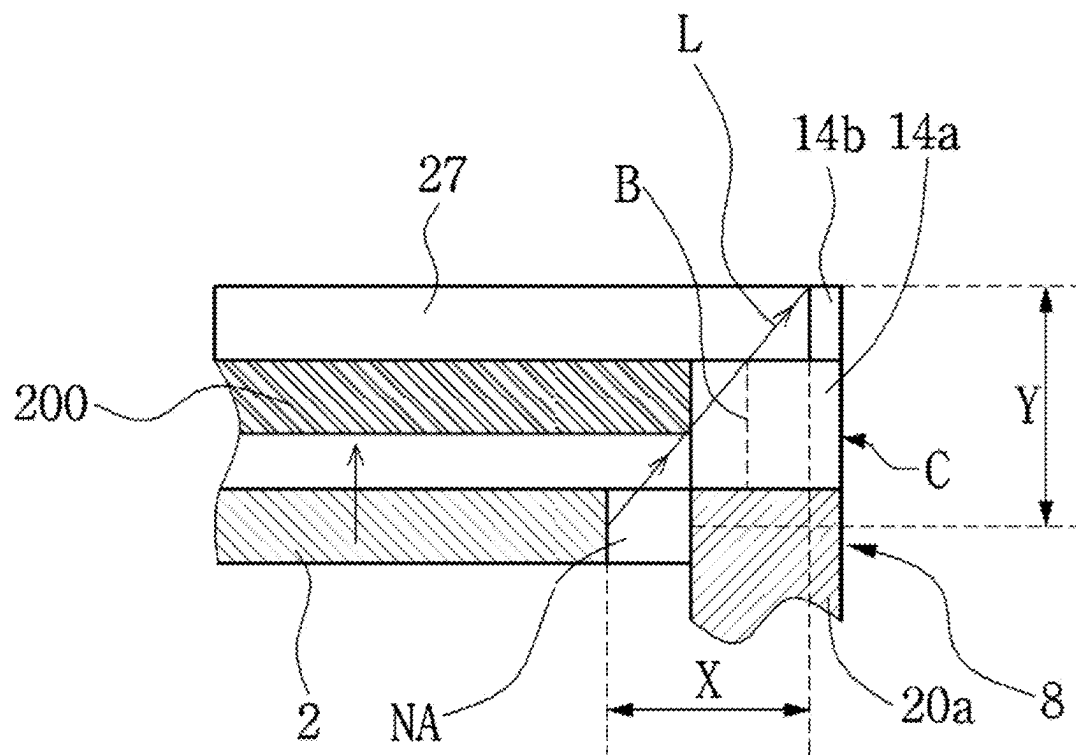


FIG. 52



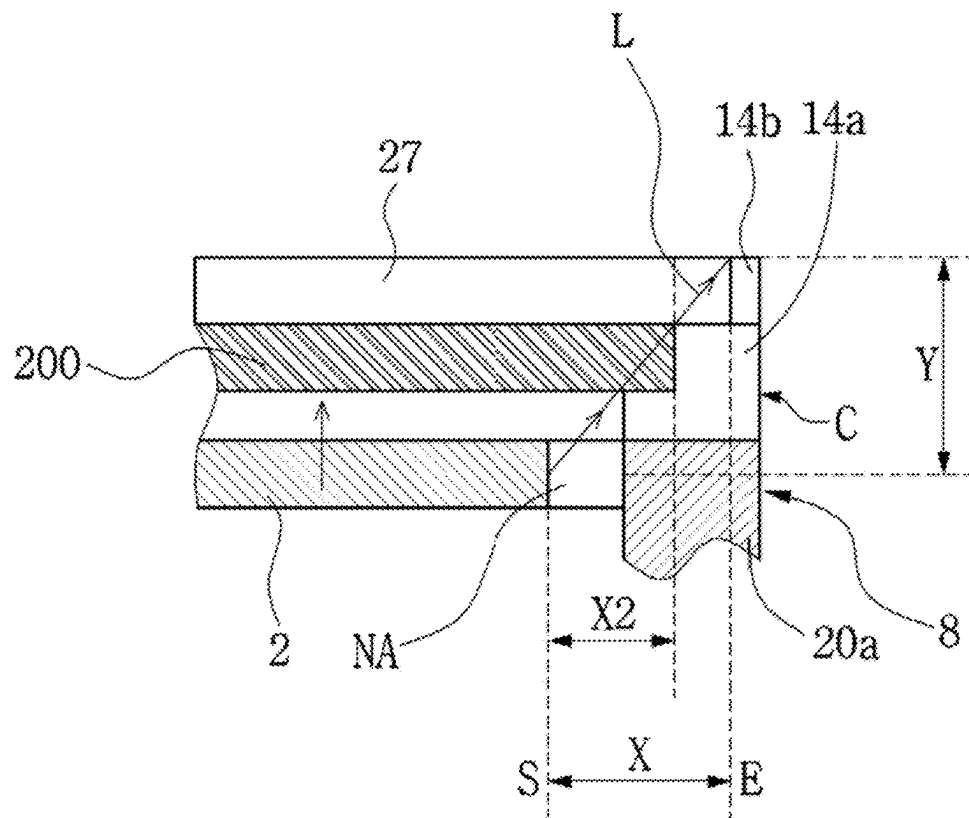


FIG. 53

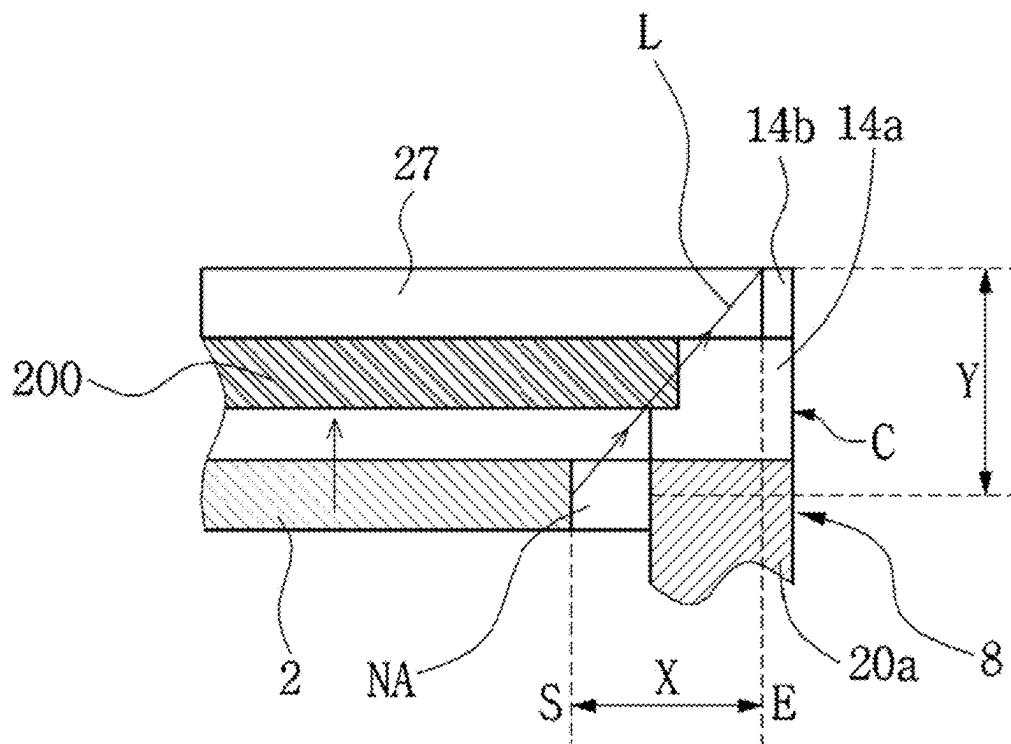


FIG. 54

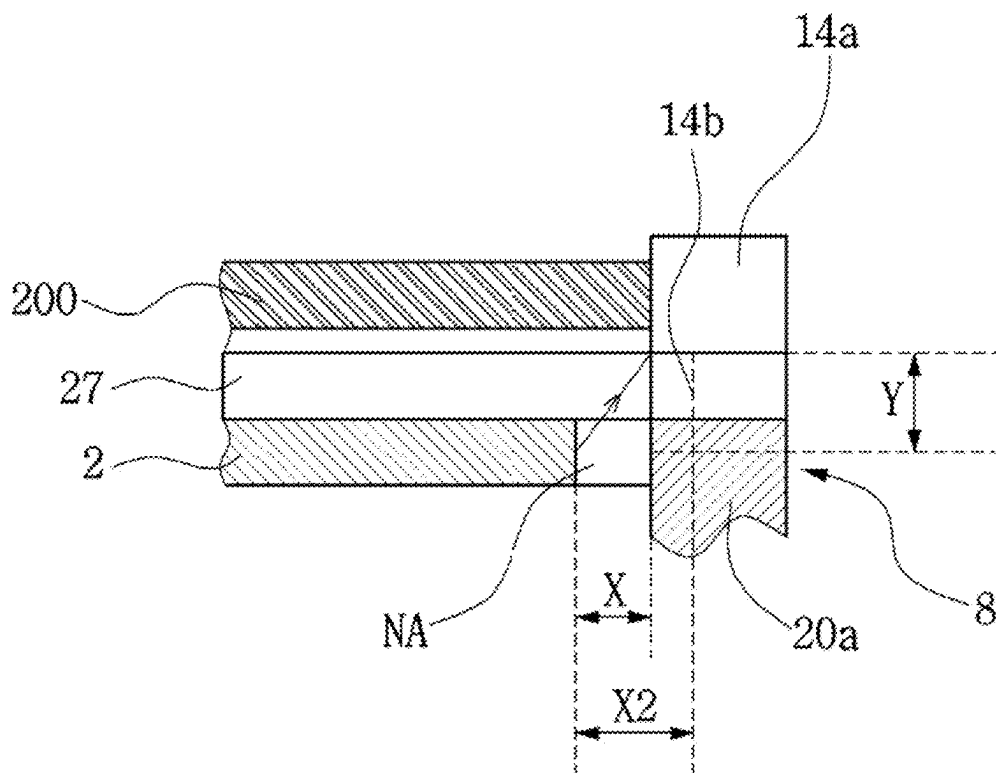


FIG. 55

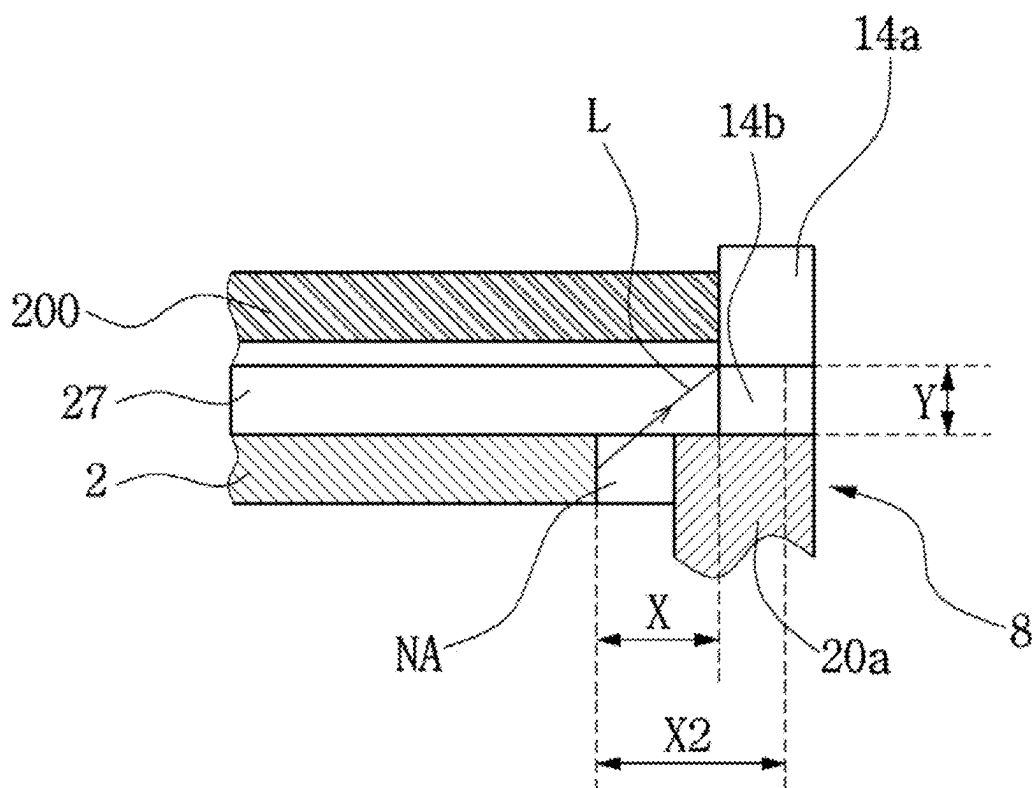


FIG. 56

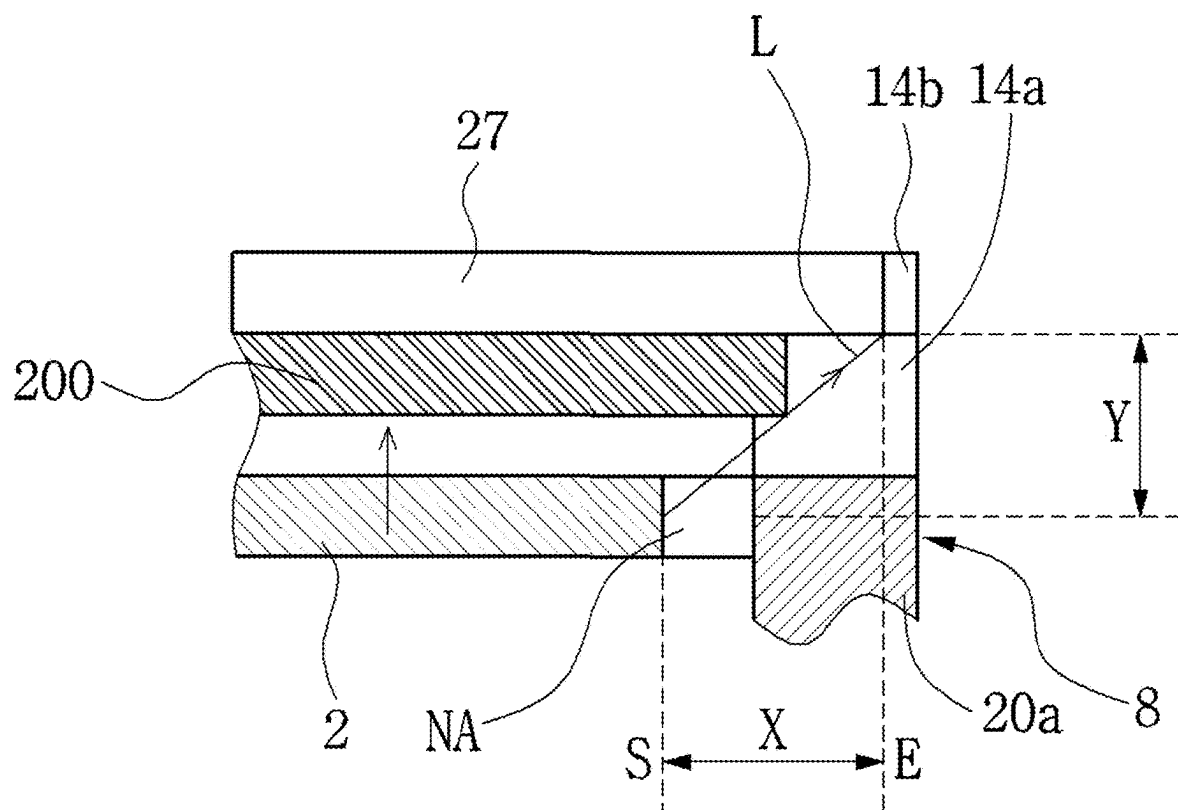


FIG. 57

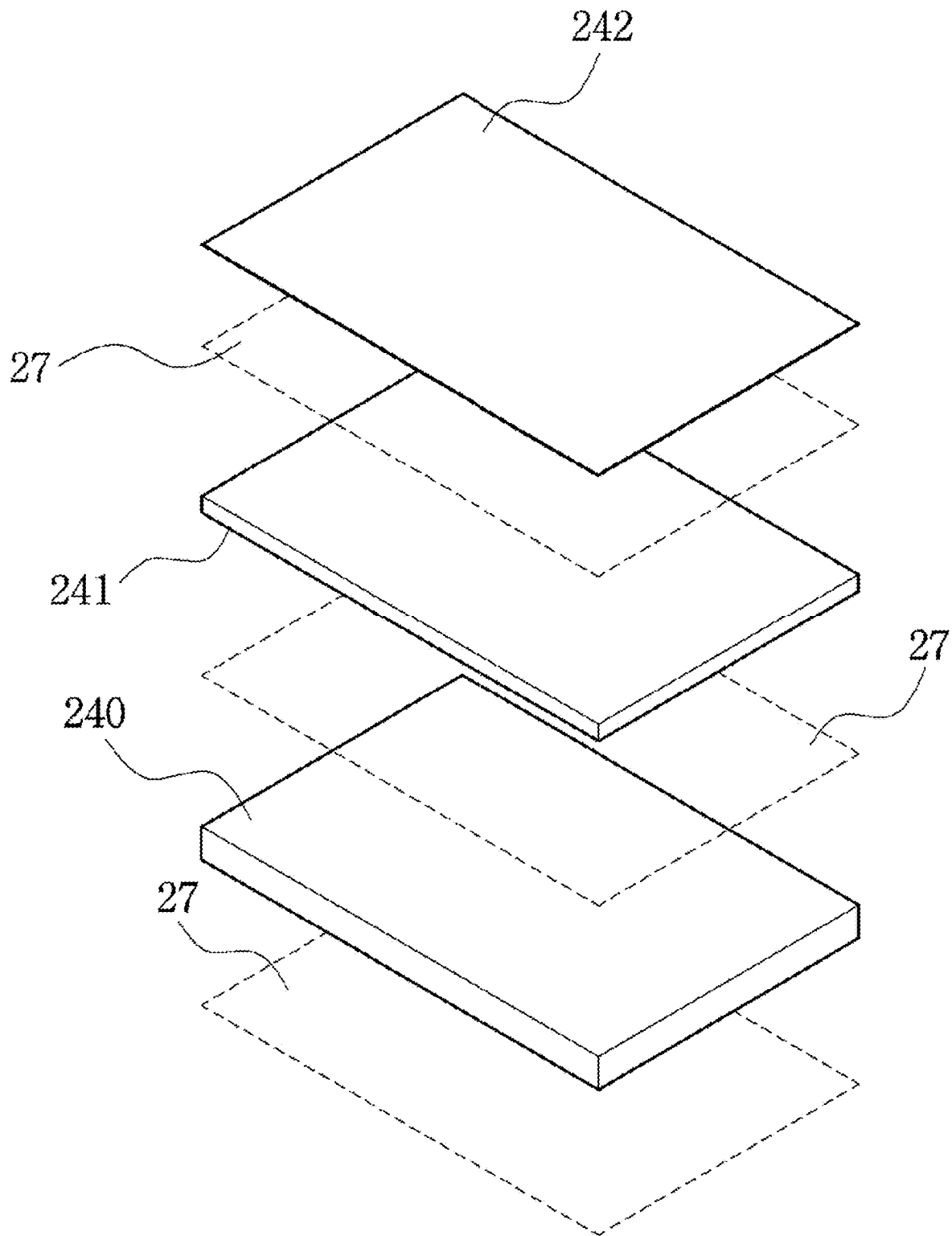


FIG. 58

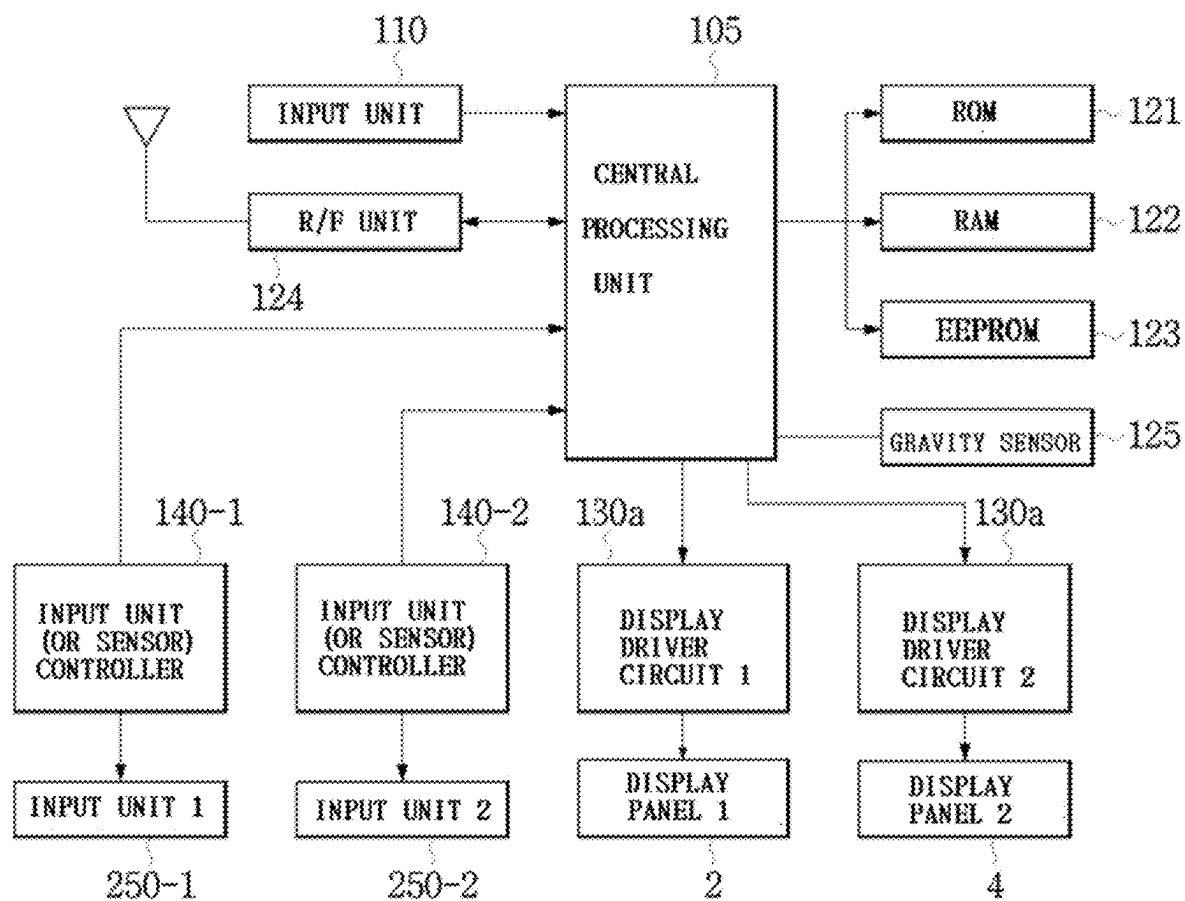


FIG. 59

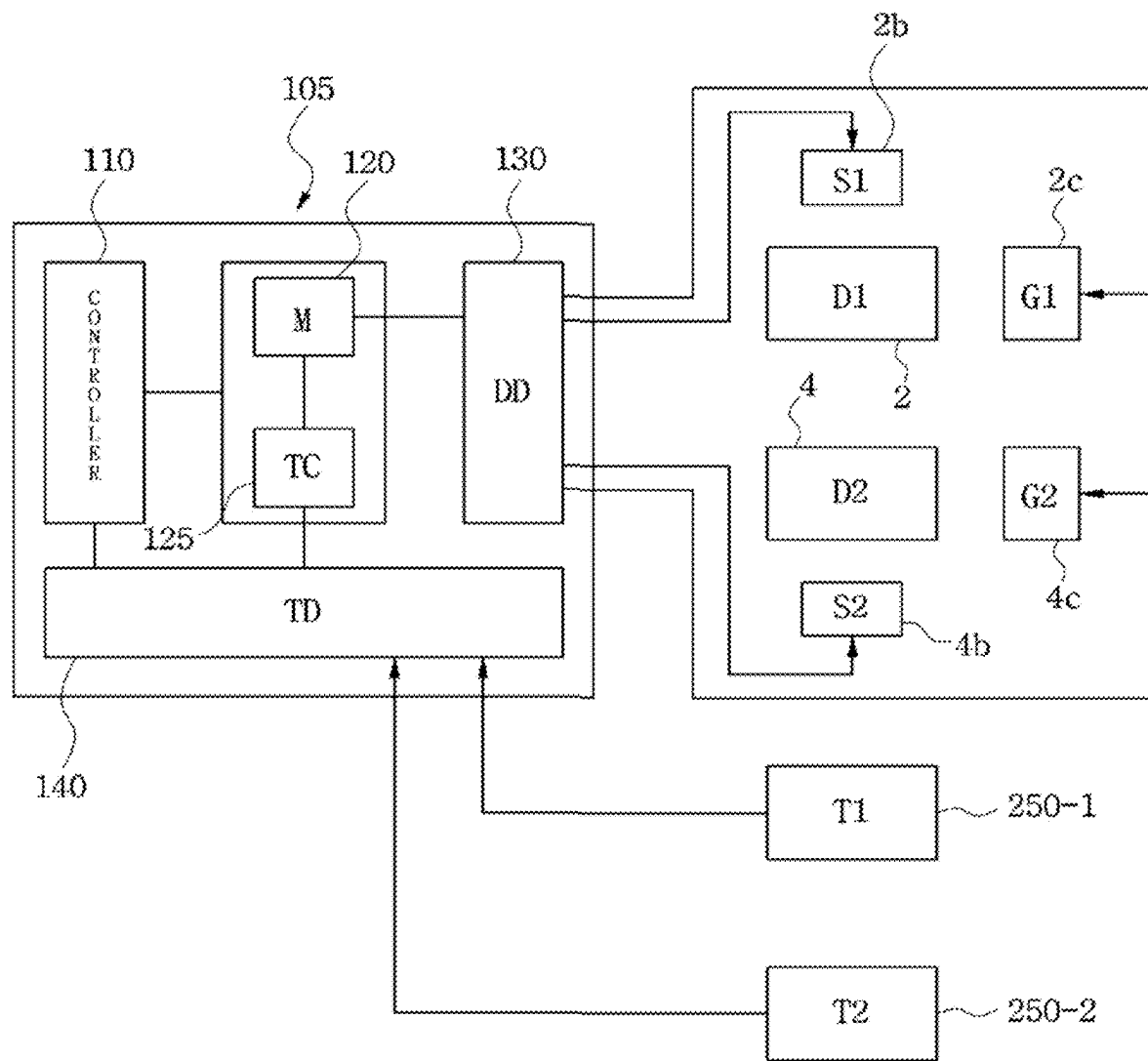


FIG. 60

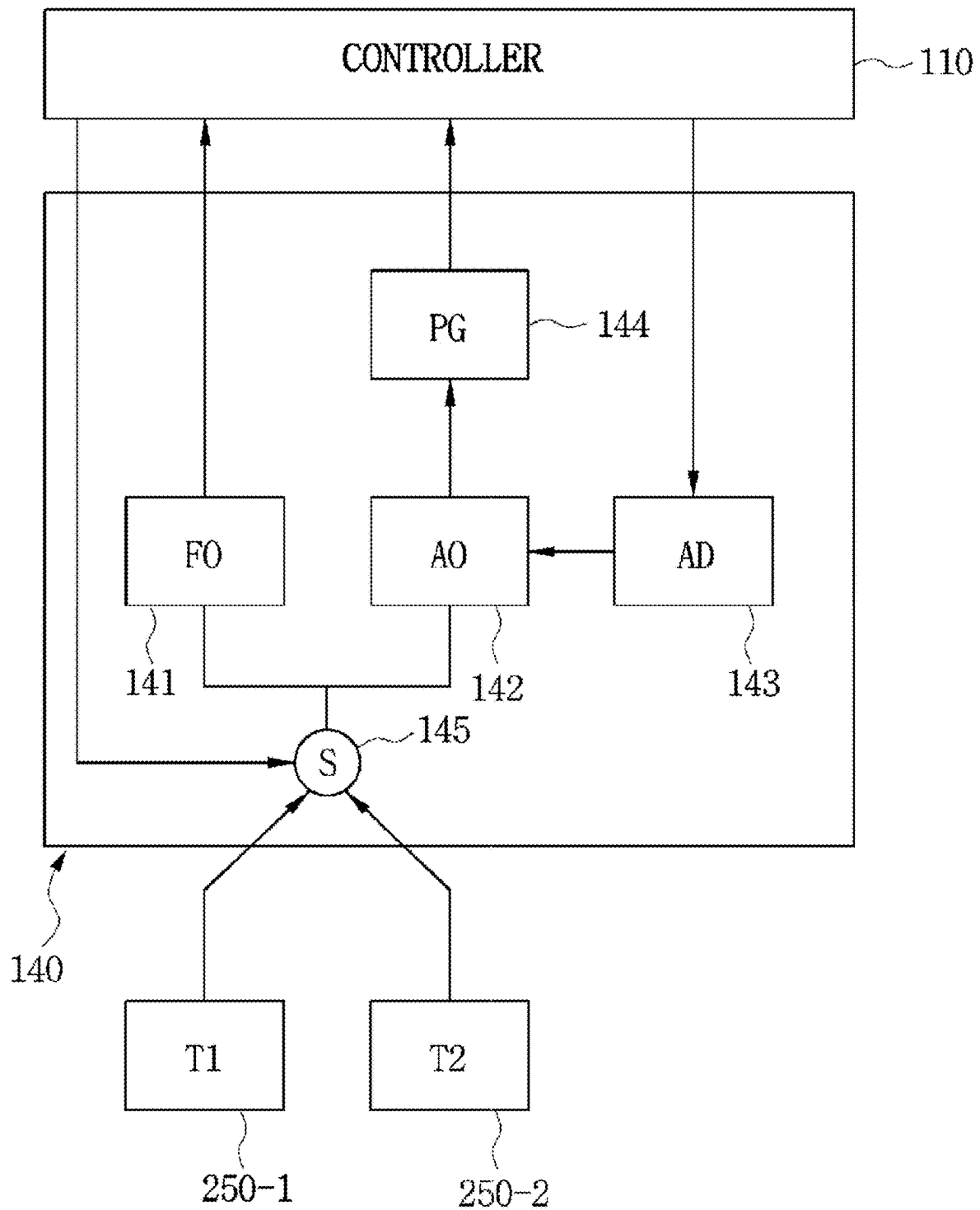


FIG. 61

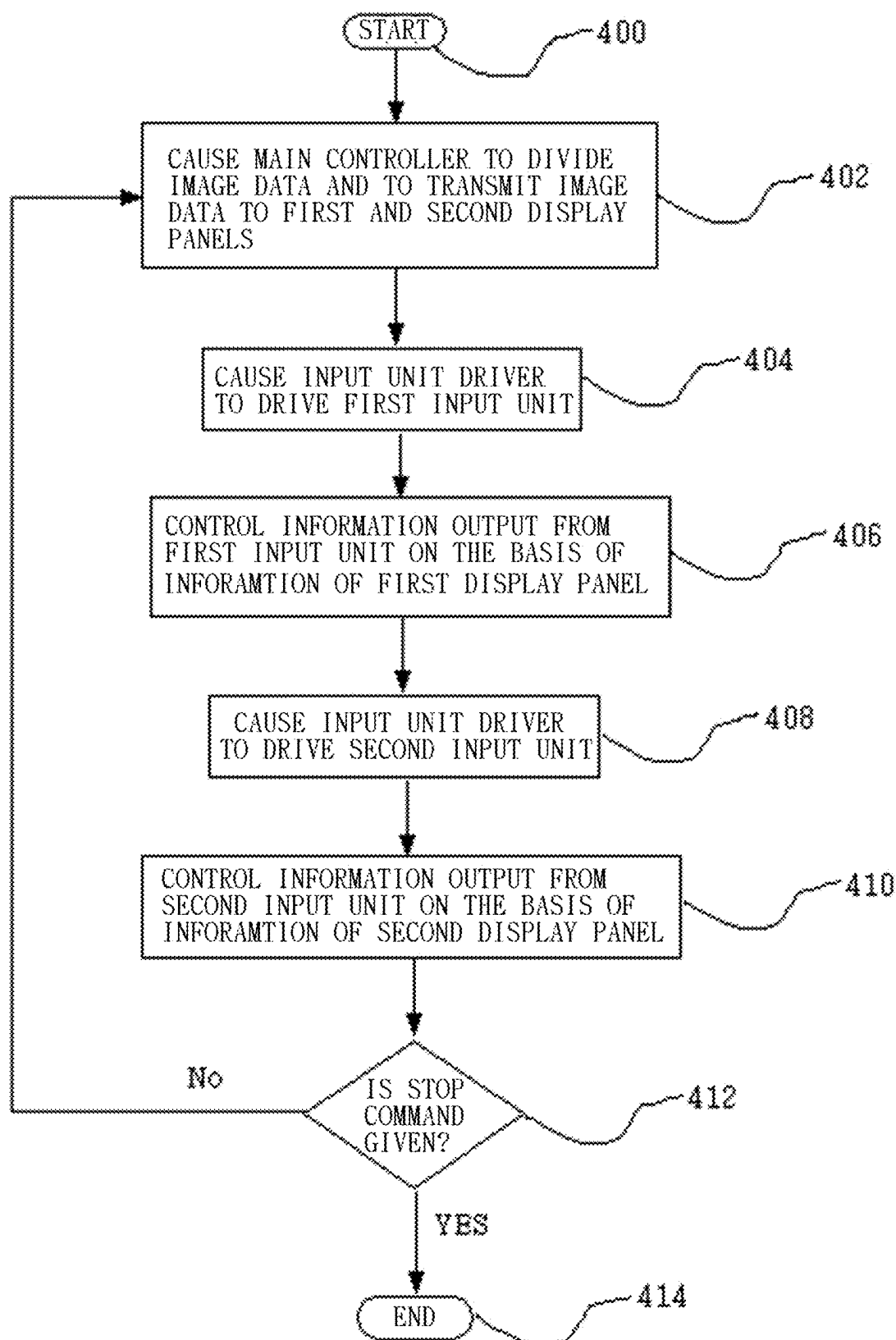


FIG. 62



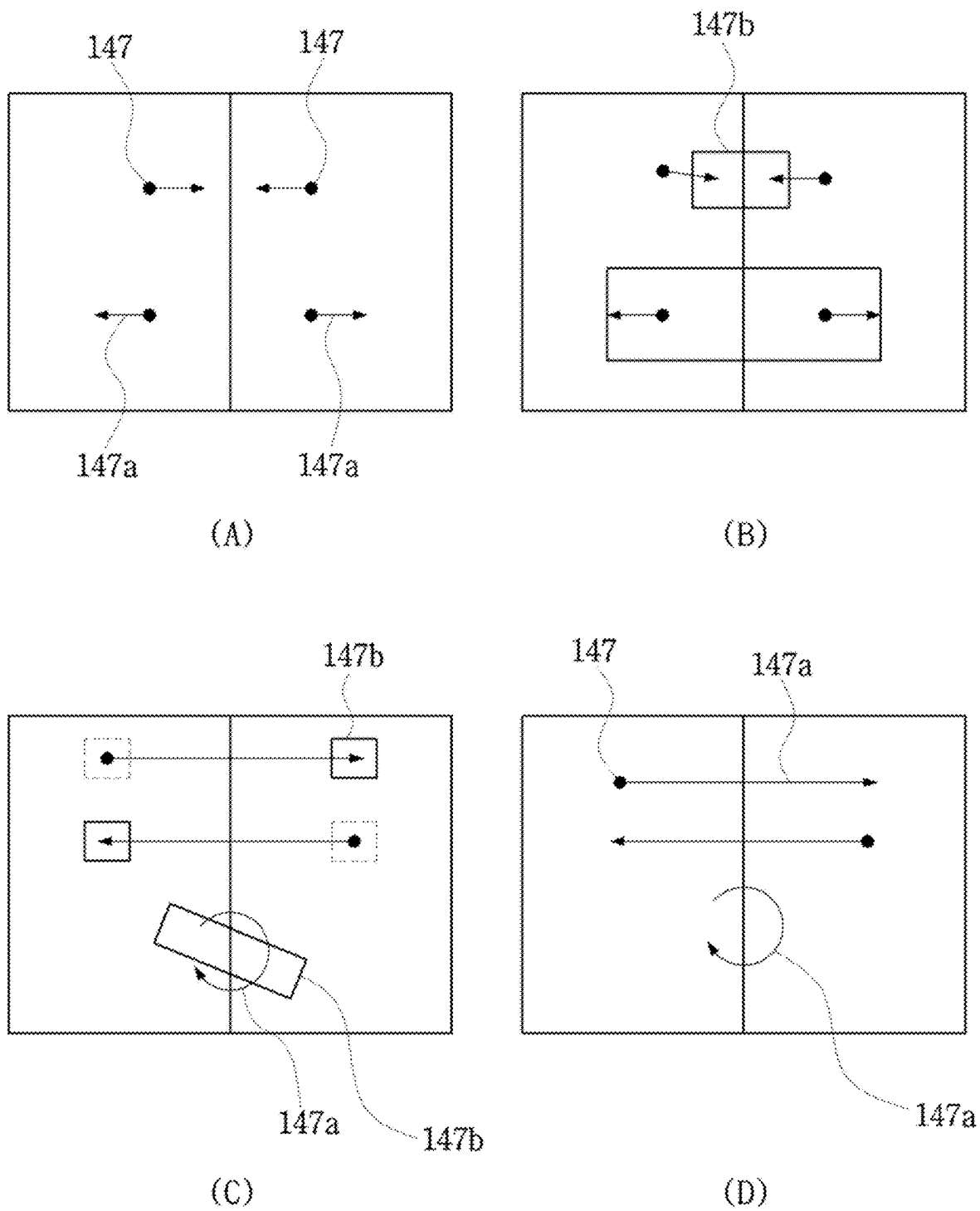


FIG. 63

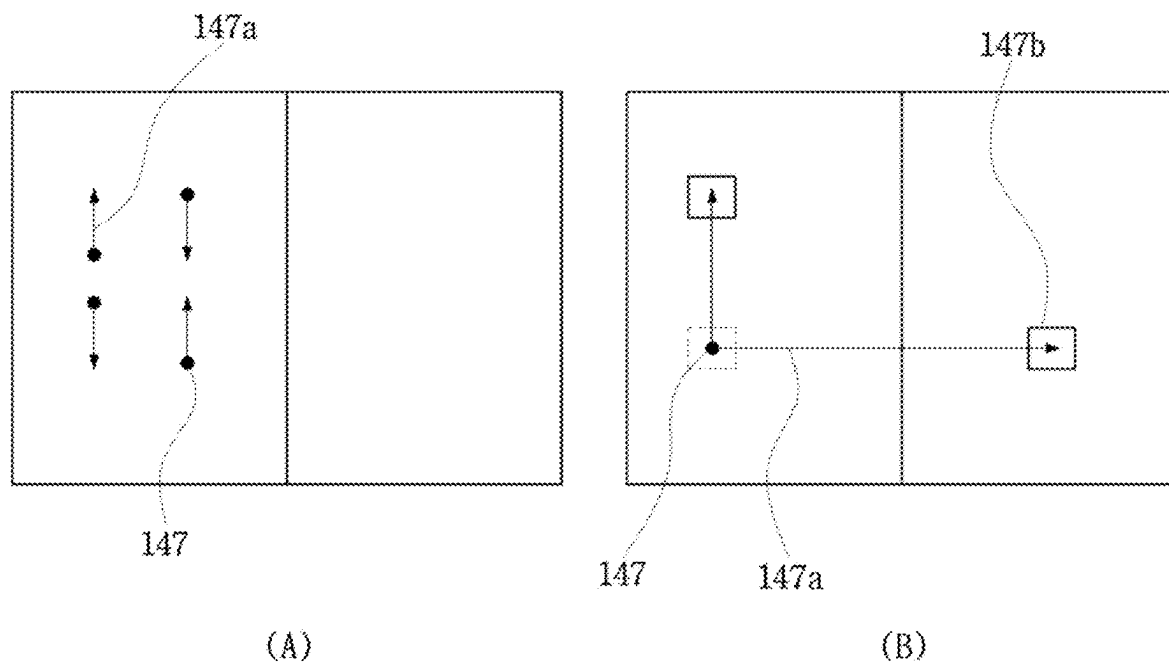


FIG. 64

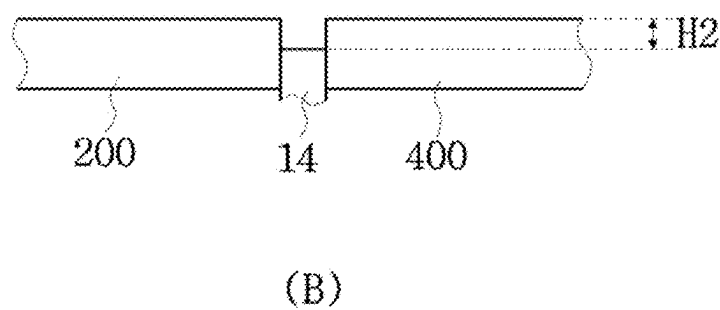
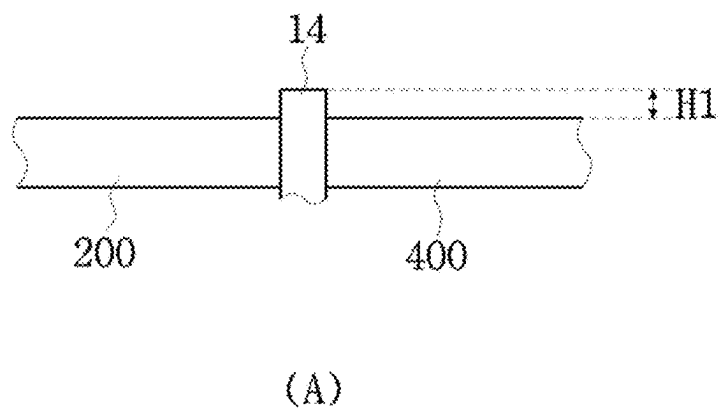


FIG. 65

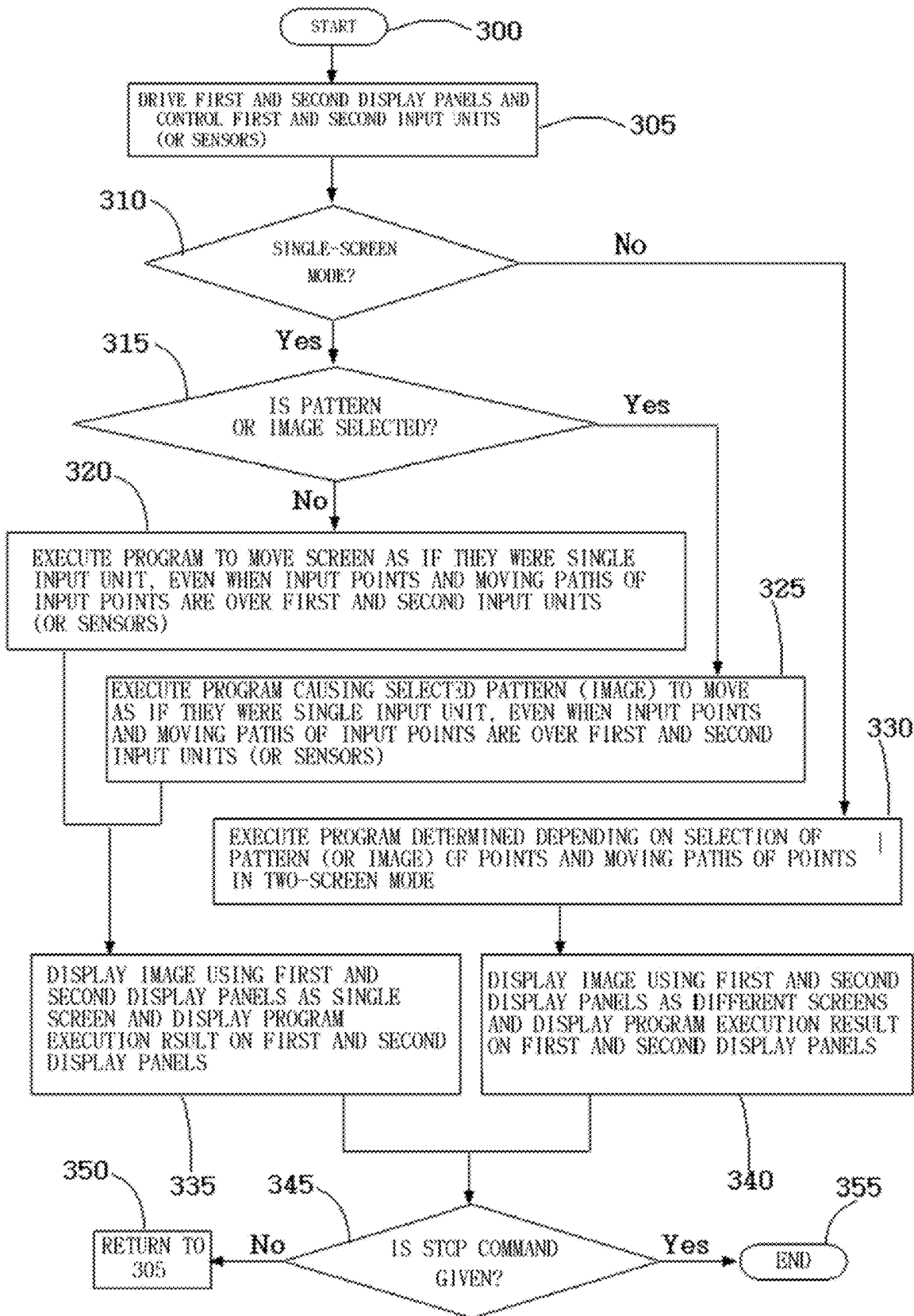
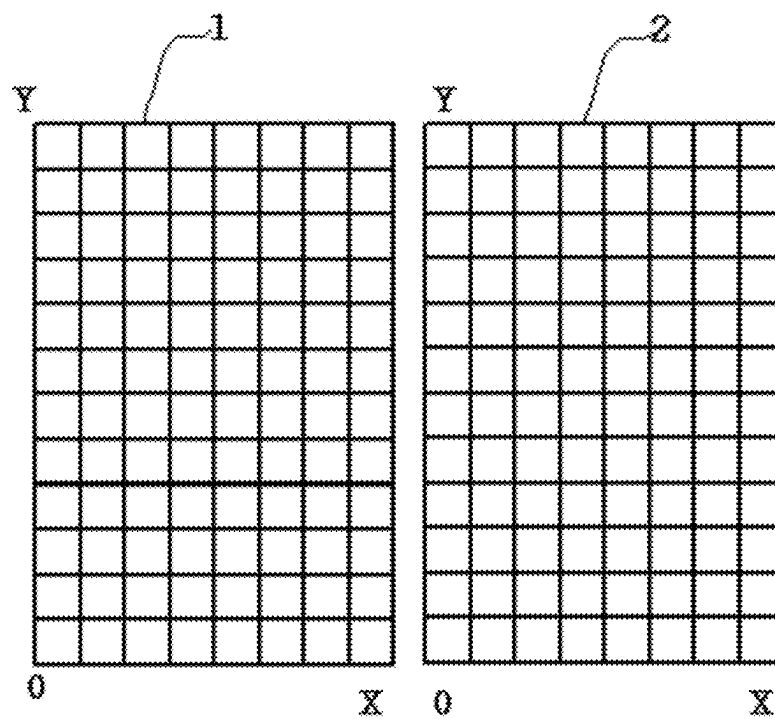
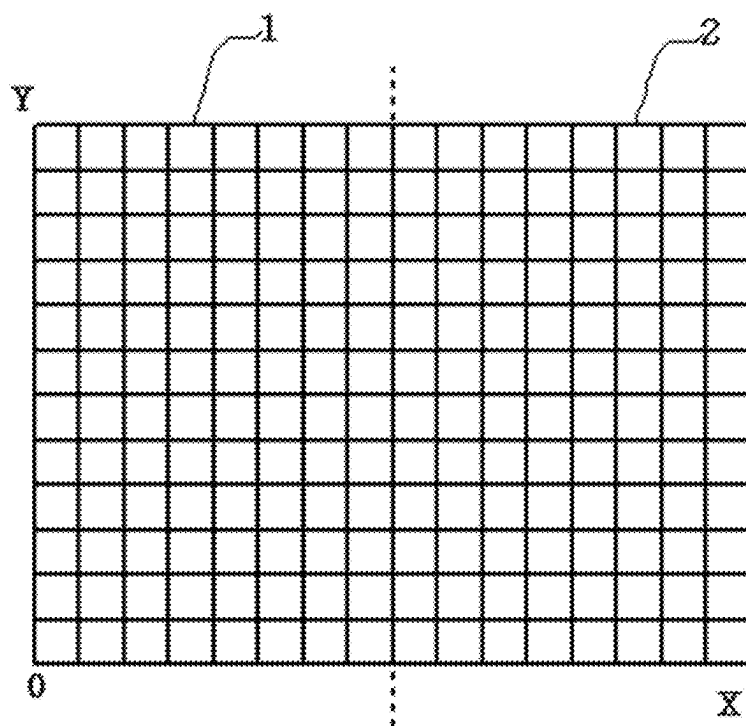


FIG. 66



(A)



(B)

FIG. 67

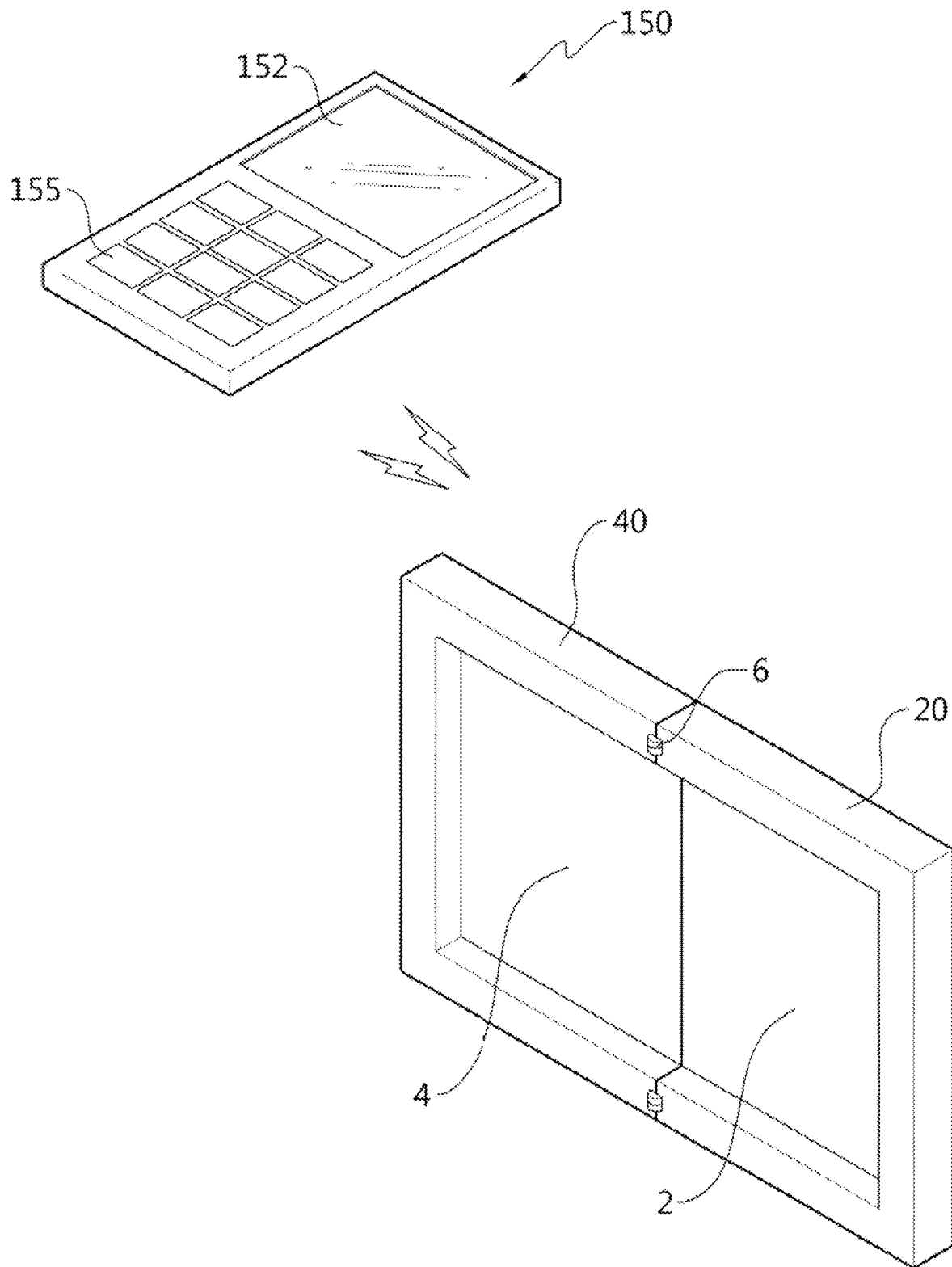


FIG. 68

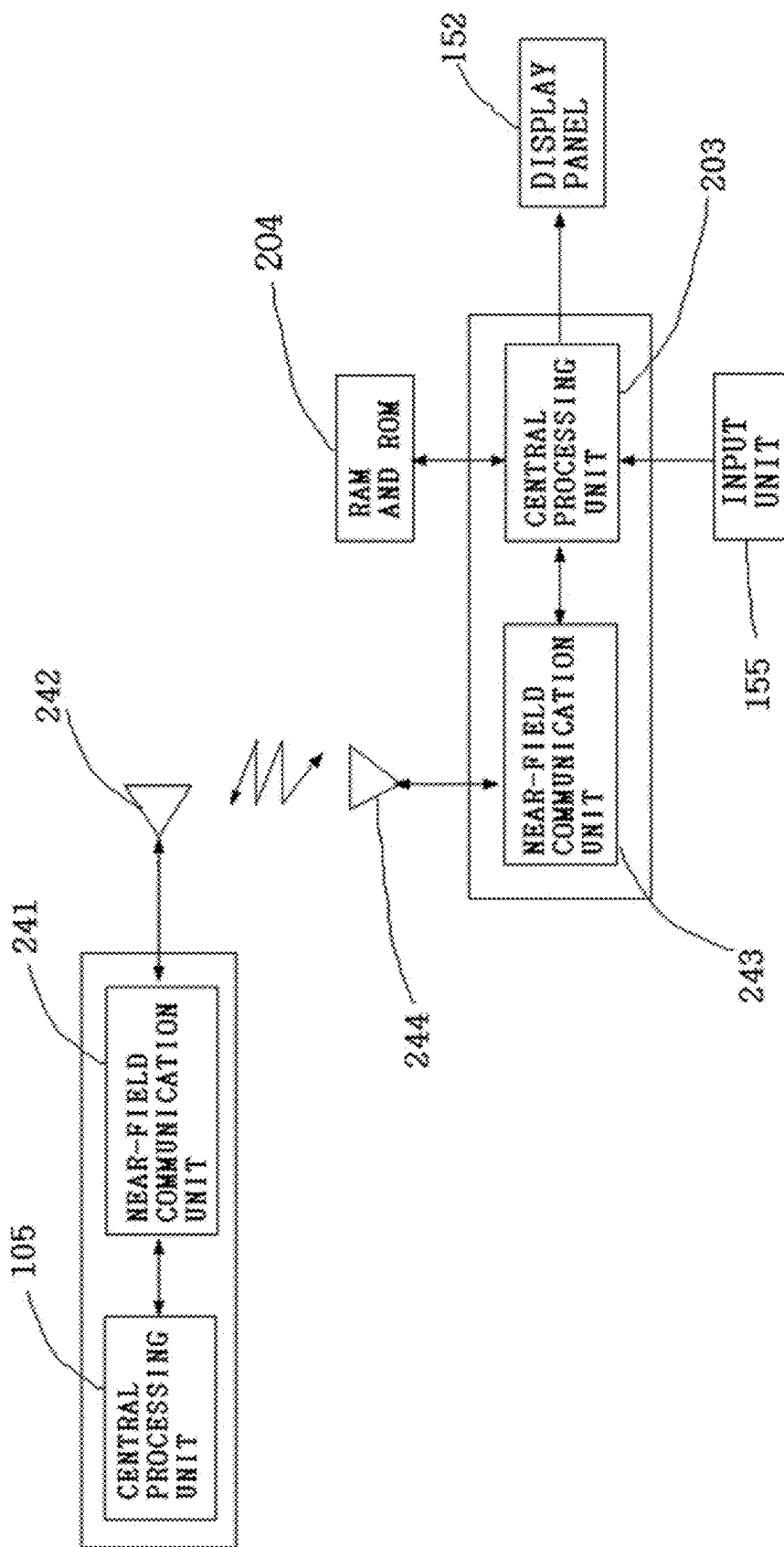


FIG. 69

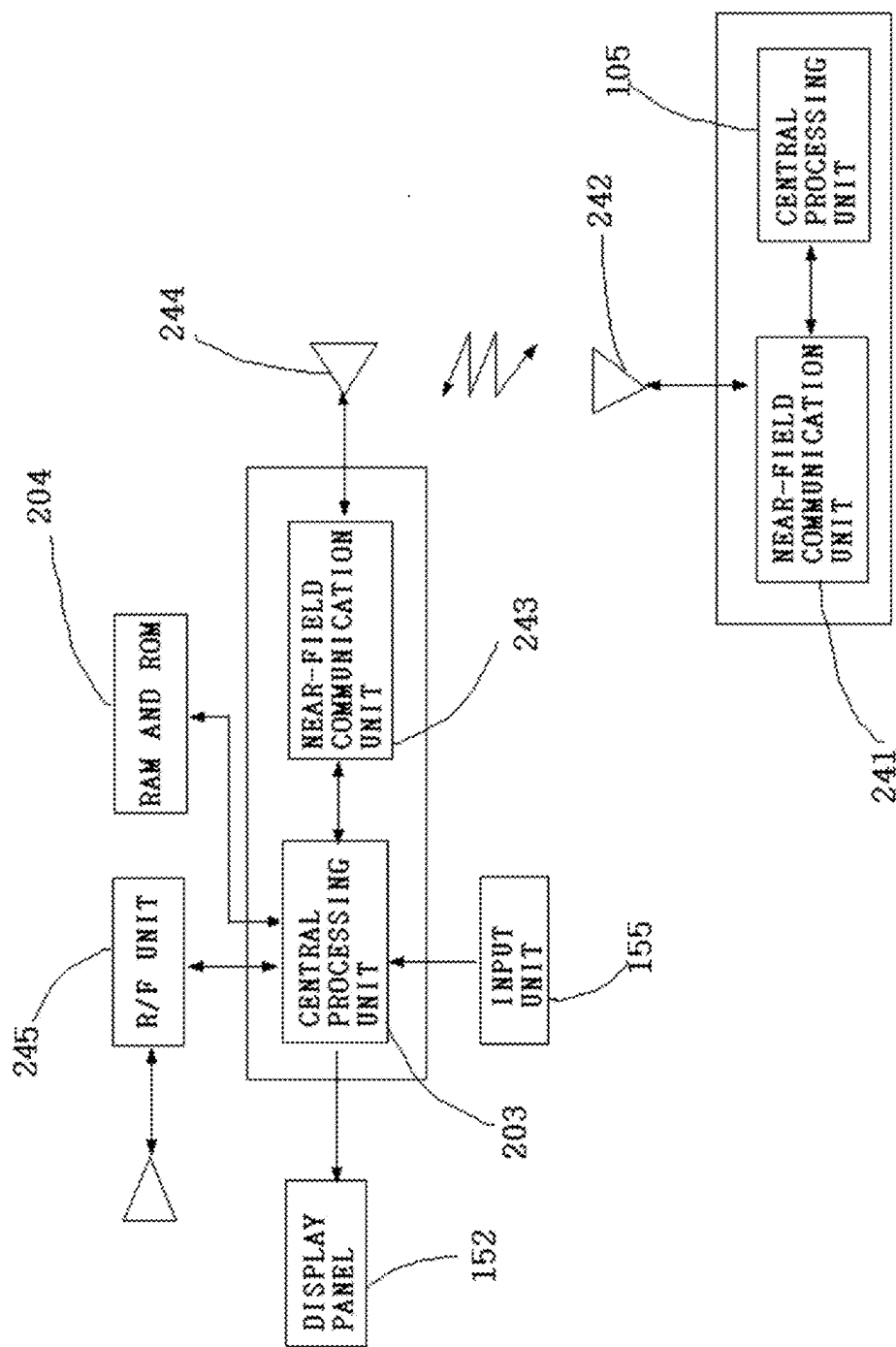


FIG. 70

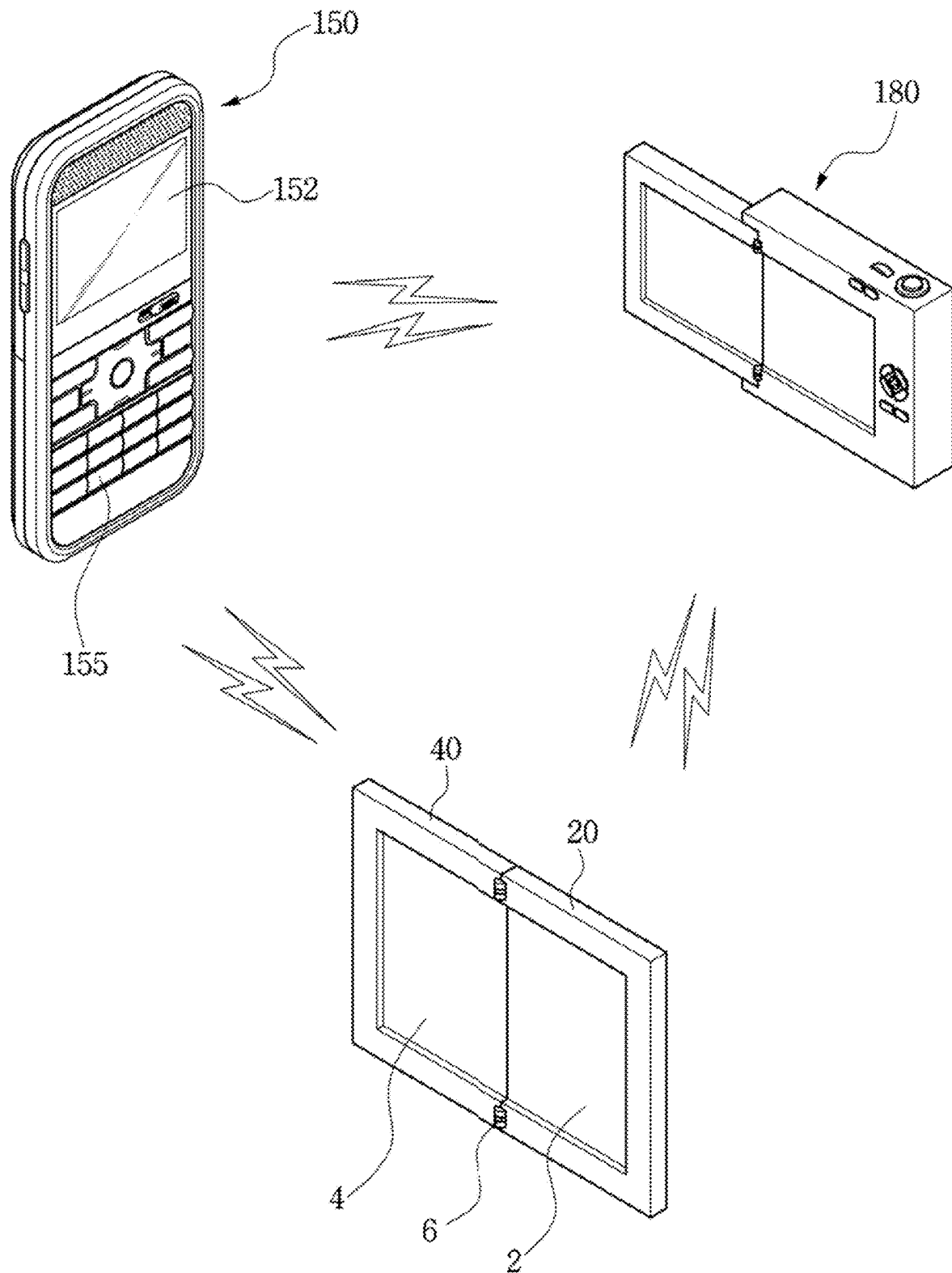


FIG. 71



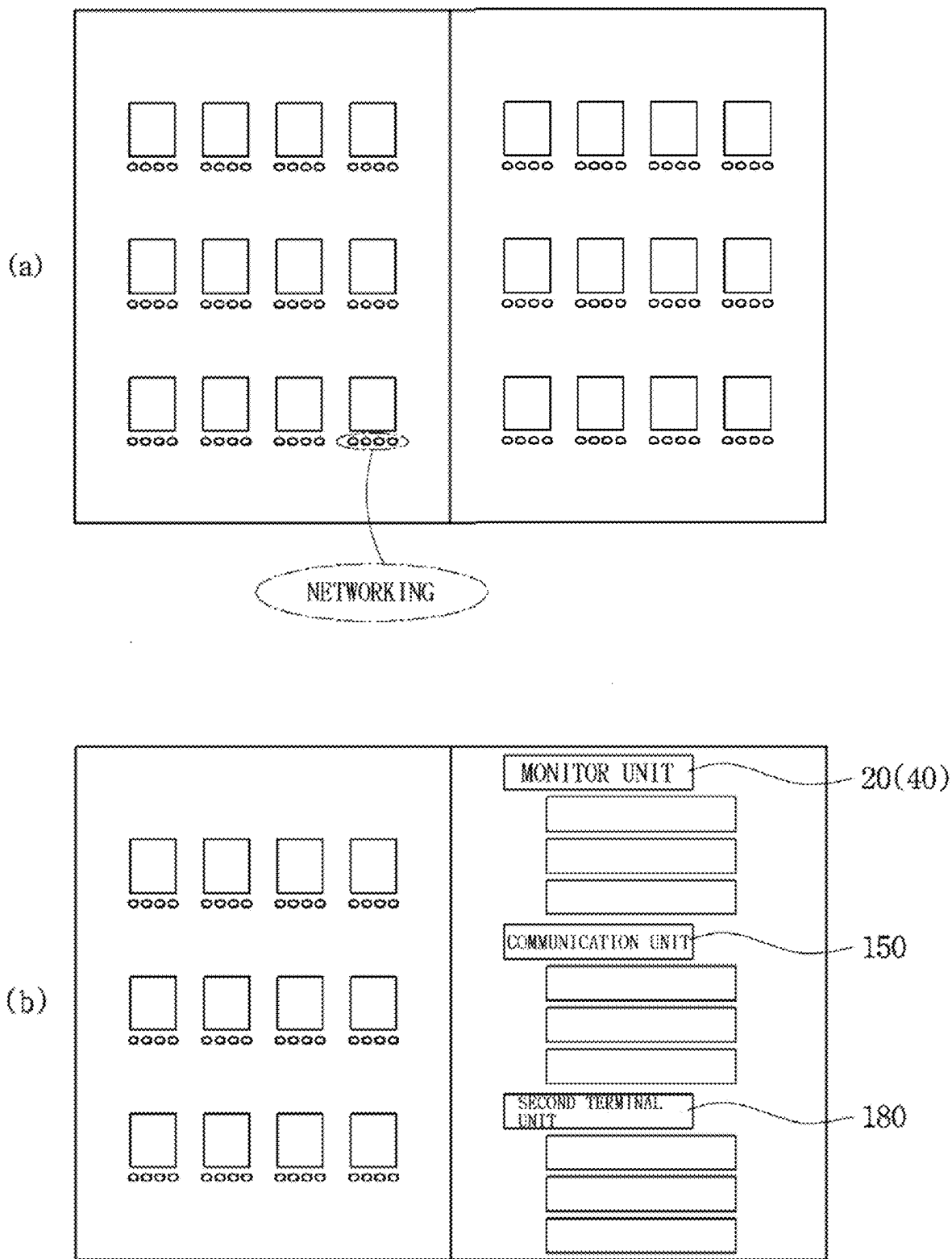
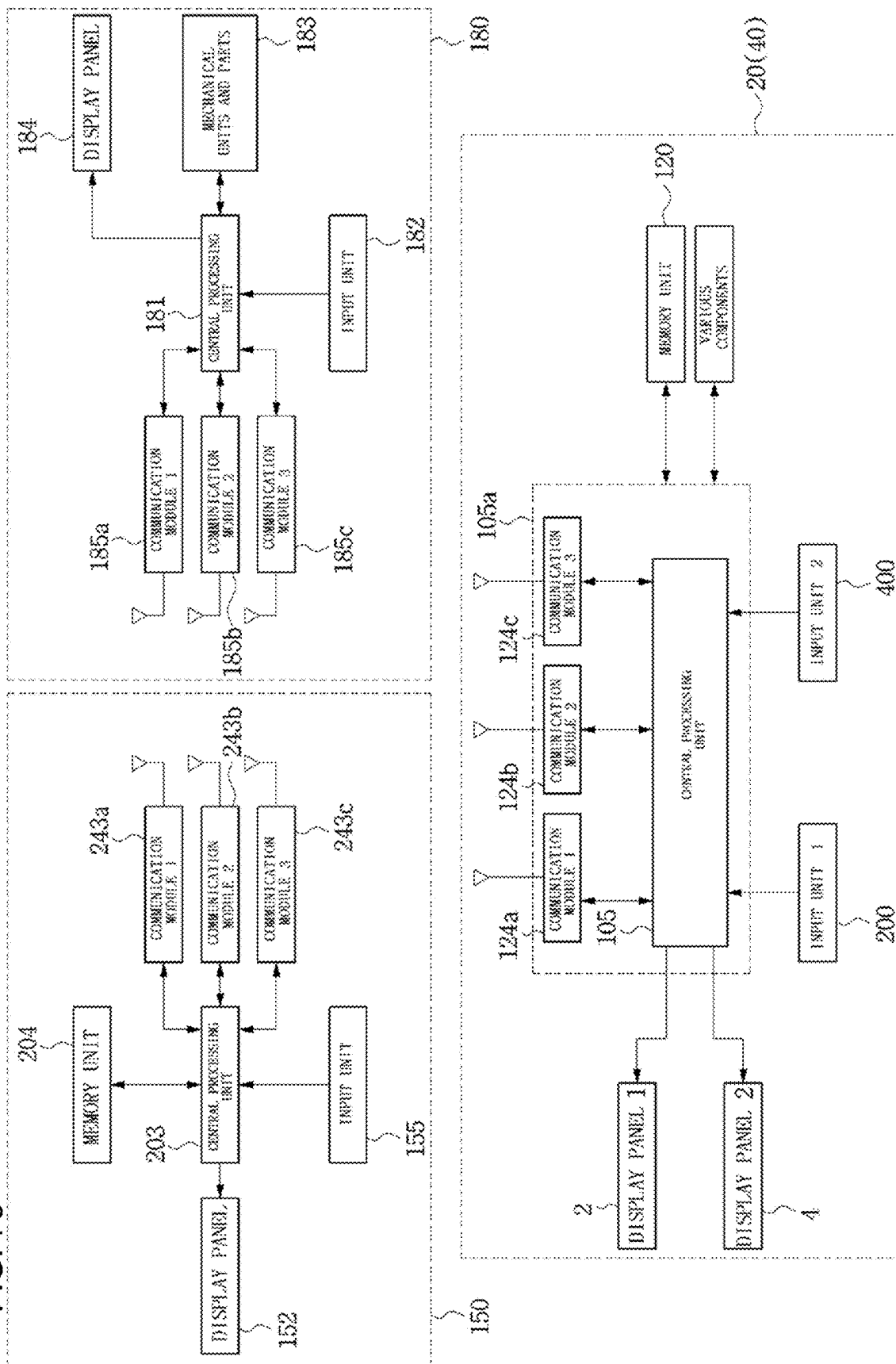


FIG. 72

FIG. 73



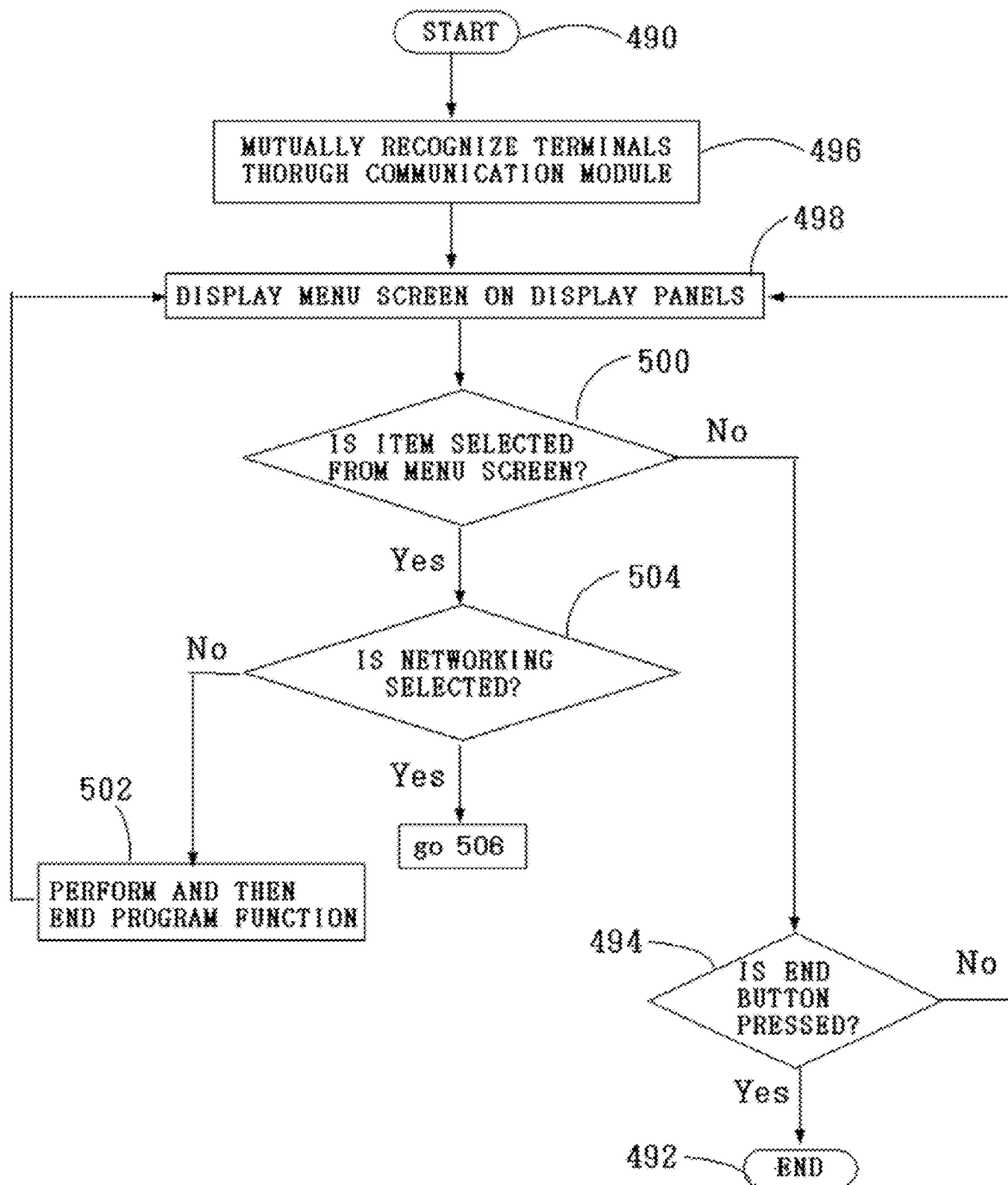


FIG. 74

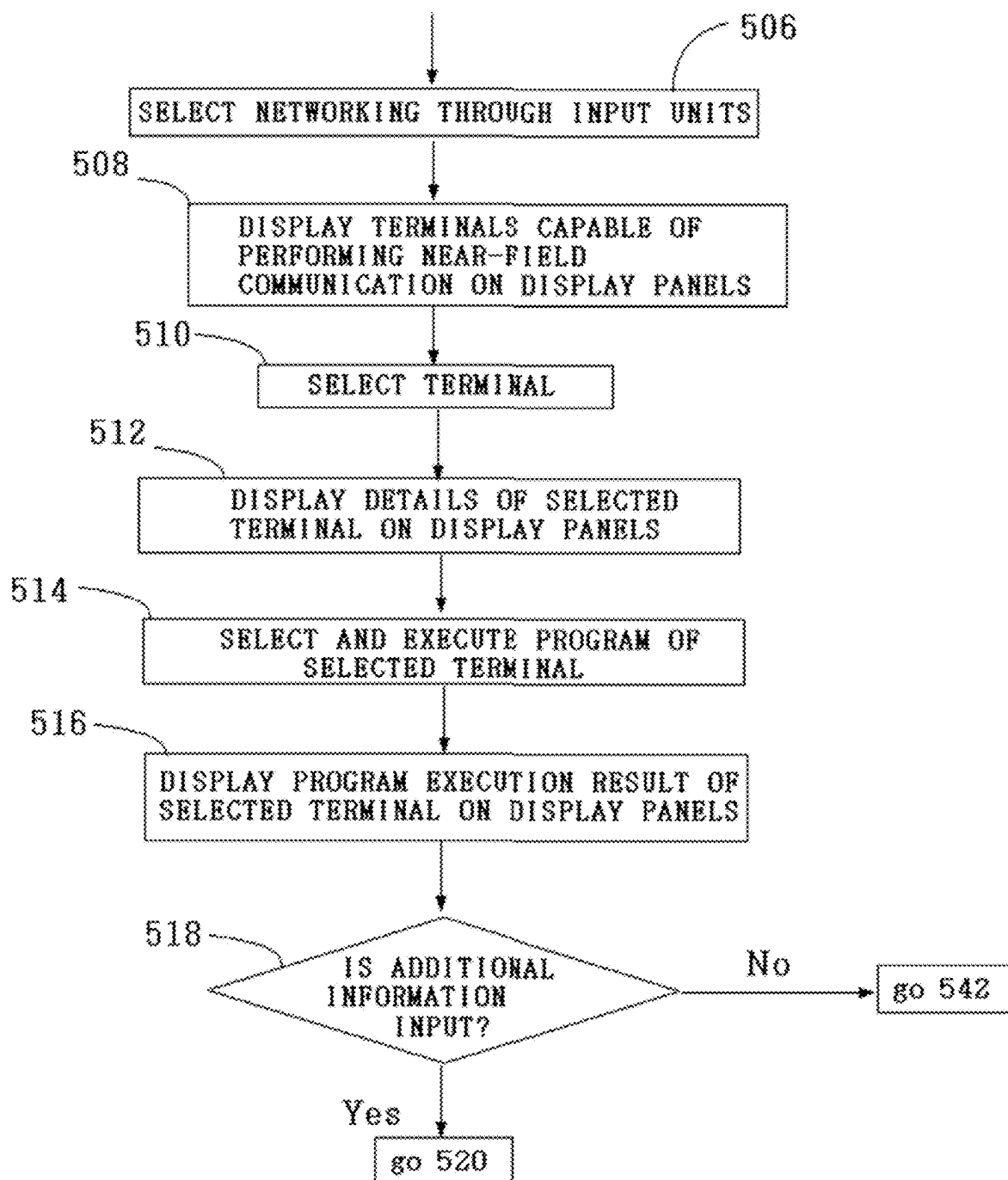


FIG. 75

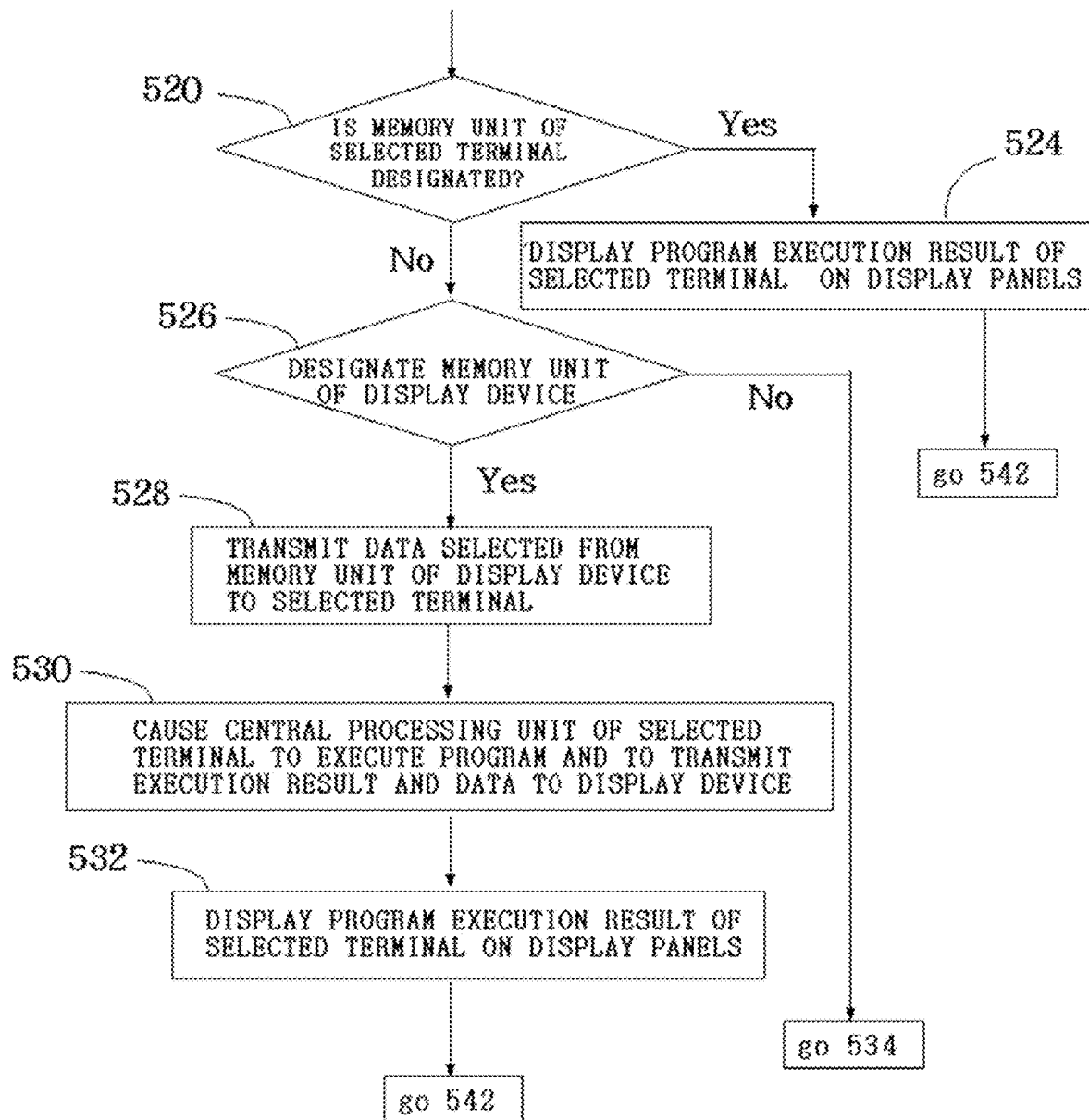


FIG. 76

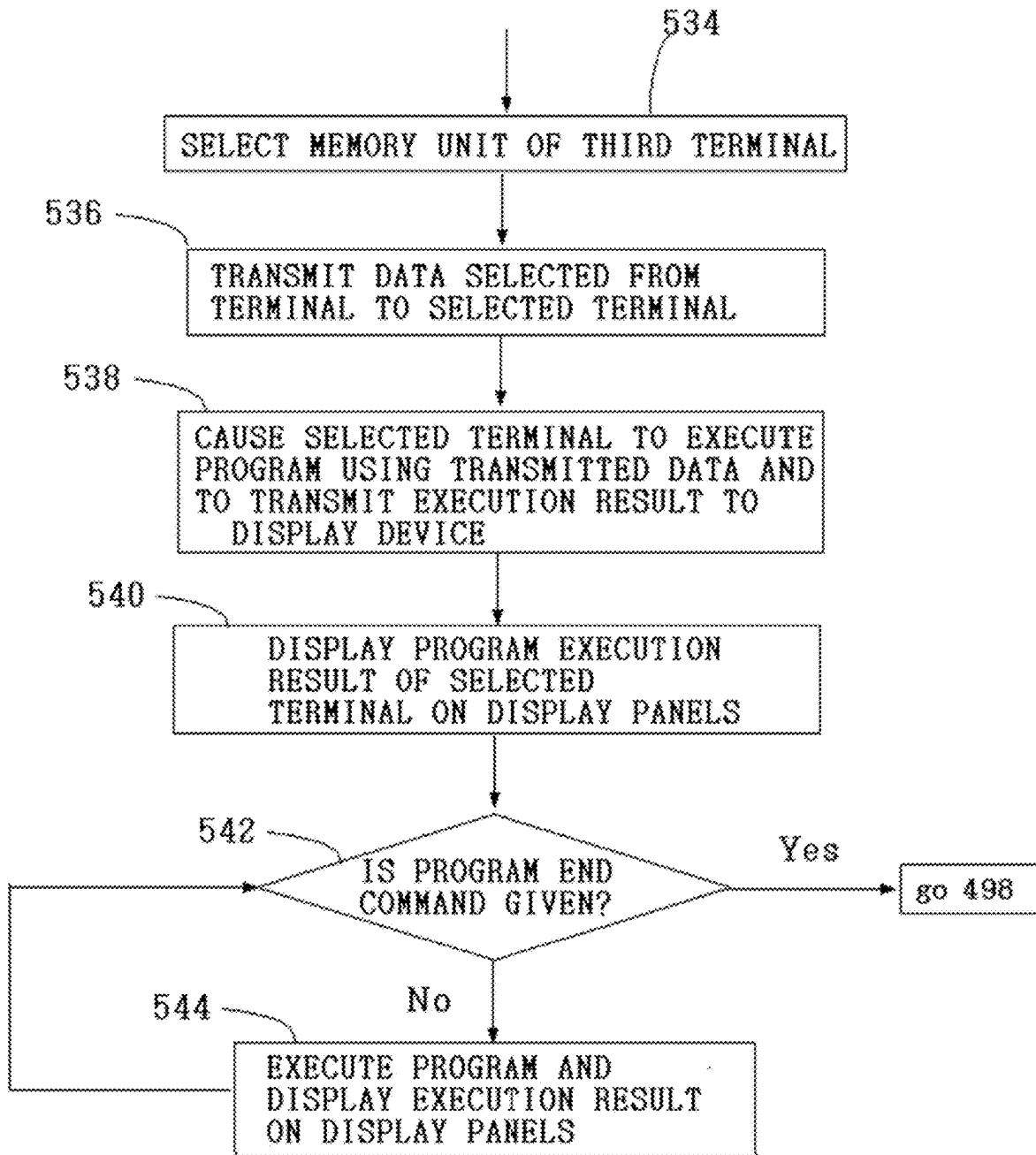


FIG. 77

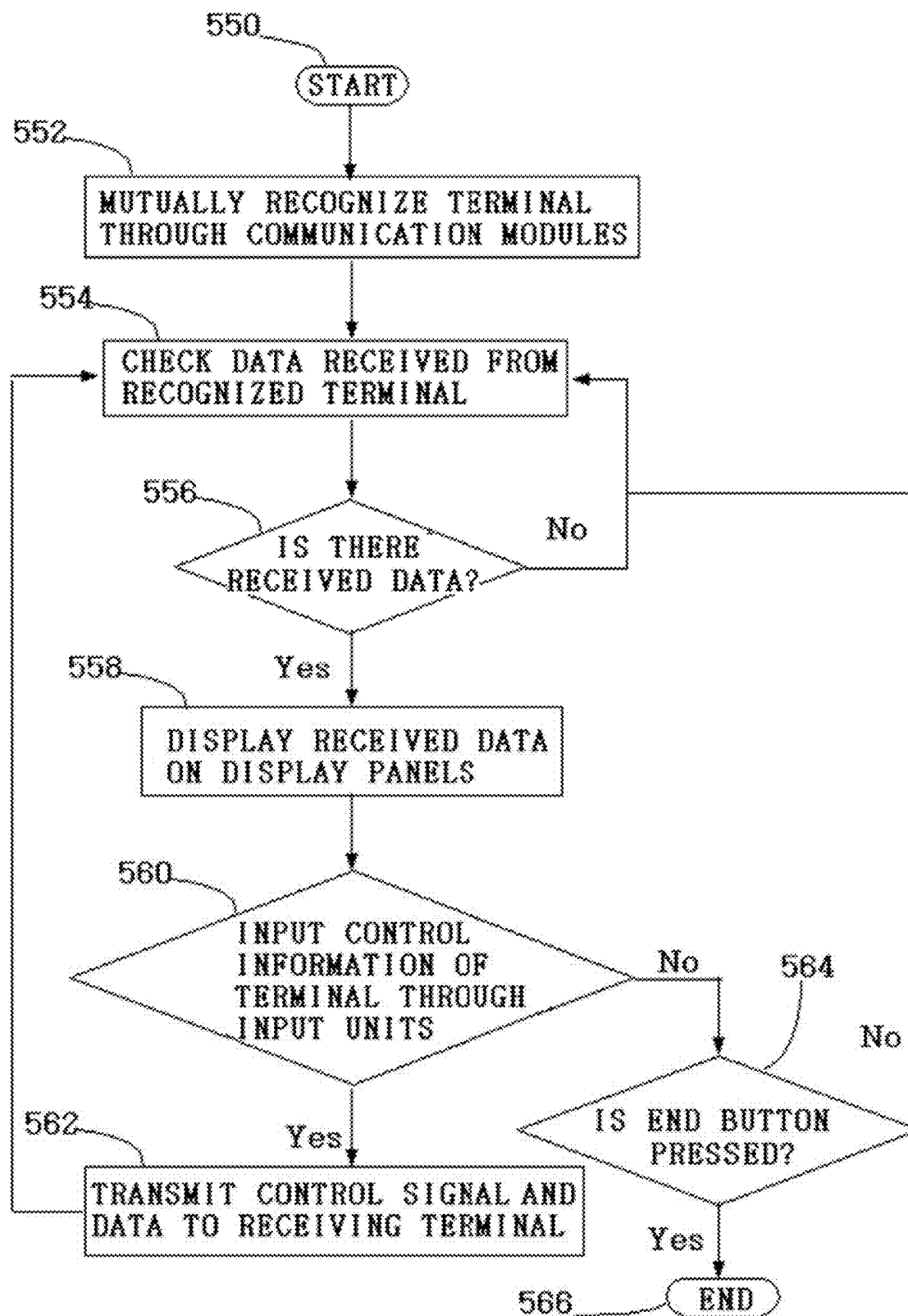


FIG. 78

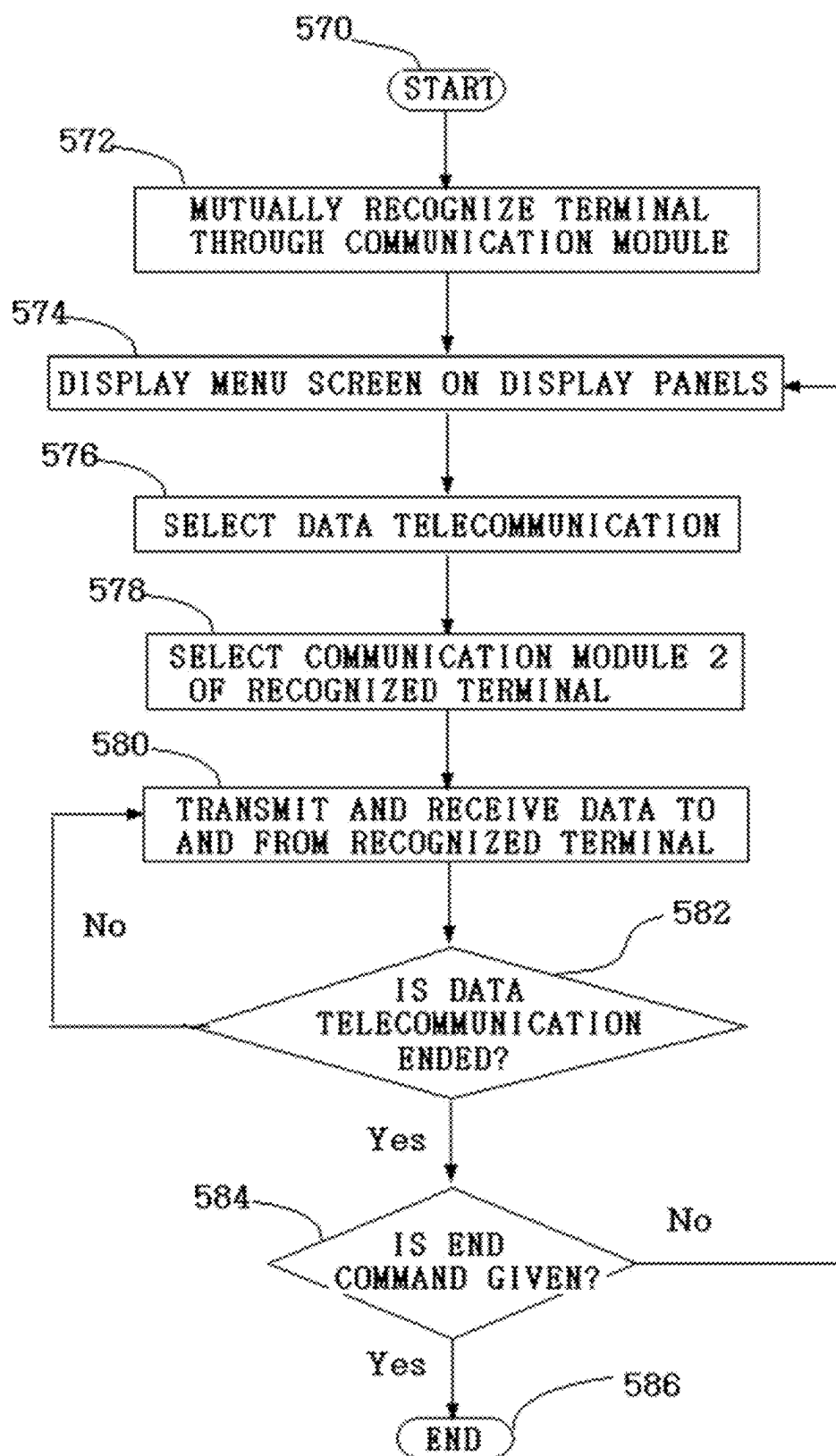


FIG. 79



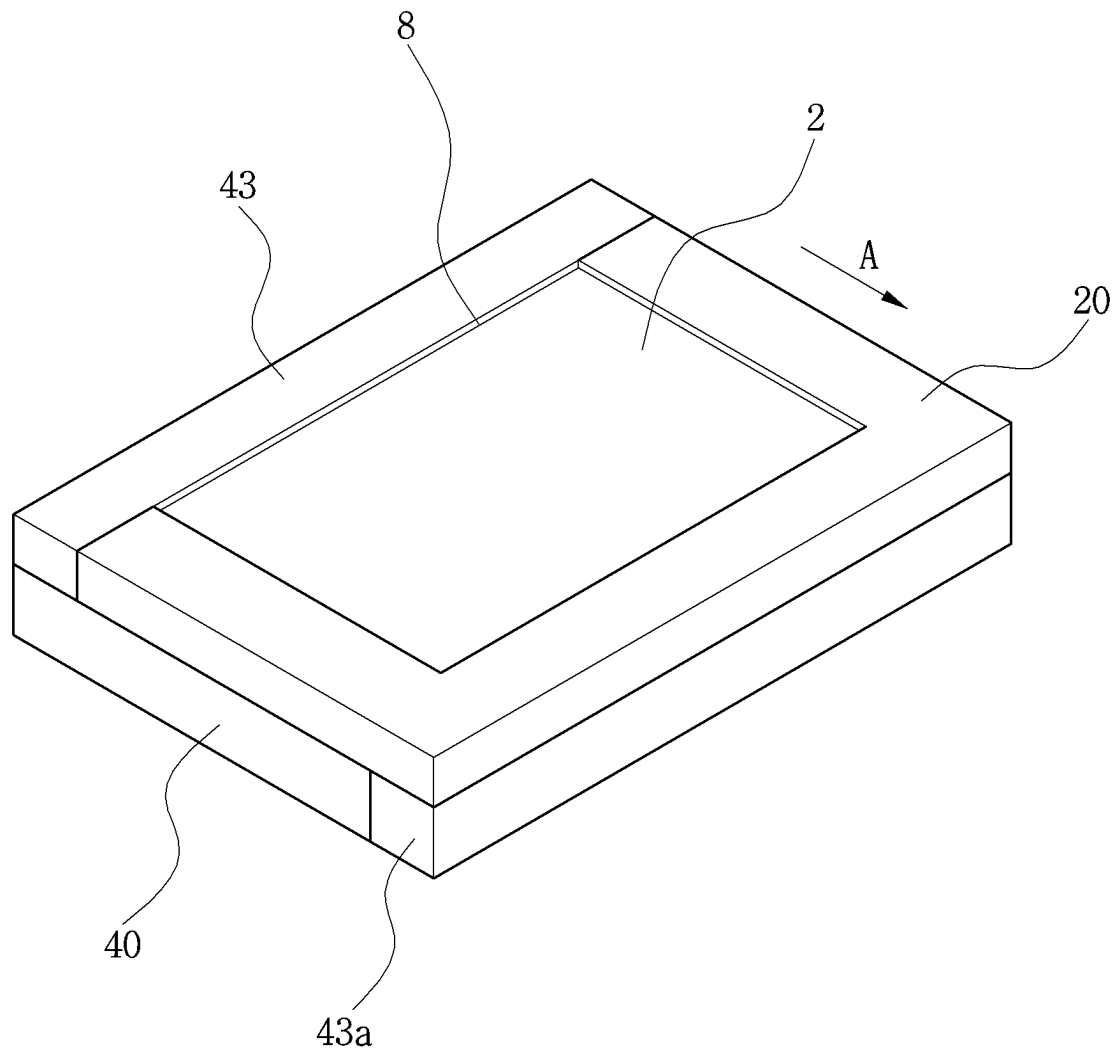


FIG. 80

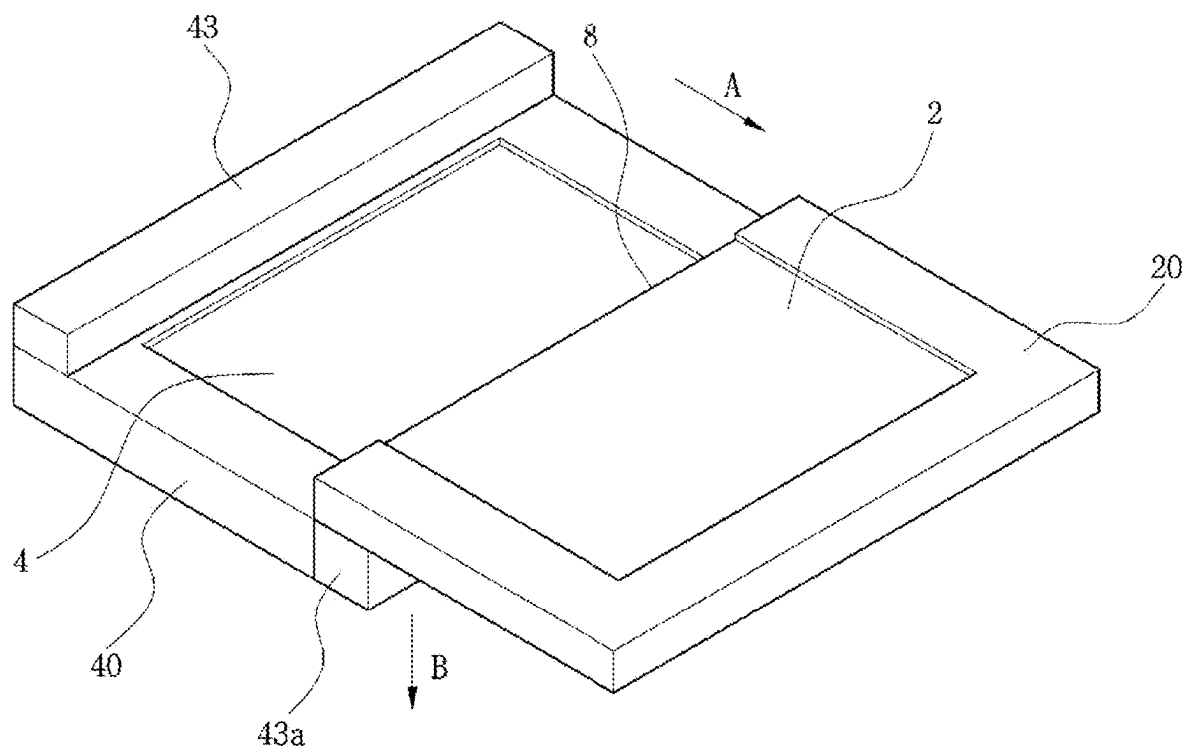


FIG. 81

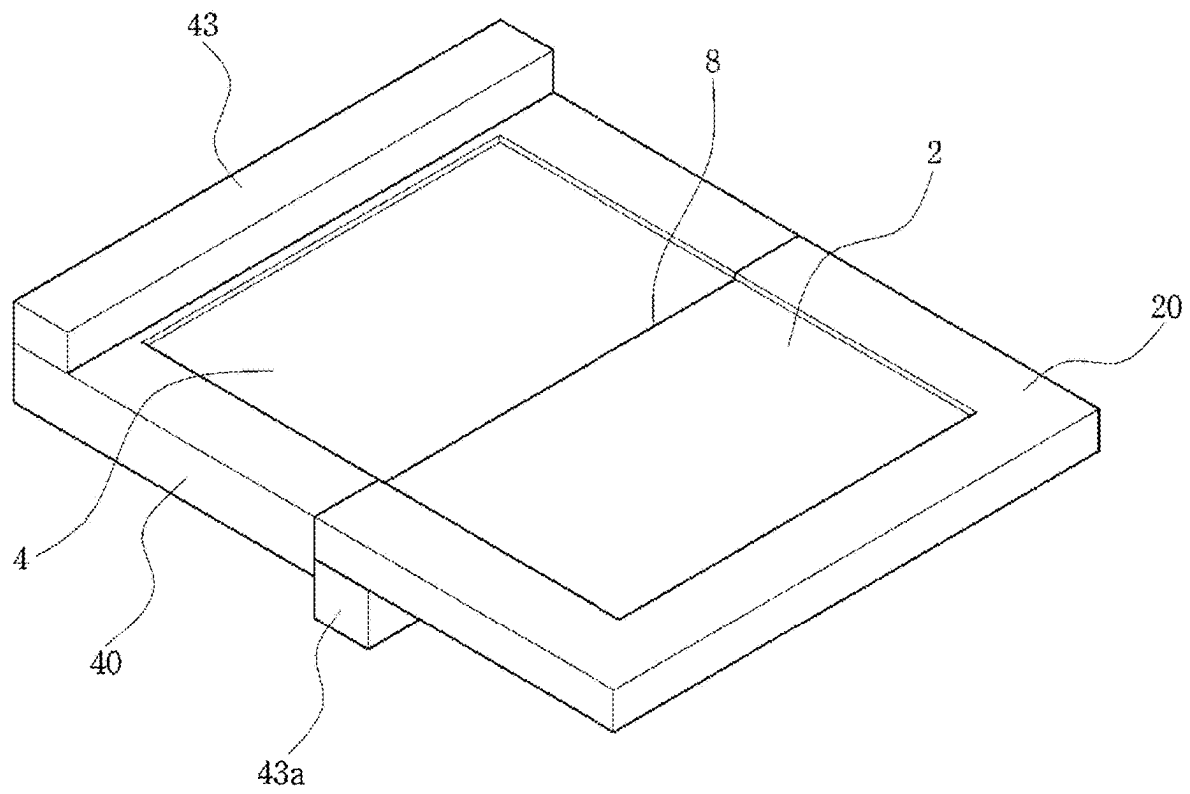


FIG. 82

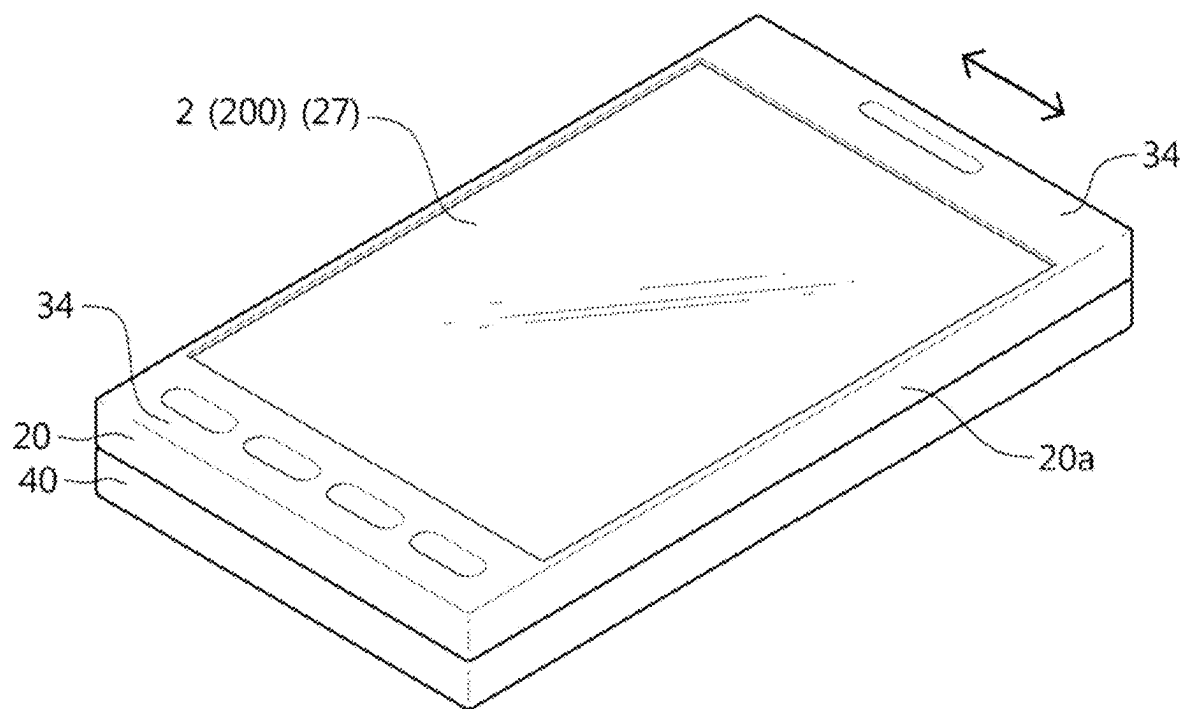


FIG. 83

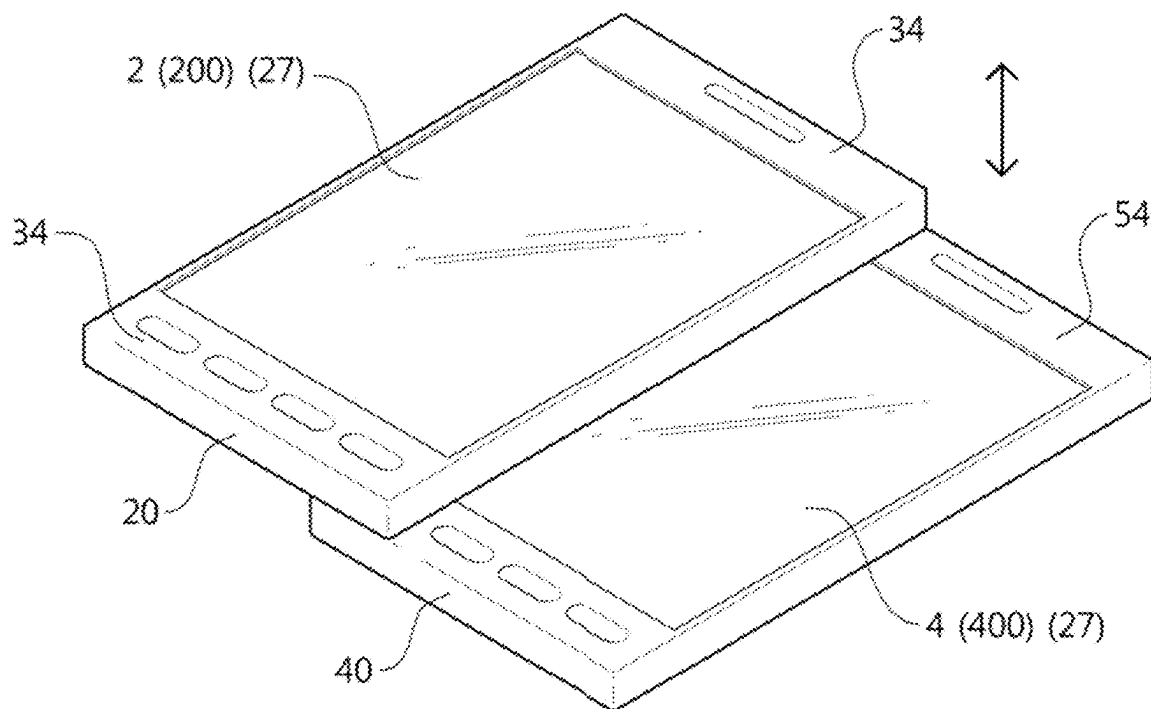


FIG. 84

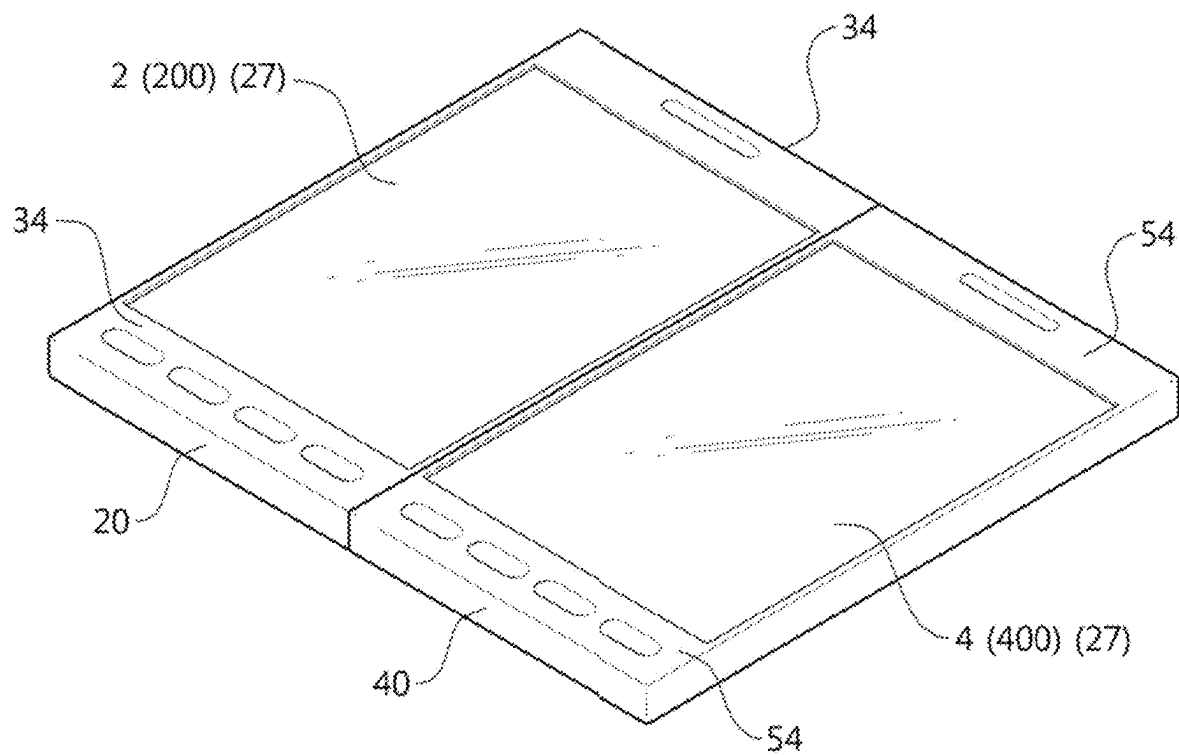


FIG. 85

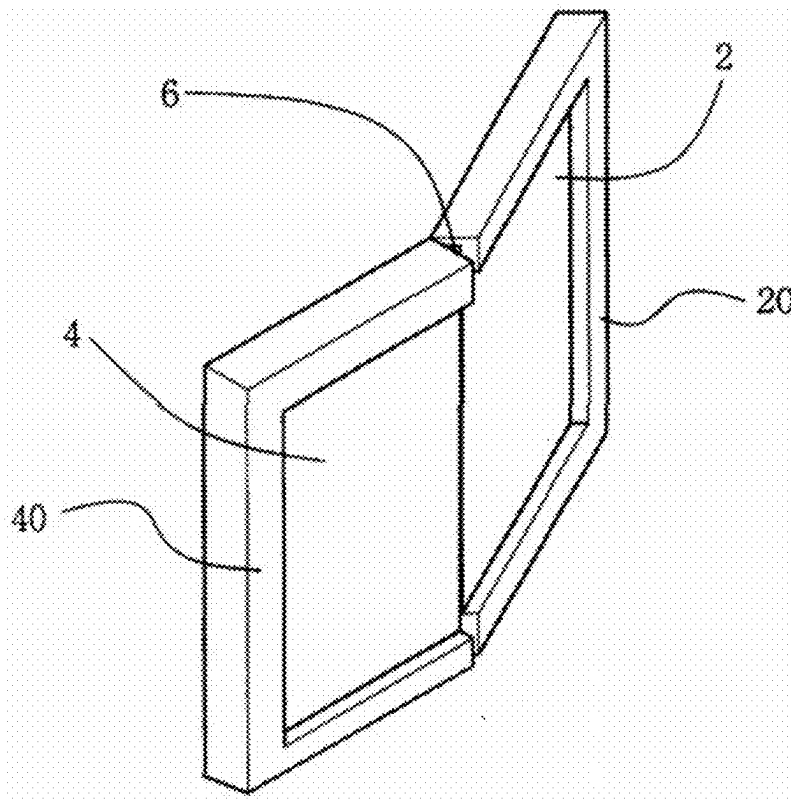


FIG. 86

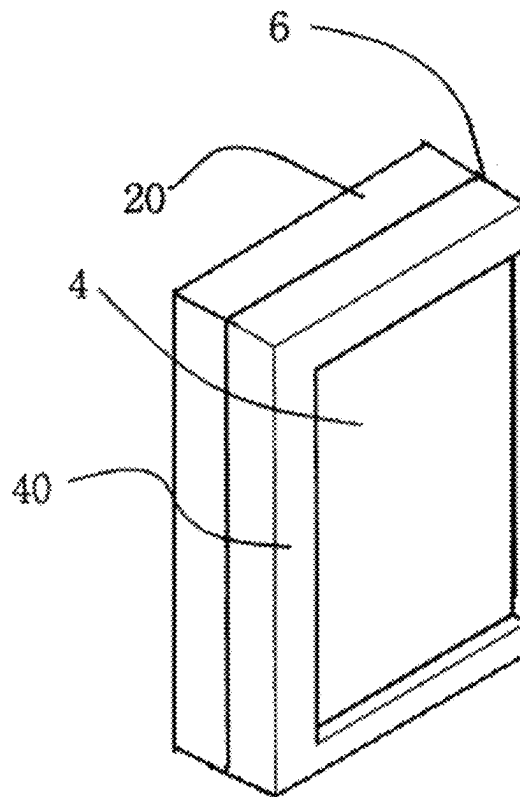


FIG. 87

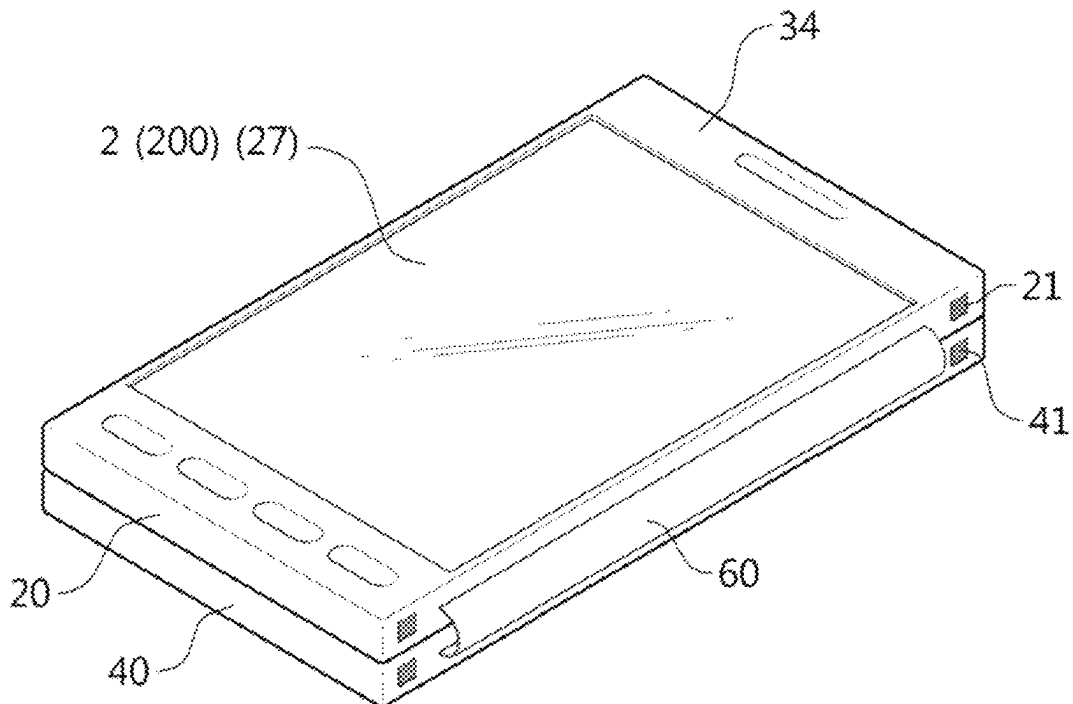


FIG. 88

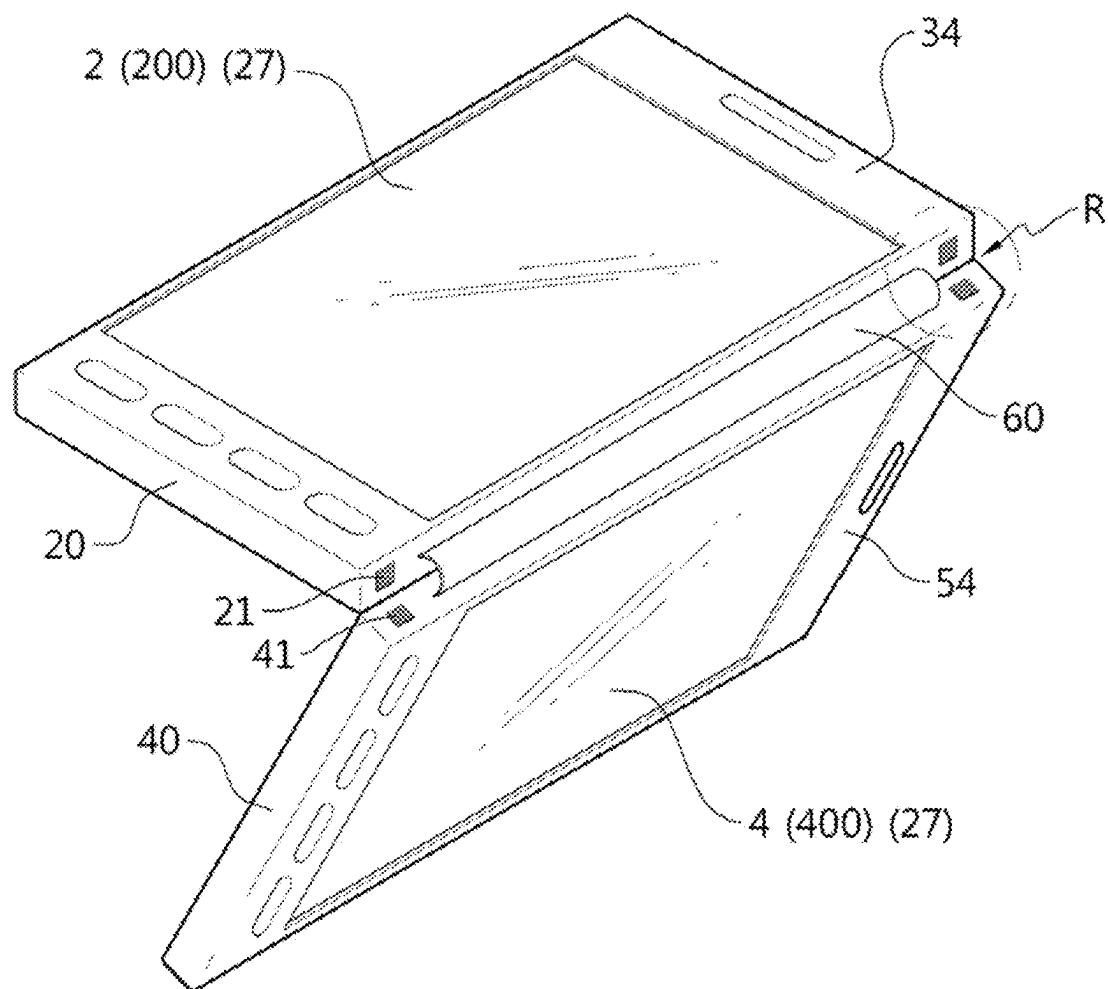


FIG. 89

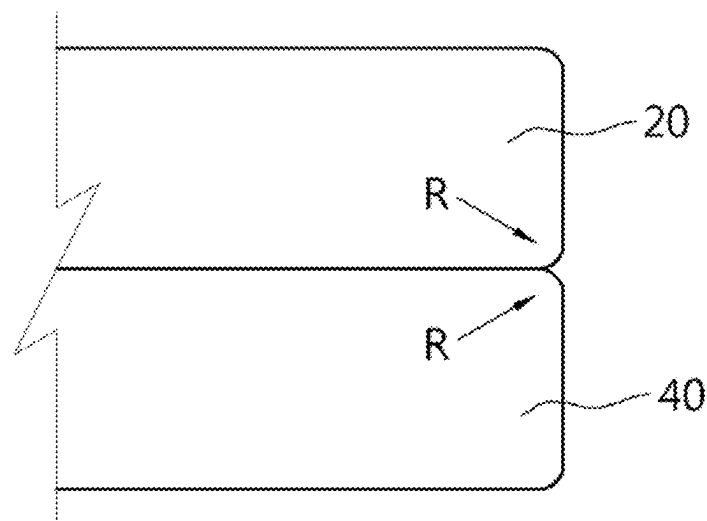


FIG. 90

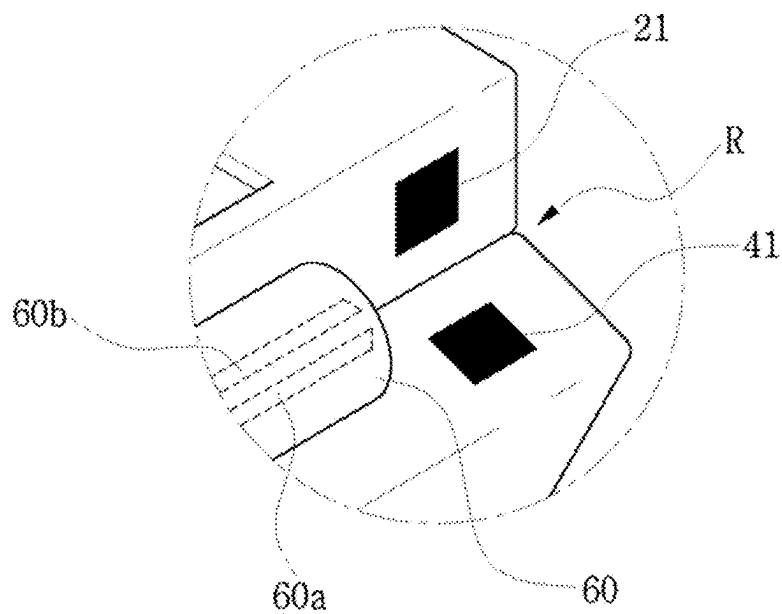


FIG. 91

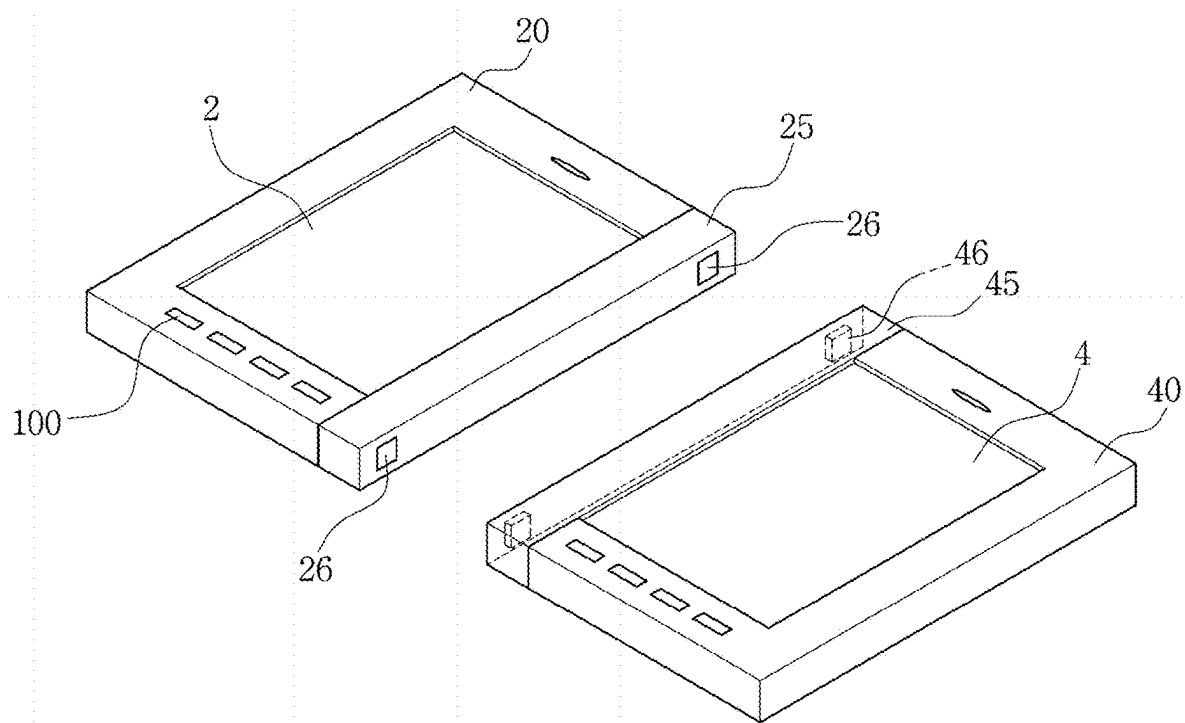


FIG. 92



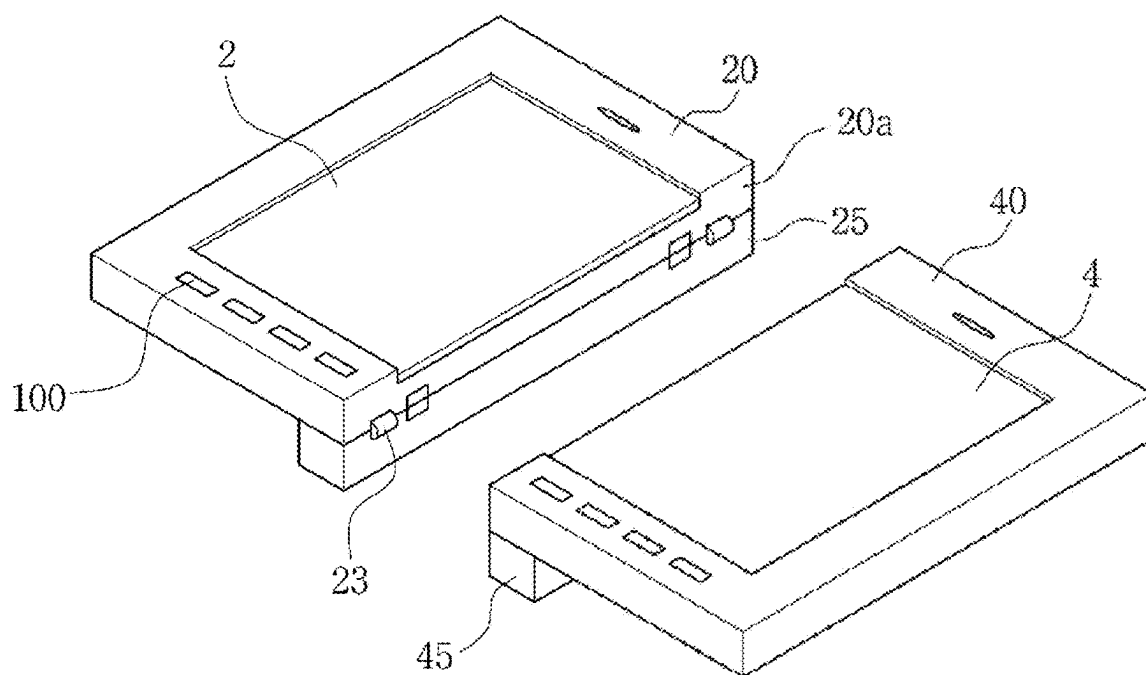


FIG. 93

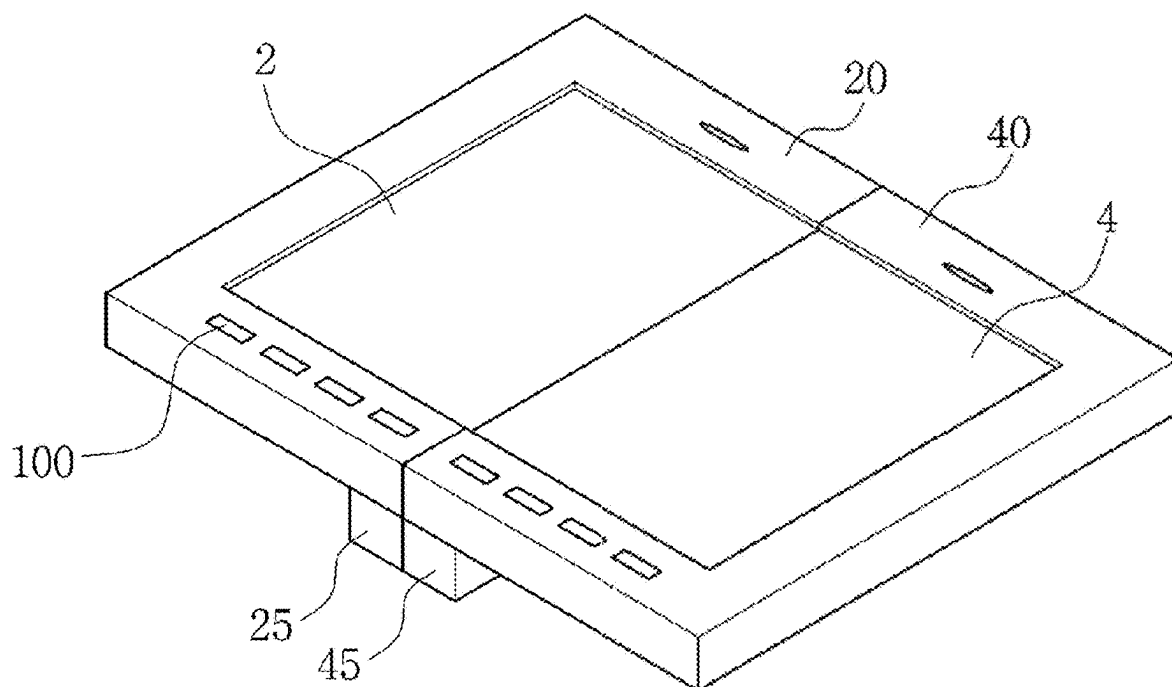


FIG. 94

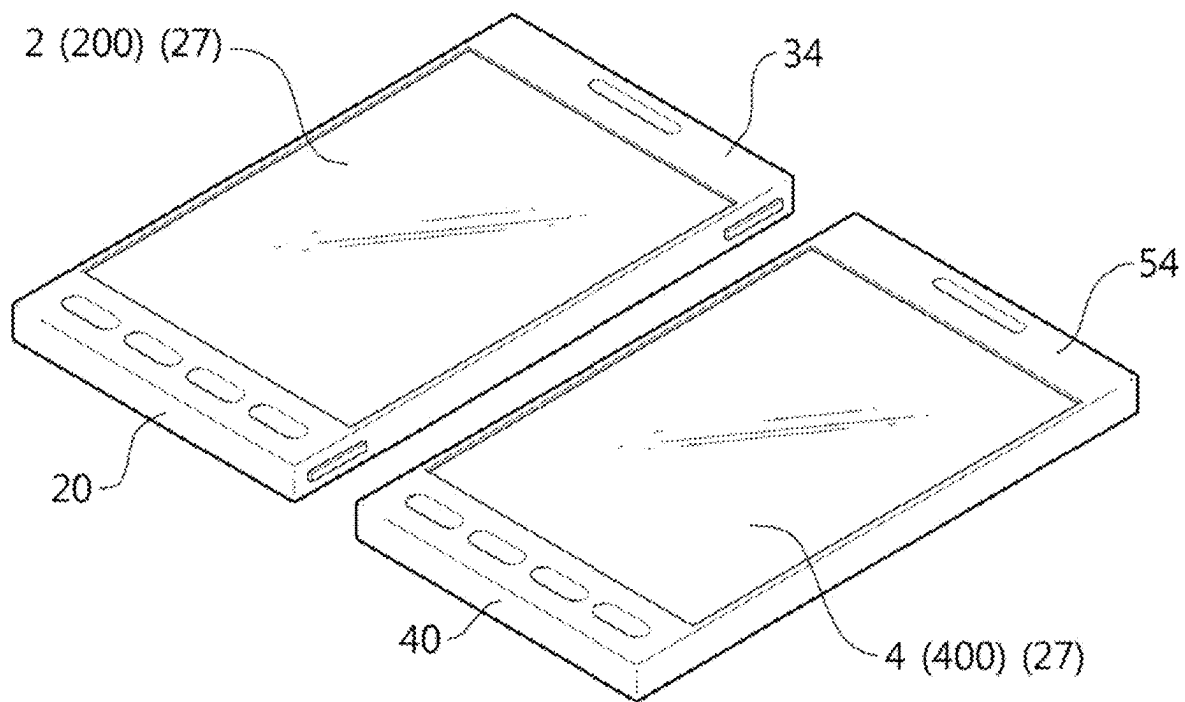


FIG. 95

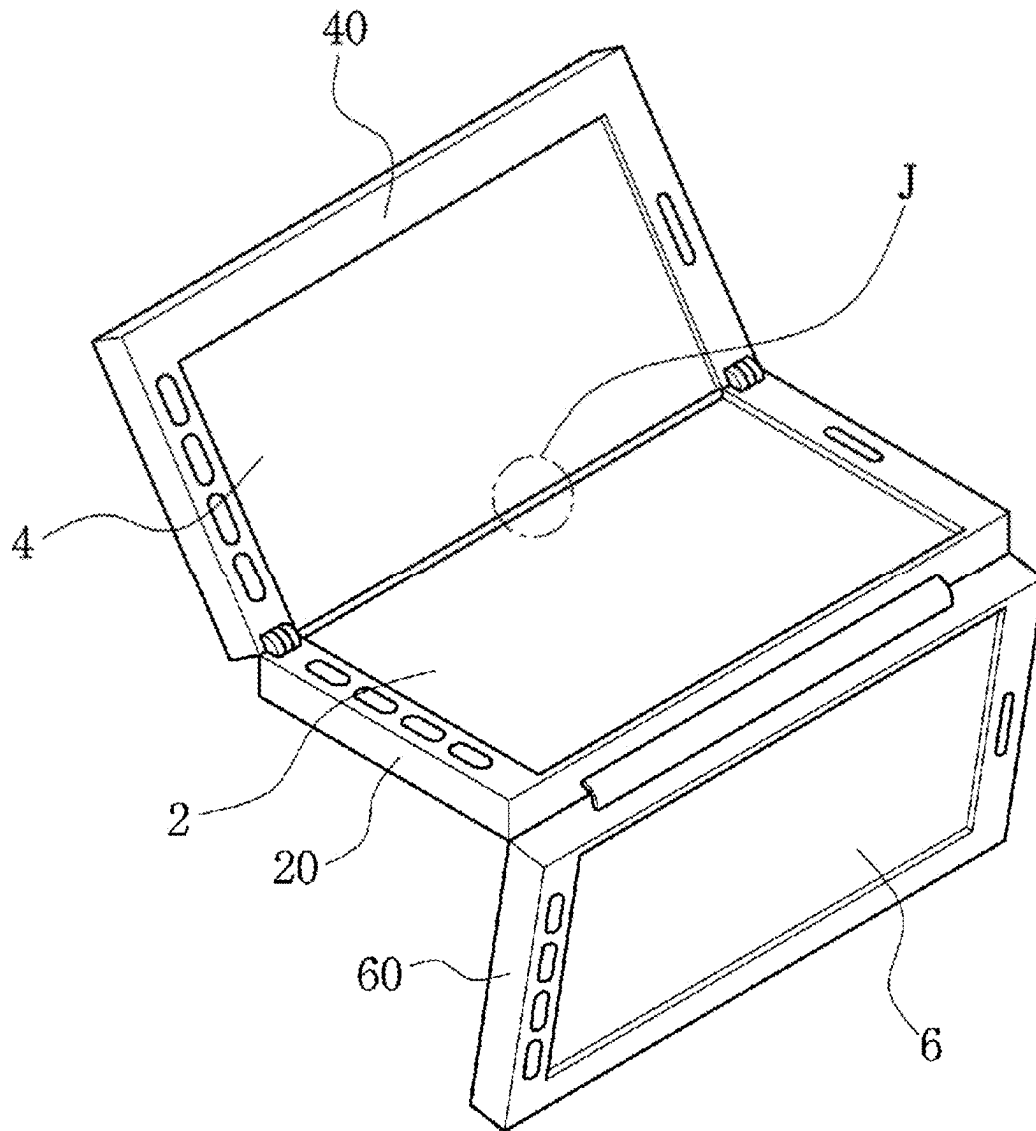


FIG. 96

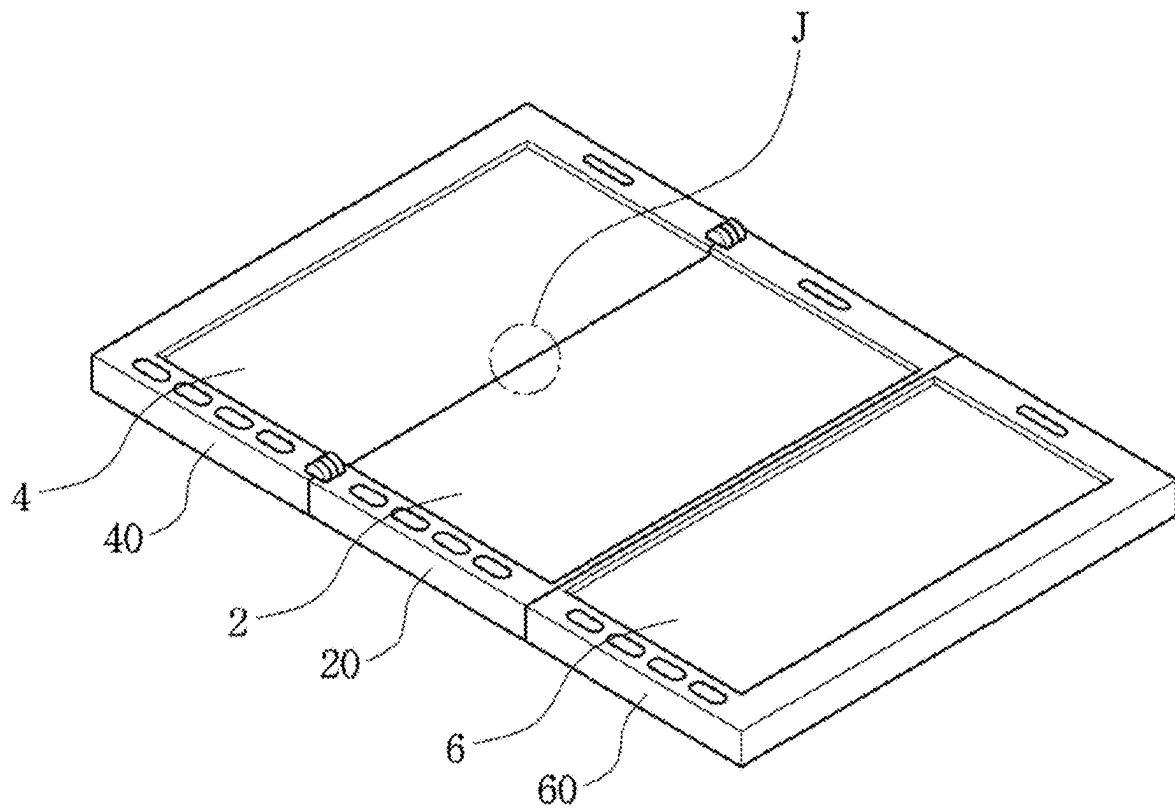


FIG. 97

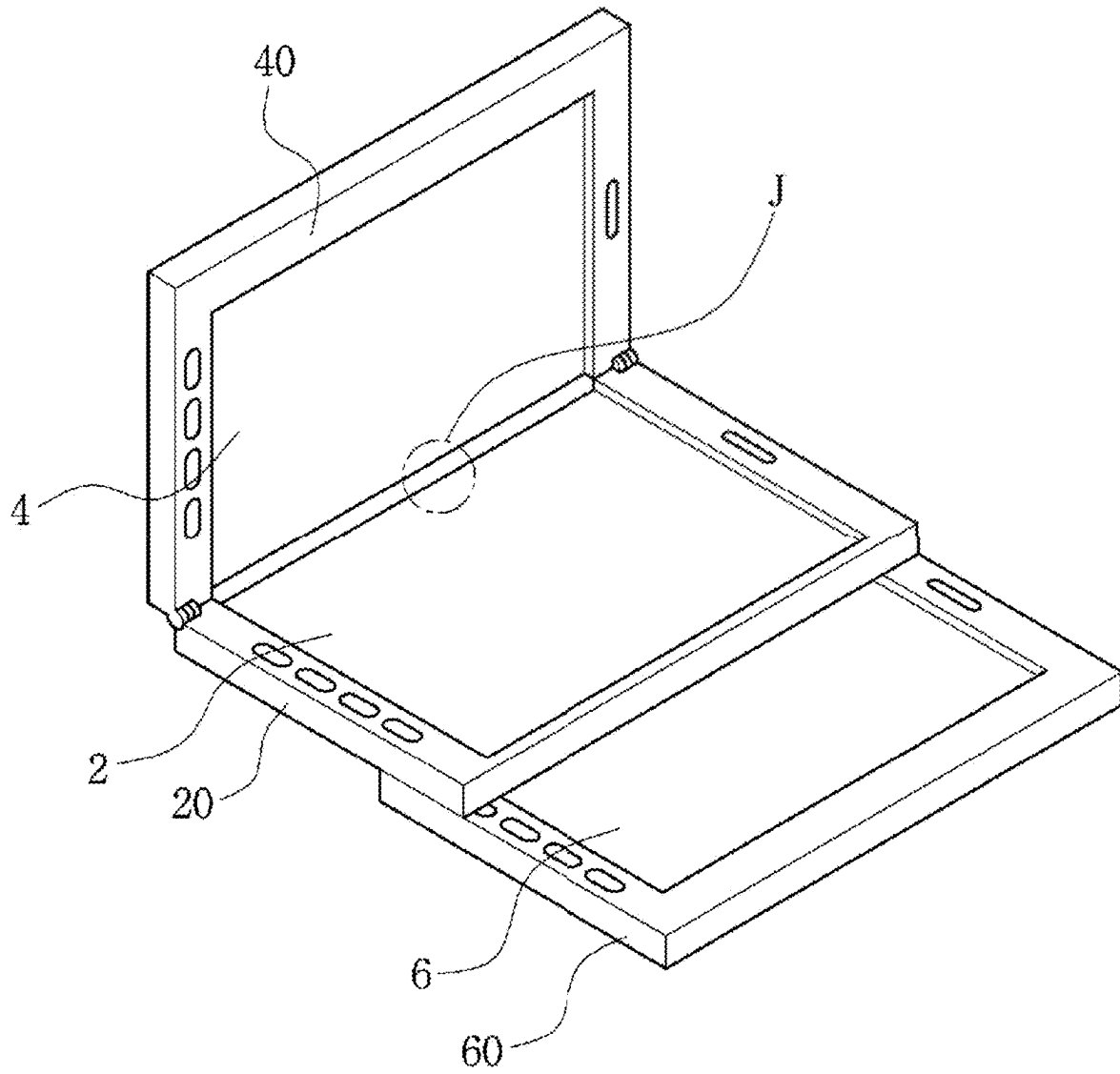


FIG. 98

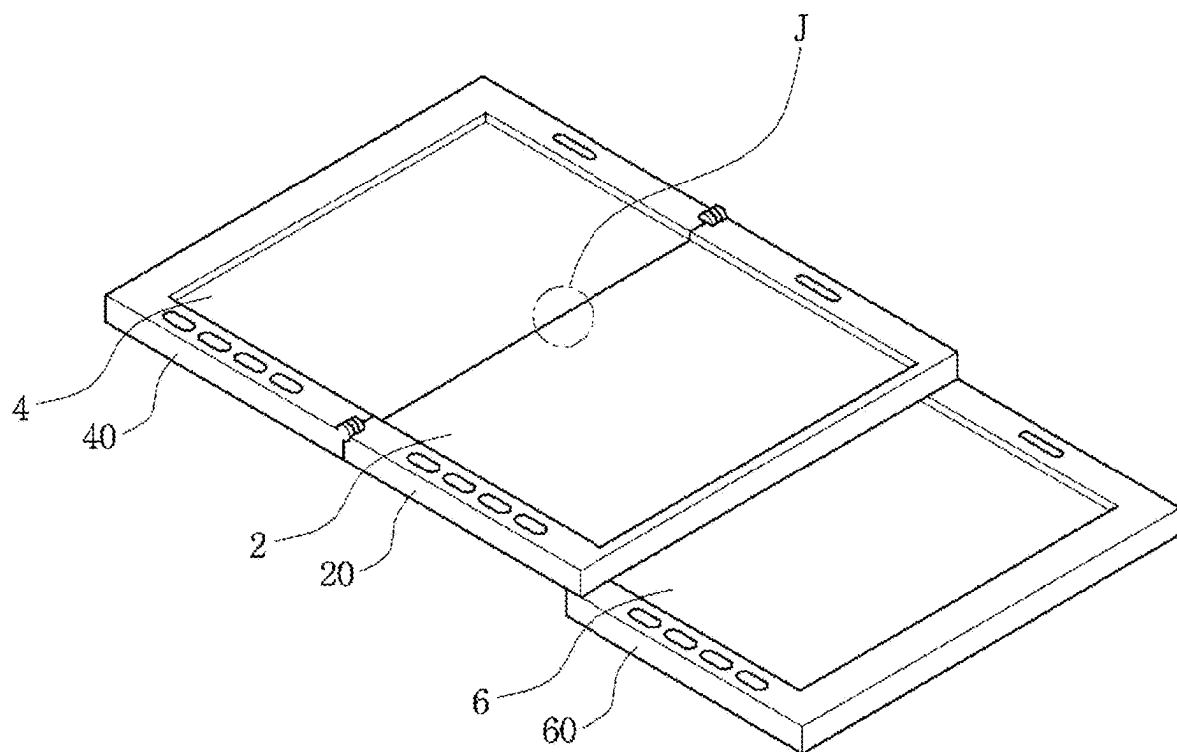


FIG. 99

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**PORTABLE DISPLAY DEVICE****CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. application Ser. No. 18/121,531, filed Mar. 14, 2023, which is a continuation of U.S. application Ser. No. 17/362,847, filed Jun. 29, 2021, now issued as U.S. Pat. No. 11,640, 186 on May 2, 2023, which is a continuation of U.S. application Ser. No. 14/557,338, filed Dec. 1, 2014, which is a continuation-in-part of U.S. application Ser. No. 13/809,355, filed Apr. 12, 2013, which is a national stage of International Application No. PCT/KR2011/002980, filed Apr. 25, 2011, which claims the benefit of priority to Korean Application No. 10-2010-0066332, filed Jul. 9, 2010 in the Korean Intellectual Property Office. All disclosures of the documents named above are incorporated herein by reference.

**BACKGROUND OF THE INVENTION****Field of the Invention**

The present invention relates to a structure for using two display panels as a single screen, and more particularly, to a structure for allowing panel housings to be unfolded from a folded state and strongly mounting display panels and input units on the panel housings.

**Description of the Related Art**

In order to enlarge a screen of a portable display device, two display panels can be joined to each other to form a single screen. Various flat display panels such as an LCD, an OLED, an FED, a PDP, and an electronic paper can be used for this purpose. This principle can be applied to any flat panel display device.

The portable display device includes plural panel housings that can be folded and unfolded and plural display panels mounted on the panel housings, respectively, and has a structure in which the display panels are contiguous to each other when the display panels are unfolded.

The area of a joint which is a non-display area formed between the display panels can be minimized by effectively designing sides of the display panels. That is, the joint is minimized by disposing the display panels, one side of which is effectively designed, to be contiguous to each other.

However, a strong shaft structure is required for effectively folding and unfolding the display panels of the portable display device and thus there is a need for a structure design for the strong shaft structure. The portable display device needs to further include an optical member for minimizing the joint and an input unit to be mounted thereon.

**SUMMARY OF THE INVENTION****Technical Problem**

An object of the invention is to provide a portable display device in which two display panels are joined to serve as a single screen when panel housings are unfolded, and a structure and a design of display panels and input units.

**Solution to the Problem**

According to an aspect of the invention, a portable display device may comprise at least two panel housings and at least

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two display panels, wherein at least one display panel is mounted on each of the panel housings, wherein the display panels are joined to each other to serve as a single screen, and wherein, when the display panels are contiguous to each other, the distance between the display panels is less than 3 mm and the distance between input units or between input sensors is less than 3 mm.

In the portable display device, the panel housing may be capable of being folded and unfolded, and the panel housings may be unfolded from a folded state. The portable display device may further include a terminal capable of performing a near field communication.

**Advantageous Effects of the Invention**

According to an aspect of the invention, it is possible to provide a portable display device in which two display panels are joined to serve as a single screen when panel housings are unfolded, and to further provide a structure in which input units or optical members can be reliably mounted on panel housings.

Additional aspects and/or advantages of the invention will be set forth in part in the description which follows and, in part, will be obvious from the description, or may be learned by practice of the invention.

**BRIEF DESCRIPTION OF THE DRAWINGS**

These and/or other aspects and advantages of the invention will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings of which:

FIGS. 1 and 2 are diagrams illustrating a foldable type portable display device according to an embodiment of the invention.

FIGS. 3 to 9 are diagrams illustrating an input unit and a display panel according to various embodiments of the invention.

FIG. 10 is a diagram specifically illustrating a boundary of display panels according to an embodiment of the invention.

FIGS. 11 and 12 are diagrams illustrating a structure on which display panels, input units, or optical members are arranged according to two possible embodiments of the invention.

FIGS. 13 and 14 are diagrams illustrating an embodiment of the invention where optical members are provided.

FIGS. 15 and 16 are diagrams illustrating another embodiment of the invention where display panels and components are arranged.

FIGS. 17 to 21 are diagrams illustrating embodiments of the invention where an input unit and optical members are disposed on display panels.

FIGS. 22 to 27 are diagrams illustrating the structure of a display module according to an embodiment of the invention.

FIGS. 28 to 36 are diagrams illustrating a mounting structure of panel housings according to an embodiment of the invention.

FIGS. 37 and 38 are diagrams illustrating panel housings on which a cover or a frame is mounted according to embodiments of the invention.

FIGS. 39 to 43 are diagrams illustrating mask prints according to embodiments of the invention.

FIG. 44 is a diagram illustrating an embodiment of the invention where a frame is formed around a shaft.

FIGS. 45 and 46 are diagrams illustrating an embodiment of the invention where a side wall extends from the frame.

FIGS. 47 to 49 are diagrams illustrating an embodiment of the invention where a cover and a frame have a side wall portion.

FIGS. 50 to 54 are diagrams illustrating an embodiment of the invention where an optical member is disposed on an input unit.

FIGS. 55 and 56 are diagrams illustrating an embodiment of the invention where an optical member is disposed between a display panel and an input unit.

FIG. 57 is a diagram illustrating a Y value according to an embodiment of the invention.

FIG. 58 is a diagram illustrating an embodiment of the invention where an optical member is disposed in an input unit.

FIGS. 59 to 61 are block diagrams of an embodiment of the invention.

FIG. 62 is a flowchart of the process of an embodiment of the invention.

FIGS. 63 to 67 are diagrams illustrating an input method according to an embodiment of the invention.

FIGS. 68 to 70 are diagrams illustrating an embodiment of the invention where a portable display device can be controlled in wired and wireless manners.

FIGS. 71 to 79 are diagrams illustrating near field communication according to an embodiment of the invention.

FIGS. 80 to 85 are diagrams illustrating embodiments of the invention where folded display panels are unfolded to be contiguous to each other.

FIGS. 86 to 91 are diagrams illustrating an embodiment of the invention where display panels are located outside when a display device is folded.

FIGS. 92 to 95 are diagrams illustrating a method of coupling two panel housings according to an embodiment of the invention.

FIGS. 96 to 99 are diagrams illustrating embodiments of the invention where three display panels and three panel housings are provided.

#### DETAILED DESCRIPTION OF THE EMBODIMENTS

Reference will now be made in detail to the present embodiments of the present invention, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to the like elements throughout. The embodiments are described below in order to explain the present invention by referring to the figures.

Aspects of the invention provide a portable display device in which at least two flat display panels are joined to each other to form a single screen. Possible flat display panels include but are not limited to an LCD, an OLED, an FED, a PDP, and an electronic paper. The principle of the invention can be applied to any display panel as long as it has a flat panel shape.

The following embodiments should not be viewed as limiting the scope of the invention, but merely as examples thereof.

#### Embodiment 1

FIGS. 1 and 2 are diagrams illustrating a foldable type portable display device according to an embodiment of the invention.

According to the embodiment of FIG. 1, a foldable type portable display device includes two display panels 2 and 4

and two panel housings 20 and 40 on which the display panels are mounted, respectively. The panel housings 20 and 40 include circuits or components used to drive or mount the display panels.

The two panel housings 20 and 40 are joined to each other with connection shafts 61 and 62 so as to be foldable and unfoldable. The connection shafts employ a general shaft, and a hinge type may be used.

Additionally, according to the embodiment of FIG. 2, an auxiliary display panel 30 may be mounted on an outer surface when the display device is folded. The auxiliary display panel 30 may also be protected with a protective window.

A button type key input unit 100 for inputting characters or numerals is further disposed on the outer surface.

A side cover 50 is disposed on the side surface of the panel housings 20 and 40 which are folded. The side cover 50 covers the side surface of the display panels when the panel housings are folded. The side cover 50 also protects the side surface of an input unit when the input unit is mounted on the panel housings.

The side cover 50 is effective when protection means 14 for protecting the side surfaces of the display panels (to be depicted in later figures) is formed thinly.

FIGS. 3 to 9 are diagrams illustrating an input unit and a display panel according to various embodiments of the invention.

Display devices can be classified into at least three types: an out-cell type in which an input unit 200 and a display panel 2 are separated from each other with a fixed gap interposed therebetween as shown in embodiment (A) of FIG. 3, an on-cell type in which the input unit 200 is located directly on the top of the display panel 2 as shown in embodiment (B) of FIG. 3, and an in-cell type in which a micro sensor 207 is inserted into the display panel 2 as shown in embodiment (C) of FIG. 3. Other configurations may also be possible without departing from the scope of the invention.

FIG. 4 is a diagram illustrating an in-cell type embodiment, where a pixel of the display panel 2 is shown. The pixel includes three sub pixels of R (Red), G (Green), and B (Blue). A TFT (Thin Film Transistor) along with a sensor is disposed in a part of the pixel.

(A) of FIG. 4 is a diagram illustrating an embodiment where a micro optical sensor 207 is built in the pixel. That is, the pixel includes the TFT and the micro optical sensor 207.

The optical sensor 207 is connected to an X line 207b and a Y line 207c, and thus the coordinates of a sensor sensing light in the display panels 2 and 4 can be acquired.

(B) of FIG. 4 is a diagram illustrating an embodiment where a sensor of a micro switch 207a type is built in a pixel. When a pressure is applied to a substrate of the display panel 2, the micro switch 207a of the portion to which the pressure is applied is turned on.

Similarly to the optical sensor 207, the micro switch 207a is also connected to an X line 207b and a Y line 207c, and thus the coordinates of a sensor sensing pressure in the display panels 2 and 4 can be acquired.

In these embodiments of the invention, since a micro sensor is disposed in each pixel of the display panels, a set of the micro sensors disposed in a first display panel is referred to as a first input sensor and a set of the micro sensors disposed in a second display panel is referred to as a second input sensor. The first input sensor has a fixed area and the second input sensor also has a fixed area.



## 5

FIG. 5 is a cross-sectional view schematically illustrating embodiments of an input unit, where (A) of FIG. 5 shows a capacitance type and (B) of FIG. 5 shows a resistive film type.

According to embodiment (A) of FIG. 5, an electrode plate **241** coated with a transparent electrode is disposed below a protective plate **240**, and the electrode plate **241** includes one or two films coated with a transparent electrode.

According to embodiment (B) of FIG. 5, two films **241a** and **241b** coated with a transparent electrode are disposed above the protective plate **240** with a fixed gap held therebetween.

A protective film (or a decoration plate) **240a** may be further disposed on the top of the input unit. The protective film **240a** is coated with a desired pattern.

FIG. 6 is a cross-section illustrating a display panel according to an embodiment of the invention, where pixel electrodes are located closer to the joint **8**. That is, an active area which is a display area is located closer to the joint.

The portable display device according to the embodiment of FIG. 6 is an OLED type, and a cross-section thereof is shown. However, similar principles, including the distance between the boundary of the display panel and a closest pixel **2n** in the joint **8** can be applied to all flat panel display devices.

In the embodiment of FIG. 6, each of the display panels **2** and **4** includes an insulating layer **53**, electrodes **54**, a coating electrode **58**, and an organic multi-layered film **53** on a substrate **51**, and may further include a moisture-absorbing layer **56** into which separation films **57** are inserted.

The substrate **51** is covered with a cover substrate **52**. A partition wall **2f** may be formed of a sealant such as an adhesive, and the partition wall may be formed through different processes or may be formed of the substrates **51** and **52**.

When the display panel **2** is mounted on the chassis **16**, the chassis **16** may cover only the side surface of the display panel in the joint or may cover only an upper part P of the display panel corresponding to a non-display area.

As already noted, this structure of the joint is not limited only to the OLED type, but can be applied to all types of flat panel display devices.

FIG. 7 is an enlarged view of a joint **8** and surroundings according to one embodiment of a display panel. The display panels **2** and **4** should be designed to reduce the joint (non-display area).

When a pixel **2n** closest to the joint **8** is considered as shown in FIG. 7, the distance J between the closest pixel **2n** and the partition wall **2f** is set to be less than 1 mm. The value of J may be 0 mm.

The thickness K of the partition wall **2f** is set to be in a range of 0.01 mm to 1 mm. The distance between the pixel **2n** closest to the joint and the boundary **2g** of each display panel **2** or **4** is set to 2 mm or less.

The partition wall **2f** generally indicates the boundary edge portion between the inside and the outside of each display panel **2** or **4**.

The display panels **2** and **4** and relevant components (including but not limited to a backlight unit or driving circuit components) are mounted on the chassis **16**. The chassis is formed of metal such as aluminum or alloy or plastic material.

The width of the chassis **16** in the joint **8** is set to a range of 0.01 mm to 1 mm.

## 6

The boundary **2g** of the display panel and the boundary of the partition wall **2f** do not have to be matched with each other, and the partition wall **2f** may be disposed within 0.5 mm from the boundary **2g**.

That is, the width of the non-display area of the joint **8** in this embodiment of the invention is the total sum of the value of J, the value of K, and the thickness of the chassis **16**. The maximum of this value is 3.0 mm, 6 mm when two display panels are joined to each other.

The chassis **16** in the joint **8** may be opened.

FIG. 8 is an enlarged view of a joint of an input unit of a capacitance type embodiment. According to the embodiment of FIG. 8, the distance between the boundary **223a** of an active area **223** which is an input area and the boundary **250g** of the input unit is set to less than 3 mm and may be set to less than 2 mm.

FIG. 9 is an enlarged view of a joint of an input unit of a resistive film type embodiment.

According to the embodiment of FIG. 9, the gap M between the bus **250n** and the partition wall **250f** in the joint **8** is set to be less than 1 mm, and the thickness L of the partition wall **250f** is set to a range of 0.01 mm to 1 mm. The gap M may be 0 mm.

The partition wall is used to couple conductor films or other plates in the input units to each other.

The distance between an electrode line **250n** and the boundary **250g** of the input unit **200** (or **400**) is set to be less than 2 mm. Alternatively, the distance between the boundary of the active area **250a** and the boundary **250g** of the input unit **200** (or **400**) is set to be less than 2 mm (or less than 2 mm).

The boundary **250g** of the joint **8** and the boundary of the partition wall **250f** do not have to be matched with each other, and the partition wall **250f** may be disposed within 0.5 mm from the boundary **250g**.

In the embodiment of FIG. 9, the input unit **200** or **400** is formed of a flat plate of plastic or glass, and can comprise all types of input units receiving information through the flat plate.

FIG. 10 is a diagram illustrating the boundary between the display panels, according to an embodiment of the invention. Details are shown in the enlarged circle.

According to the embodiment of FIG. 10, the non-display area in the joint **8** includes the side walls of the panel housings **20** and **40**, protective films **14** and **16** (where the side walls of the chasses or members may be used instead of the protective films), and the partition walls **2f** and **4f**. The non-display area for display panel **2** is represented by B in the drawing and the entire non-display area doubles the area of B.

In another embodiment (not depicted), the side walls of the panel housings **20** and **40** may be removed leaving only the protective film **14**.

The thickness of the protective film **14** is generally set to be less than or equal to 0.5 mm though it may be set to be less than or equal to 1 mm in some cases.

The thickness of the partition wall is generally set to be less than or equal to 1 mm, preferably to be less than or equal to 0.5 mm, and more preferably to be less than or equal to 0.3 mm.

The distance between the partition walls **2f** or **4f** and the closest pixels **2n** or **4n** is set to a range of 0.1 mm to 0.3 mm.

In the joint **8**, the display panel may be in close contact with the side wall of the panel housing or the distance between the display panel and the side wall of the panel housing may be set to be less than 1 mm.

In this embodiment of the invention, the distance between the display panels **2** and **4** and the distance between the input units **200** and **400** are minimized. A structure in which a micro sensor is built in each pixel of the display panels **2** and **4**, instead of using the input units **200** and **400**, may also be employed. That is, in the case of an embodiment with an in-cell type input unit, the distance between the micro sensors in the joint **8** can be minimized.

A sensing area for sensing an input in the input units or the input sensors can be considered.

Therefore, when an area in the first display panel **2** is defined as a first sensing area and an area in the second display panel **4** is defined as a second sensing area, the distance between the first sensing area and the second sensing area in the joint **8** is not greater than the width of the non-display area. An error therebetween may be less than 1 mm.

#### Embodiment 2

FIGS. **11** and **12** are diagrams illustrating a structure in which display panels and input units or optical members are arranged, according to two possible embodiments of the invention.

The optical members in the embodiments of FIGS. **11** and **12** are optical plates having a function of reducing a non-display area in the boundary portion between two display panels. A lens, a micro lens, a micro prism, or any other structure which refracts light can be used as an optical member. That is, any means for shifting an optical path of light to the boundary of the display panels can be used as optical members.

FIG. **11** is a diagram illustrating an embodiment where the distance between the display panels and the distance between the input units are equal to each other.

According to the embodiment of FIG. **11**, a part defined as “V” is disposed between the display panels **2** and **4** or between the input units **200** and **400**. This part “V” may be protective member such as a film, which protects the side surfaces of the display panels and the side surfaces of the input units.

Alternatively, the part “V” may have a thickness similar to that of the side walls **20** and **40** of the panel housings, and may form a body along with the side walls of the panel housings. The thickness of the part “V” is set to be less than 1.5 mm.

When the part “V” is a protective film, the protective film may be formed of metal or resin. When the protective film is formed of resin, the resin film is coated with metal or inorganic material. In this case, the thickness of the part “V” as the protective film is preferably set to a range of 0.005 mm to 0.5 mm. When the protective film is formed of metal, the thickness is preferably set to a range of 0.2 mm to 1.5 mm,

On the other hand, the protective film may be a coating film formed on the side surfaces of the display panels or on the side surfaces of the input units. In this case, the thickness thereof may be set to be less than several tens of micro meters.

The distance between the part “V” and the display panels may be set to 0 mm or may be set to about 1.5 mm.

Therefore, when the thickness of the part “V” is 1.5 mm and the panel housings are unfolded, the distance between the display panels and the distance between the input units are 3 mm. However, even when the thickness of the part “V” does not reach 1.5 mm, the distance between the display panels and the distance between the input units may be 3

mm. In this case, there is a gap between the part “V” and the display panels and there is a gap between the part “V” and the input units.

FIG. **12** is a diagram illustrating an embodiment where the distance between the input units is smaller than the distance between the display panels.

When a part disposed between the input units **200** and **400** is defined as “V1” and a part disposed between the display panels **2** and **4** is defined as “V2”, the thickness of the part “V1” is smaller than the thickness of the part “V2”.

The parts “V1” and “V2” may form a body along with the side walls of the joint of the panel housings **20** and **40**.

Alternatively, only the part “V2” may form a body along with the side wall of the joint of the panel housings **20** and **40**, and the part “V1” may form a separate protective film or protection means. When the part “V1” is a protective film or protection means, the protection means or the protective film described in the embodiment of FIG. **11** is employed.

As yet another alternative, the parts “V1” and “V2” may both form protection means separated from the panel housings.

The maximum thickness of the part “V2” is set to be less than 1.5 mm, and the thickness of the part “V1” may be set even smaller. That is, when the distance between the display panels **2** and **4** is less than 3 mm, the distance between the input units **200** and **400** may be smaller than the distance between the display panels.

The thickness of the part “V1” may be 0 mm. That is, when the panel housings are unfolded, the input units **200** and **400** may come in contact with each other. Since the input units **200** and **400** may include a protective plate **240** (see FIG. **5**), this structure can be embodied by forming the protective plate out of reinforced glass or crystal.

#### Embodiment 3

FIGS. **13** and **14** show an embodiment where an optical member is provided.

According to the embodiment of FIG. **13**, parts “V3” and “V4” are disposed between the optical members **27** and between the display panels **2** and **4** and serve to protect the side surfaces of the display panels and the optical members. The parts “V3” and “V4” may be formed in a body along with the side walls of the panel housings **20** and **40**, only the part “V4” may be formed in a body along with the panel housings, or both parts may be formed as protective members separated from the panel housings **20** and **40**.

The protective members are formed of transparent or opaque plastic resin or metal such as alloys.

The possible thickness of the part “V4” is the same as that of part “V2” in the embodiment of FIG. **12**, and the possible thickness of the part “V3” is the same as that of part “V1”. That is, the distance between the display panels is preferably 3 mm and the distance between the optical members **27** is less than 3 mm.

FIG. **14** shows an embodiment where protection means **14** is disposed between the display panels **2** and **4** and between the optical members **27**. According to the embodiment of FIG. **14**, the distance between the display panels **2** and **4** is equal to the distance between the optical members **27**.

As shown FIG. **14**, protective films are used as the protective means **14** to reduce the distance as much as possible. The thickness of the protective film is less than 0.5 mm, that is, the same as that of part “V” in the embodiment of FIG. **11**.

#### Embodiment 4

FIGS. **15** and **16** show another embodiment of the arrangement of the display panels and the components.

According to the embodiment of FIG. 15, the thicknesses of the panel housings 20 and 40 (which mean display device cases in the invention) may be different from each other.

Out of two display panels 2 and 4, a bottom plate (which is indicated by "0" in the drawing) of the panel housing 20 is located just below one display panel 2 and a main controller, a central processing unit, and a battery are disposed below the other display panel 4.

The input units 200 and 400 or the optical members 27 are disposed above the display panels 2 and 4.

According to the embodiment of FIG. 16, a gap exists between the bottom plate and the display panel. The gap may be 0 when the bottom plate is display panel 2 and a main controller, a central processing unit, and a battery are disposed below the other display panel 4.

Although not shown in the drawing, a display panel support or a sheet-like shock absorbing plate absorbing impact may be inserted into the gap between the display panel and the bottom plate. In FIG. 15, by contrast, only a support 26 is shown.

#### Embodiment 5

FIGS. 17 to 21 show embodiments where an input unit and an optical member are disposed on a display panel.

FIGS. 17 and 18 show an embodiment where the optical members 27 are disposed on the display panels 20 and 40 and the input units 200 and 400 are disposed thereon.

According to the embodiment of FIG. 17, a part "V5" is disposed between the display panels 2 and 4, between the optical members 27, and the input units 200 and 400.

The possible thickness and material of the part "V5" are the same as that of part "V" in the embodiment of FIG. 11.

FIG. 18 shows an embodiment where protection means having a different thickness is provided. According to the embodiment of FIG. 18, a part "V6" is disposed on the side surfaces of the display panels 2 and 4 and a part "V7" is disposed on the side surface of the input units 200 and 400 and the optical members 27.

The possible materials and thicknesses of the parts "V6" and "V7" are the same as those of parts "V1" and "V2" in the embodiment of FIG. 12. The distance between the input units is smaller than the distance between the display panels, and the distance between the optical members is smaller than the distance between the display panels.

According to FIG. 18, the part "V7" can be separated into a part "V7-1" disposed on the side surfaces of the input units 200 and 400 and a part "V7-2" disposed on the side surfaces of the optical members 27. The thicknesses of the parts "V7-1" and "V7-2" may be different from each other.

FIG. 19 is a diagram illustrating the side surface of the protection means according to an embodiment of the invention.

According to the embodiment of FIG. 19, the outer surface (which is indicated by an arrow in the drawing) of the protection means or the protective film protecting the side surfaces of the display panels 2 and 4, the input units 200 and 400, and the optical members 27 may be coated with a colorant.

That is, a color film may be attached to the outer surface (which is indicated by an arrow in the drawing), or the outer surface may be coated with a colorant. The same color as the panel housings may be selected.

FIGS. 20 and 21 show embodiments where the optical member is disposed on the uppermost surface.

According to the embodiment of FIG. 20, a part "V8" is the same as part "V5" in the embodiment of FIG. 17, except that the optical members 27 are disposed on the top of the input units 200 and 400.

According to the embodiment of FIG. 21, parts "V9" and "V10" are the same as parts "V7" and "V6" in the embodiment of FIG. 18, except that the positions of the optical members 27 and the input units 200 and 400 are exchanged.

The distance between the display panels 2 and 4, the distance between the optical members 27 and 27, and the distance between the input units 200 and 400 are the same as in the above embodiments.

The part "V10" can be separated into a part "V10-1" disposed on the side surfaces of the optical members 27 and a part "V10-2" disposed on the side surfaces of the input units. The thicknesses of the parts "V10-1" and "V10-2" may be different from each other.

#### Embodiment 6

FIGS. 22 to 27 are diagrams illustrating the structure of a display module according to an embodiment of the invention.

FIG. 22 is a diagram illustrating a possible arrangement of the display panels and the input units. FIG. 22 shows an on-cell type embodiment (see FIG. 3), where the input units 200 and 400 are disposed just above the display panels 2 and 4. The same principle may be applied to an out-cell type.

According to the embodiment of FIG. 22, flexible printed circuits (FPC) 2c, 4c, and 250c are disposed to connect the display panels and the input devices to connectors 2d, 4d, and 250d. The FPCs 2c, 4c, and 250c are disposed on a side opposite to the joint.

The connector 250d of the input unit 200 can be electrically connected to junctions 2e and 4e of the display panels 2 and 4. Accordingly, both the display panels 2 and 4 and the input units 200 and 400 can be controlled by a main unit (including a central processing unit and the like).

The display signal line FPC may be located on the side opposite to the joint, or the signal line (FPC) of the input unit 200 may be located on one side among three sides other than the joint 8.

FIG. 23 is a diagram illustrating a possible electrode structure of a display panel.

According to the embodiment of FIG. 23, when four sides of the display panel are defined as first, second, third, and fourth sides and the first side is used as the joint, a driver 2k is mounted on the third side opposite to the joint.

When a signal supply line 2i connected to a panel electrode 2f is drawn out from the second side or the fourth side, the signal supply line is connected to the third side via the edge of the display panel. The signal supply line drawn out from the third side is connected to the third side.

Accordingly, signals can be sent to the display panel through the use of only the driver 2k mounted on the third side.

FIGS. 24 to 26 are diagrams illustrating other embodiments of the display panel.

According to the embodiment of FIG. 24, when the first side is used as the joint, the driver 2k is disposed on the fourth side. When the joint is disposed on the right side (left side in some cases), the driver 2k is disposed on the upper side or the lower side.

When a signal supply line 2i connected to a panel electrode 2f is drawn out from the first side or the third side, the signal supply line is drawn to the fourth side via the edge of

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the display panel. The signal supply line drawn out from the fourth side is connected to the fourth side.

Accordingly, by mounting the driver on the fourth side, the display panel can be driven.

The numbers of signal supply lines drawn out from the first side and the third side may not be equal to each other. For example, the number of signal supply lines drawn out from the third side may be more. This is to further reduce the width of the joint. In some cases, the number of signal supply lines drawn out from the first side may be more.

FIG. 25 shows an embodiment which is similar to the embodiment shown in FIG. 24 and in which signal supply lines are drawn out from only the third side and no signal supply line is drawn out from the joint (the first side).

According to the embodiment of FIG. 25, a signal supply line **2i** is an electrical wire connected to the driver **2k** so as to supply a drive signal therefrom. In the invention, the electrode **2f** may be a pixel electrode supplying a drive signal to a TFT (Thin Film Transistor).

FIG. 26 is a diagram illustrating an embodiment where two display panels are arranged. FIG. 26 shows an in-cell type embodiment in which the joint **8** is located on the right side (left side in some cases) and the FPCs **2c** and **4c** or the external connectors **2d** and **4d** connected to the driver are located on the lower side (upper side in some cases). Similarly, the FPC **250c** and the connector **250d** for connection of the input unit are located on the lower side. The connector for connection of the input unit may be mounted on three sides other than the joint.

According to the embodiment of FIG. 26, the driver is installed on one selected side of the display panel. For example, the driver may be disposed on two sides out of three sides other than the joint.

FIG. 27 is a diagram illustrating an embodiment where a display panel is mounted on a chassis.

According to the embodiment of FIG. 27, the display panel **2** is mounted on the chassis **16**, and the connectors and the signal lines are not shown for the purpose of convenience.

A fixing portion **16c** is disposed in the chassis **16** so as to strongly fix the chassis to the panel housing **20**. Two display panels are symmetric in a state where they are mounted on the chasses.

## Embodiment 7

FIGS. 28 to 36 are diagrams illustrating a mounting structure of the panel housings **20** and **40**, according to the embodiment of the invention.

FIG. 28 shows an on-cell type embodiment.

According to the embodiment of FIG. 28, the display panels **2** and **4** and the input units **200** and **400** are mounted on the panel housing in a state where an input unit is located just on a display panel.

As shown in the drawing, the display panel **2** or **4** is mounted on the panel housing, and the result is covered with a cover frame **24** having a shape in which one side of a rectangle is opened. That is, the edges which are a non-display area of the display panel **2** or **4** is covered with the cover frame. Input buttons **110** or a speaker **101** may be mounted on the cover.

FIGS. 29 to 32 are diagrams illustrating an embodiment where an input unit is mounted on the cover frame.

According to the embodiment of FIGS. 29 and 30, the cover frame **24** has a shape in which one side of a rectangle is opened, and the joint **8** is located on the opened side.

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The input unit includes signal supply lines **250c** and a connector **250b**. A groove **24g** is formed at a position corresponding to the signal supply lines and the connector.

The input unit **200** is mounted on a support **24c** disposed in the cover frame **24**. FIG. 29 is a disassembled view of this embodiment, and FIG. 30 is an assembled view in which the input unit is mounted on the support **24c**.

When the input unit **200** is mounted on the cover frame **24**, the height of the surface of the cover frame **24** is larger than the height of the surface of the input unit **200**. The difference therebetween is preferably set to be less than 1.5 mm.

A cutout **24k** is formed to fit a shaft.

According to the embodiment of FIG. 31, a cover side wall **24d** may be attached to the opened portion (the joint **8**) of the cover frame **24**. The cover side wall **24d** protects the side surface of the input unit. The height of the cover side wall **24d** may be equal to or less than the height of the cover frame **24**. When the cover side wall is lower than the cover frame, the difference therebetween is preferably set to be greater than or equal to 2 mm.

The thickness of the cover side wall **24d** is preferably set to be greater than or equal to 2 mm.

The cover side wall has the same function as the protection means or the side wall of the panel housing in the embodiments of FIGS. 17 to 21. The height of the cover side wall **24d** may be equal to the height of the surface of the cover frame **24**. The height of the cover side wall may be equal to the height of the input unit.

The cover frame **24** and the cover side wall **24d** may be formed as a single body. A shaft is disposed in the cutout **24k**.

FIG. 32 shows an embodiment where the input unit **200** is mounted from the downside of the cover frame **24**. Such a portable display device might not include an input unit. In this case, a protective plate is disposed at the position corresponding to the input unit.

FIGS. 33 to 35 are diagrams illustrating an embodiment where an input unit is mounted on a frame.

According to the embodiment of FIG. 33, a frame **20b** includes a frame support **20c** and the input unit **200** is mounted on the frame support **20c**. Similarly, the frame **20b** has a shape in which one side is opened. The joint **8** is disposed in this opened portion.

A mask print **16d** is formed in the input unit **200** mounted on the frame **20b**. The mask print **16d** has an effect of causing the edge of the input unit to be opaque. The mask print may be formed on a protective film disposed on the input unit. The protective film and the input unit may be formed as a single body.

The mask print **16d** serves to cause the edges other than the edge corresponding to the joint to be opaque. Holes in which an opening for a camera or a speaker may be located in the mask print.

A cutout **16f** is disposed at both corners (or one corner) of each input unit **200** or **400**. A connection shaft is disposed in the cutout **16f**.

FIG. 34 is a diagram illustrating an embodiment where the input unit **200** is mounted on the frame **20b**, and FIG. 35 is a diagram illustrating an embodiment where a frame side wall **24d** is disposed in the side of the frame **24** corresponding to the joint **8**. The thickness of the frame side wall **24d** is equal to that of the cover side wall described in the above-mentioned embodiment.

In a display device not including the input unit **200**, a protective plate formed of glass or plastic may be disposed in place of the input unit **200**.

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FIG. 36 shows the mounting of the frame on a panel housing, according to an embodiment of the invention.

According to the embodiment of FIG. 36, the cover frame 24 including the input unit 200 is mounted on the panel housing 20. When a shaft is disposed in the panel housing, the cover frame 24 includes cutouts 24k.

The opened side of the cover frame 24 is located to correspond to the joint 8 of the display panel 2.

The opened side of the frame 20b and the joint of the input unit 200 are located to correspond to the joint 8 of the display panel 2.

## Embodiment 8

FIGS. 37 and 38 are diagrams illustrating panel housings having a cover or a frame mounted thereon, according to embodiments of the invention.

FIG. 37 shows an embodiment where two covers 24 and two input units 200 and 400 are mounted. FIG. 38 shows an embodiment where two frames 20b and 40b are mounted.

According to the embodiments of FIGS. 37 and 38, when two panel housings 20 and 40 are unfolded, the display panels 2 and 4 are made to be contiguous to each other and the input units 200 and 400 on the display panels are made to be contiguous to each other.

The panel housings 200 and 400 can be folded and unfolded through the use of a shaft 6.

FIGS. 39 to 43 are diagrams illustrating mask prints according to embodiments of the invention.

FIG. 39 shows an embodiment where a mask print 16d is formed on the joint 8. The width of the mask print corresponding to the joint is larger than that on the side opposite to the joint. The width of the mask print 16d to be formed on the joint 8 should not be greater than 3 mm.

FIGS. 40 and 41 are diagrams illustrating an embodiment where buttons are disposed in a frame.

According to the embodiment of FIG. 40, a button plate 110 is formed in the frame 20b and buttons 100 are disposed in the button plate. The buttons 100 are connected to a central processing unit 105 to control the portable display device.

Cutouts on which a shaft 6 is mounted are formed in the button plate 110. The shaft is attached to the button plate and the input units 200 and 400.

FIG. 41 shows another embodiment of the mask print 16d of the input unit.

According to the embodiment of FIG. 41, when the button plate 110 is provided, the mask print 16d may be formed on only the side opposite to the joint 8 and the upper side in the input unit 200.

The mask print 16d may be formed in the joint. In this case, the width of the mask print formed in the joint is set to be less than 3 mm.

FIG. 42 is a diagram illustrating an embodiment where the mask print 16d is formed on only the side opposite to the joint and a cover 24 is formed on the upper side and the lower side of the frame 20b.

FIG. 43 is a diagram illustrating an embodiment of a structure in which a shaft is not used. FIG. 43 shows an embodiment of the cover 24, but various structures in which no shaft is formed in the frames 20b described above in the invention can be provided.

FIG. 44 is a diagram illustrating an embodiment where a frame is formed around a shaft.

According to the embodiment of FIG. 44, frames 20b-1 and 20b-2 are formed around a shaft. In the frame 20b of which the side corresponding to the joint 8 is opened, the

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coupling force between the frame 20b and the panel housing 20 in the joint may be weakened. The frame is formed to extend around the shaft for the purpose of the weakening of the coupling force.

## Embodiment 9

FIGS. 45 to 49 are diagrams illustrating embodiments where a side wall is formed in a frame. These figures show a cross-sectional view taken along a direction B as shown in FIGS. 30 and 34. The cover side wall 24d further extends in direction B.

According to the embodiment of FIG. 45, a joint side wall 24f extending from the cover side wall 24d is formed in the joint and an extension side wall 24e is formed on the side wall opposite to the joint.

The joint side wall 24f extending in the joint cover a part or all of the side surface of the display panel 2.

Only one of the extension side wall 24e and the joint side wall 24f may be formed.

The extension side wall 24e may be formed as a single body along with the cover and the joint side wall 24f may be formed as a single body along with the cover side wall 24d.

According to the embodiment of FIG. 46, an extension side wall 20e and a joint side wall 24f are formed in the frame 20b and may have the same structure as shown in FIG. 45.

FIGS. 47 to 49 are diagrams illustrating an embodiment where a side wall is formed in the cover and the frame.

According to the embodiment of FIGS. 47 to 49, the sides of the cover frame 24 and the frame 20b corresponding to the joint may be opened. Accordingly, the coupling force to the panel housing may be weakened and the frame side wall 24h is provided to solve this problem. The frame side wall 24h is formed as a single body along with the frame.

According to the embodiment of FIG. 47, the side wall 24h is formed in the cover frame 24.

According to the embodiment of FIG. 48, a display panel is mounted on the side wall 24h of the frame 20b.

According to the embodiment of FIG. 49, a display panel is disposed on the side surface of the side wall 24h of the frame 20b.

The thickness, material, level difference, and the like of the side walls of the panel housings 20 and 40 in the joint can be applied to the side wall of the embodiment of FIGS. 45 to 49.

## Embodiment 10

FIGS. 50 to 54 are diagrams illustrating an embodiment where an optical member is disposed on an input unit.

According to the embodiment of FIG. 50, a non-display area (represented by NA in the drawing) is shown in the display panel 2. The non-display area of the joint in the portable display device includes the non-display area NA in the display panel 2 or 4 and a joint portion 20a (which may further include a chassis or protection means which is excluded to explain an embodiment where the non-display area is optically compensated for).

The non-display area of the portable display device is compensated for by the optical member 27.

In FIG. 50, the non-display area of the display panel 2 is compensated for.

FIGS. 51 and 52 show an embodiment where the thickness of the protection means 14b protecting the side surface of the optical member is further reduced.

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According to the embodiment of FIG. 51, a path L of light is shown.

The non-display area of the display panel 2 and a part of the joint portion 20a (which may further include the side wall of a panel housing, protection means, and an extension of a cover or a frame) are compensated for.

The distance between the display panel and the optical member is defined as Y. Since an image is displayed at the center of the display panel, the distance Y is measured from the center of the display panel. X is a value compensated for by the optical member. Conclusively, the optical member is designed on the basis of the X value and the Y value.

According to the embodiment of FIG. 52, the protection means 14a for protecting the side surface of the input unit 200 is formed of a transparent material. Here, a light path L passes through the protection means 14a for protecting the side surface of the input unit. Then, the distance X compensated for by the optical member 27 is as shown in the drawing.

When the protection means 14a is formed of a transparent material, the side surface (represented by C in the drawing) of the protection means is coated with a specific colorant or a thin film having a specific color. The color with which the side surface of the protection means 14a is coated is preferably similar to the color of the side wall of the panel housings, but may be different therefrom depending on the design.

The part (represented by a boundary B in the drawing) of the protection means 14a through which the light path L passes can be formed of a transparent material such as reinforced glass or reinforced plastic and the other part can be formed of a reinforcement material such as alloy, thereby enhancing the reliability.

FIG. 53 shows an embodiment where the protection means 14a for protecting the side surfaces of the input units 200 and 400 is thinner. That is, the protection means 14b for protecting the side surface of the optical member 27 is thinner than the protection means 14a for protecting the input unit 200. In this case, the X value and the Y value are the same as shown.

When the thickness of the protection means 14b for protecting the optical member 27 is equal to the thickness of the protection means 14a for protecting the input unit 200, the X value is X2.

FIG. 54 is a diagram illustrating an embodiment where the light path L passes through the protection means 14a for protecting the input unit 200. In this embodiment, the X value and the Y value are the same.

The protection means 14a may again be transparent, similarly to the above-mentioned embodiments.

FIGS. 55 and 56 are diagrams illustrating an embodiment where an optical member is disposed between a display panel and an input unit.

According to the embodiment of FIG. 55, the thickness of the joint portion 20a protecting the side surface of the display panel is equal to the thickness of the protection means 14a for protecting the side surface of the input unit. In this case, the X value and the Y value are determined as shown.

Alternatively, the thickness of the protection means 14b for protecting the side surface of the optical member 27 may be smaller than that of the joint portion 20a. In this case, the X value is X2.

FIG. 56 is a diagram illustrating another embodiment of FIG. 55. According to this embodiment, the thickness of the protection means 14a for protecting the side surface of the input unit 200 is smaller than that of the joint portion 20a

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protecting the side surface of the display panel 2. In this case, the X value and the Y value are determined as shown.

On the other hand, the thickness of the protection means 14b for protecting the optical member may be smaller than that of the protection means 14a for protecting the side surface of the input unit. In this case, the X value is X2.

In this disclosure, the expression "a thickness is equal to another thickness" includes an error of +1 mm. The error is preferably less than +0.1 mm. However, even when the error increases to +1 mm, it does not mean that it exceeds the scope of the invention.

FIG. 57 is a diagram illustrating an example of the Y value according to an embodiment of the invention.

When the optical member is formed of a single plate, a micro lens having an optical function may be located on the plate or may be located under the plate, as shown in the embodiment of FIG. 57. Then, the Y value is measured up to the bottom of the optical member 27. This may also be true of the embodiments of FIGS. 50 to 56.

When the thickness of the optical member is less than 0.3 mm or is several micrometers, the Y value does not vary, without depending on whether a tilted structure is located on or under the optical member.

FIG. 58 is a diagram illustrating an embodiment where an optical member is disposed in an input unit.

According to the embodiment of FIG. 58, the optical member can be disposed in the form of a film in the input unit including several plates or films. The positions in which the optical member can be disposed is represented by dotted lines.

That is, a protective plate may be disposed in the input unit and the optical member may be disposed on the surface of the protecting plate.

#### Embodiment 11

FIGS. 59 to 61 are block diagrams of a portable display device according to an embodiment of the invention.

According to this embodiment, the portable display device includes two display panels 2 and 4 and two input units 250-1 and 250-2, and input units are disposed on the display panels.

The input units may employ the input units 200 and 400 described various previous embodiments, or may be the input sensor shown in FIG. 4.

FIG. 59 is a block diagram illustrating the configuration of the portable display device according to an embodiment of the invention and shows a control flow of the display panels and the input units.

According to the embodiment of FIG. 59, a central processing unit 105 is a control unit that controls the overall operation of the portable display device. A ROM 121 controls programs executed by the display device, a RAM 122 stores data generated at the time of executing the programs, and an EEPROM 123 stores data necessary for a user or data necessary for processing the data.

The R/F unit 124 tunes an RF channel as a radio frequency, amplifies an input sound signal, and converts an RF signal received via an antenna into an intermediate frequency signal. An input unit 110 includes various input devices, ten keys, a menu key, and a selection key.

Display driving circuits 130a serve to drive the display panels on the basis of the output of the central processing unit 105. The first and second display panels 2 and 4 display information on the screens on the basis of the output signal of the driving circuits.

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The display device further includes a gravity sensor **235** that determines whether two display panels **2** and **4** are located vertically or horizontally. That is, the central processing unit **105** determines whether two display panels **2** and **4** are arranged vertically or horizontally on the basis of the input signal from the gravity sensor **235**. This determination is performed on the basis of the output of the gravity sensor by the use of a predetermined program.

The central processing unit controls the first and second input units **250-1** and **250-2** through the use of first and second input unit controllers **140-1** and **140-2**. That is, since two input units **250-1** and **250-2** are disposed, two input unit controllers **140-1** and **140-2** are also disposed to control the corresponding input units.

In the embodiment shown in FIG. **4**, the input unit controllers **140-1** and **140-2** are replaced with sensor controllers. The first and second input units are replaced with first and second sensors. The first and second sensors are sets of sensors disposed in the first display panel and the second display panel.

FIG. **60** shows another block diagram according to an embodiment of the invention.

According to the embodiment of FIG. **60**, a central processing unit or a main controller **105** includes a control unit **110**, a memory **120**, a time controller **125**, a display driver **130**, and an input unit driver **140**.

That is, a single component can be manufactured in which elements serving as the control unit **110**, the memory **120**, the time controller **125**, the display driver **130**, and the input unit driver **140** are all disposed in the main controller **105**.

The display driver **130** supplies data, which an image is divided into, to the first display panel **2** and the second display panel **4**. The input unit driver **140** corrects and adjusts signals to be output to the first input unit **250-1** and the second input unit **250-2**.

To display an image on the display panels **2** and **4**, source units **2b** and **4b** supplying data signals and gate units **2c** and **4c** supplying line selection signals are provided.

When dividing an image, a source signal and a gate signal can be divided. The signals may be divided at a ratio of 1:1, or may be divided at various ratios such as 2:1, 1.5:1, and 3:1. That is, the number of lines of the first display panel and the number of lines of the second display panel may be different from each other.

FIG. **61** is a diagram illustrating the input unit driver **140** in detail according to an embodiment of the invention.

According to the embodiment of FIG. **61**, a calibration function executing unit **141** executes a function of calibrating the input units **250** at the time of starting up the system. A panel (part of the input unit to which information is directly input) signal corresponding to a correct coordinate value in the input units **250** is selected by the calibration function executed by the calibration function executing unit **141**.

That is, the signal of the input unit **250** corresponding to the coordinate value is selected depending on the resolution (the numbers of source and gate lines) of the display panels **2** and **4**, and the selected signal is supplied to the control unit **110**. Accordingly, the control unit **110** stores and manages the coordinate value corresponding to the panel signal.

An average detection number adjusting unit **143** adjusts the number of times of detecting an average value for the panel signals output from the input unit **250** on the basis of the screen resolution information of the display panels **2** and **4** supplied from the control unit **110**.

An average value detector **142** detects the average value of the panel signals transmitted from the input units **250** on

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the basis of the number of times of detecting an average value, which is adjusted by the average detection number adjusting unit **143**. The detected average value is transmitted to a panel signal generator **144**.

The panel signal generator **144** generates an adjusted panel signal on the basis of the changed screen resolution of the display panels **2** and **4** supplied from the control unit **110** or the position information of the display screen changed by a virtual scroll and the average value of the presently-input panel signal.

Since there are the first input unit **250-1** and the second input unit **250-2** in the invention, the input unit driver **140** performs the signal control on the first input unit **250-1** and the signal control on the second input unit **250-2**.

That is, under the control of the time controller **125** and the control unit **110**, a switching element **145** alternately connects the first input unit **250-1** and the second input unit **250-2** to the input unit driver **140**.

Instead of using the switching element **145**, input unit drivers corresponding to the number of the input units may be provided.

FIG. **61** shows an input unit which is used in an embodiment of the invention. However, the invention is not limited to the method performed by the input unit in FIG. **61**, but can use any method of controlling the input unit as long as it is a method of controlling the input unit or input means.

FIG. **62** is a flowchart illustrating the flow of processes according to an embodiment of the invention.

According to the embodiment of FIG. **62**, when the portable display device is started up, the main controller **105** is started up to control the units thereof (**400**).

Accordingly, the main controller **105** divides an image and transmits the corresponding image data to the first display panel and the second display panel (**402**).

Then, the input unit driver **140** drives the first input unit **250-1**. That is, information output from the first input unit is controlled (**404**). The information output from the first input unit is controlled on the basis of the image display information of the first display panel **2** (**406**).

Similarly, the input unit driver controls the output signal of the second input unit (**408**), and controls the output signal of the second input unit on the basis of the image display information of the second display panel (**410**).

When a stop command is not given, the main controller again divides an image and drives the respective input units (returning to **402**). When a stop command is given, such functions are ended (**412** and **414**).

That is, the input unit driver in the invention controls the information output from the first input unit on the basis of the screen information of the first display and controls the information output from the second input unit on the basis of the screen information of the second display. By causing the input unit driver to repeatedly perform such control, two input units can be controlled as if they are a single input unit.

## Embodiment 12

FIGS. **63** to **67** are diagrams illustrating an example of an input method according to an embodiment of the invention.

FIGS. **63** and **64** show an embodiment where information is inputted to the surface of the display panels **2** and **4** of the display device. FIG. **63** shows an embodiment where information is inputted in a single-screen mode in which two screens are used as a single screen. FIG. **64** shows an embodiment where information is inputted in a two-screen mode in which two screens are individually used.

According to the embodiment of FIG. 63, a point means a coordinate point in an input unit which is generated by designating a specific point in a screen to input information.

Specifically, in (A) of FIG. 63, two or more input points 147 are recognized and paths 147a in which the points 147 move are also recognized. Even when two points are present in each screen, the points are recognized as if they were present in a large single screen.

It is assumed that the screen is reduced when two points move inward and the screen is enlarged when two points move outward, in accordance with a predetermined program. Then, the central processing unit 105 or the control unit 110 transmit screen information, which indicates that the screen is enlarged or reduced with the movement 147a of two points 147, to the display drivers 130a and 130a. That is, two screens form a single screen and the single screen is enlarged or reduced.

In (B) of FIG. 63, a pattern (or an image) 147b which are present over two screens is selected using two points.

It is assumed that the pattern (or image) 147b is reduced when two points used to select the pattern or image move inward and the pattern (or image) is enlarged when two points move outward, in accordance with a predetermined program. Then, the central processing unit 105 or the control unit 110 transmits screen information, which represents that the pattern (or image) is enlarged or reduced with the movement of two points 147, to the display drivers 130a and 130a.

In (C) of FIG. 63, a pattern (or image) 147b is selected using a point 147, and the pattern (or image) 147b moves along the moving path of the point 147 when the point moves.

Similarly, when the movement is programmed, the central processing unit 105 or the control unit 110 causes the pattern (or image) 147b to move with the movement of the point, as if the movement of the pattern (or image) 147b occurred in a single screen.

In (C) of FIG. 63, the point 147 moves horizontally and rotates. At the time of rotation, the selected pattern (or image) 147b can be made to rotate.

In (D) of FIG. 63, a single point 147 moves. Even when the moving path 147a is present over two screens, the point is displayed to move in a single screen.

When a screen is programmed to change depending on the moving path of a point, the central processing unit 105 or the control unit 110 transmits information on the previous screen or the subsequent screen based on the movement of the point to the display drivers 130a and 130a so as to display a screen.

When the moving path 147a of the point 147 rotates, the screen can also be made to rotate.

FIG. 64 is a diagram illustrating an embodiment where information is inputted in a two-screen mode.

According to the embodiment of (A) of FIG. 64, when a point 147 moves on one screen, the result is displayed on only one screen along the moving path 147a.

On the screen, two points 147 can move inward or outward. The screen can be reduced or enlarged with the movement of two points. When a pattern (or image) is selected, the selected pattern (or image) 147b can be reduced or enlarged.

The movement result of the points is displayed on one screen.

According to the embodiment of (B) of FIG. 64, one pattern (or image) 147b is selected on one screen. The

pattern (or image) 147b can move in one screen or can move to the other screen along the moving path 147a of the pattern 147.

FIG. 65 is a diagram illustrating an embodiment where the height of the protection means 14 between two input units 200 and 400 (or two protective plates) is defined.

According to the embodiment of FIG. 65, the side walls of the panel housings 20 and 40 may be used instead of the protection means 14, or the cover side wall 24d or the side wall extension 24a extending from the cover frame (or frame) may be used.

That is, the height H1 in the drawing represents the height of the protection means (the cover side wall or the side wall extension). In the invention, when information is inputted onto the display panels 2 and 4, an input point moves over two display panels 2 and 4. Accordingly, when the height H1 is excessively large, it causes inconvenience to the information input.

When the protection means (or the side wall or the side wall extension) has a height smaller than that of the input unit (or the protective plate), the difference therebetween is defined as H2.

In this embodiment, the value of H1 or H2 is preferably set to be less than 2 mm or 3 mm, respectively. Both values are more preferably set to 0 mm or a value closer to 0 mm.

FIG. 66 is a flowchart illustrating the process flow of inputting information through the use of the input unit according to an embodiment of the invention.

According to the embodiment of FIG. 66, when the display device is started up (300), the first and second display panels are driven and the first and second input units (or input sensors) are controlled (305).

A single-screen mode or a two-screen mode is selected (310).

When a single-screen mode is selected, the first and second display panels are combined into a single screen, and a position corresponding to a coordinate value of the combined single screen is compared with a coordinate value input through the input unit (sensor).

In the single-screen mode, it is determined whether the coordinate value input through the input unit (or sensor) is a coordinate value in a pattern (or image) displayed on the display panels (315).

When the input coordinate value is not located in the pattern (or image) and the input point and the moving path of the input point are located in the first and second input units (or sensors), a program is executed as if the information were input through a single input unit, and the screen moves (320).

When the input coordinate value is located in the pattern (or image) and the input point and the moving path of the input point are located in the first and second input units (or sensors), a program is executed as if the information were input through a single input unit (325).

When a pattern is selected in a single-screen mode as described in the above-mentioned embodiment, the central processing unit or the control unit executes a predetermined program depending on the number of points or the moving path of a point.

When the two-screen mode is selected in 310, a predetermined program is executed depending on the selection of a pattern (or image) in the two-screen mode, the number of input points, and the moving path thereof (330).

The central processing unit or the control unit transmits a drive signal to the display driver in accordance with the



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program control result in the single-screen mode, and the first and second display panels display the result as if they were a single screen (335).

On the other hand, when a program is executed using the first and second display panels as a single screen, a program, determined depending on the input points and the moving paths of the points, is executed, and the execution result is displayed on the first and second display panels (340).

Once the program execution result is displayed on the screen, the process flow goes to 305 when a stop command is not given (350), and the process flow is ended when a stop command is given (355).

FIG. 67 is a diagram illustrating the principle for using two screens as a single screen according to an embodiment of the invention.

According to the embodiment of FIG. 67, the input units or the display panels have corresponding coordinate systems. In the invention, since the input units or the display panels are actually separated from each other, the corresponding coordinate systems are separated as shown in (A) of FIG. 67.

When two input units are controlled as a single input unit, the coordinate systems which are actually separated are connected as shown in (B) of FIG. 67. That is, the control unit or the central processing unit connects the separated coordinate systems and controls information in a single coordinate system.

## Embodiment 13

FIGS. 68 to 70 are diagrams illustrating an embodiment where the portable display device can be controlled in wired and wireless manners.

According to the embodiment of FIG. 68, when the display device is large, a particular terminal 150 serving as a mobile phone may be further provided. The particular terminal 150 includes an input unit 155 allowing numerals or characters to be inputted and a display unit 152 displaying simple information.

FIG. 69 is a block diagram illustrating an embodiment where the display device including two screens includes a telecommunication module.

According to the embodiment of FIG. 69, the central processing unit 105 shown in the drawing controls two screens as in the embodiments of FIGS. 59 and 60. The other components of FIGS. 59 and 60 are not depicted in FIG. 69. A near-field communication unit 241 and an antenna 242 which are additionally provided are depicted.

The near-field communication unit 241 can be unified with the central processing unit 105 to form a single module.

The particular terminal 150 includes a near-field communication unit 243 and an antenna 244 and may further include a RAM, a ROM, and an EEPROM 204. A particular central processing unit 203 used to control the memory 204, the display unit 152, and the input unit 155 (which may have a button type or a touch panel type) is provided.

Although not shown in the drawing for the purpose of facilitating the explanation, a microphone and a speaker for voice communication may be further provided.

The telecommunication module may be a normal communication module used to transmit and receive information to and from a wide area network. The near-field communication unit 241 may be a communication module for communication between the display panels 2 and 4 and the particular terminal 150.

Possible local area networks include but are not limited to all near-field communication means such as Bluetooth,

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UWB (Ultra-Wide Band), Zigbee, Wibree, binary CDMA, infrared communication, and RS-232 port.

The central processing unit 203 and the near-field radio communication unit 243 of the particular terminal 150 may be formed in a module.

At the time of transmitting information, the information is encoded, the encoded information is converted, and the converted information is sent via the antenna by a signal transmitting unit.

At the time of receiving information, the information received via the antenna is converted into a signal by a signal receiving unit and the signal is then decoded.

That is, the near-field radio communication units 241 and 243 include a signal transceiver, a signal converter, a signal decoder, and a signal encoder.

A signal distributor instead of the central processing unit 203 may be disposed in the terminal 150 so as to distribute and send the signal output from the radio communication unit 243 to the speaker or the display panel. That is, the central processing unit is not necessarily provided.

FIG. 70 is a diagram illustrating an embodiment where a telecommunication module is disposed in the particular terminal.

According to the embodiment of FIG. 70, a signal output from the telecommunication R/F unit 245 of the particular terminal 150 is transmitted to the central processing unit 203.

Among the information processed by the central processing unit 203 of the particular terminal 150, information to be displayed on the two display panels is transmitted to the large-screen display panels 2 and 4 via the near-field communication units 241 and 243. Then, the central processing unit 105 controls the display drivers to display the information on the display panels 2 and 4.

Information input via the input unit 110 and the input units 250-1 and 250-2 shown in the block diagram of FIG. 59 is transmitted to the central processing unit 203 of the particular terminal 150 via the near-field communication units 241 and 243.

The display device including two display panels 2 and 4 may serve as only a simple monitor. In this case, a signal distributor may be provided instead of the central processing unit 105 shown in FIG. 59. The information output from the near-field communication unit is transmitted to the corresponding display panel or the corresponding speaker.

The portable display device may not include the RAM, the ROM, and the EEPROM. In this case, the central processing unit 203 of the particular terminal 150 substantially controls two display panels 2 and 4 and two input units 250-1 and 250-2. The RAM, the ROM, and the EEPROM of the particular terminal 150 serve as high-capacity memories.

In an alternative embodiment, the display device may connect the two display panels and the particular terminal to each other in a wired manner, instead of a wireless manner.

## Embodiment 14

FIGS. 71 to 79 are diagrams illustrating an example of the near-field communication according to an embodiment of the invention.

FIG. 71 shows an embodiment of a terminal that can perform near-field communication with the display device.

According to the embodiment of FIG. 71, a mobile phone 150 and a digital camera 180 can perform near-field communication with each other and various computers and various terminals can perform near-field communication

with each other. The near-field communication is performed in the same way as in the embodiments of FIGS. 69 and 70.

FIG. 72 shows an embodiment of a menu screen displayed on the display panels 2 and 4 of the invention.

That is, as shown in (a) of FIG. 72, a menu screen is displayed in the form of icons and functional names are marked below the corresponding icons. When "networking" is selected from the menu screen, information on terminals capable of performing near-field communication is displayed on the display panels 2 and 4 as shown in (b) of FIG. 72.

Since two display panels are present, the menu screen is displayed on one display panel and detailed information of the selected icon is displayed on the other display panel. That is, program information or stored data of a terminal capable of performing near-field communication with the portable display device according to the embodiment of FIG. 72 is displayed.

FIG. 73 is a block diagram illustrating multiple configurations of a terminal capable of performing near-field communication with the display device according to an embodiment of the invention.

The display panels 2 and 4 according to the embodiment of FIG. 73 are based on FIG. 59 and further include a first communication module 124a, a second communication module 124b, and a third communication module 124c. These communication modules can be formed as a single module 105a along with a central processing unit. The single module may further include a memory unit 120. The single module may further include an additional communication module if necessary.

A terminal 150 embodied in a mobile phone includes a central processing unit 203, an input unit 155, a display panel 152, and a memory unit 204 and further includes a first communication module 243a, a second communication module 243b, and a third communication module 243c for communications.

In the mobile phone terminal 150, the first communication module may be a mobile-phone communication module such as a CDMA, a GSM, and a TDMA and the second communication module may be a communication module related to high-speed data communication such as a WiFi and a WiBro. The third communication module is a communication module related to near-field communication. The third communication module is not limited to the near-field communication module described above, but may employ any module capable of performing near-field communication.

A terminal embodied in a digital camera 180 includes a central processing unit 181, an input unit 182, a display panel 184, and a mechanical units and parts 183. The mechanical units and parts are associated with a digital camera, are not associated with the subject matter of the invention, and thus will not be described in detail. Similarly, the digital camera includes a first communication module 185a, a second communication module 185b, and a third communication module 185c.

The above-mentioned three terminal embodiments necessarily include the third communication module capable of performing near-field communication, but do not necessarily include the first communication module and the second communication module. Other embodiments are also possible without departing from the scope of the invention.

Any of these terminals can perform telecommunication through the near field communication with a terminal capable of performing telecommunication.

FIGS. 74 to 79 are flowcharts illustrating process flows of a near-field communication method according to embodiments of the invention.

FIGS. 74 to 77 are flowcharts illustrating embodiments of process flows of near-field communication in accordance with programs. According to the embodiment of FIG. 74, when a terminal is turned on (490), a terminal capable of performing near-field communication through the third communication module can be recognized (496).

A program system in which a protocol is set up in advance and communication environments are set up should be constructed to mutually recognize the terminal capable of near-field communication and this is typically known.

A menu screen indicating the functions as shown in FIG. 72 is displayed on the display panels 2 and 4 (498). When an item is selected from the menu screen (500), a function performing operation is performed. When an item is not selected from the menu screen, it is determined whether the process flow should be stopped (494). When an end button is pressed, the process flow is ended (492). Otherwise, the process flow is returned to an execution operation.

When a normal program is selected from the menu screen, the process flow is returned to the initial operation after performing a function (502). When a networking function is selected (504), a networking operation is prepared and the process flow goes to 506.

According to the embodiment of FIG. 75, when a networking function is selected through the use of the input units 200 and 400 of the invention, a control command corresponding to a coordinate value determined by the input unit driver is carried out and this function is performed by the input unit driver and the central processing unit 105. The central processing unit 105 displays its results on a terminal capable of performing near-field communication on the display panels 2 and 4 (508).

Here, when a specific terminal, for example the mobile phone terminal 150, is selected (510), information on the selected terminal 150 is displayed in detail on the display panels 2 and 4.

Details of the selected terminal can be displayed on the basis of information stored in advance through the following process. That is, when information is inputted through the input units 200 and 400, the central processing unit 105 processes information and transmits a command via the third communication module 105a which is a near-field communication module.

Then, the third communication module 243c of the terminal 150 receives the information and sends the received information to the central processing unit 203. The central processing unit 203 sends the information on the terminal 150 stored in the memory unit 204 to the display panels 2 and 4 of the invention. The central processing unit 105 of the display panels 2 and 4 display the information received from the terminal 150 on the display panels 2 and 4.

One of the functions of the terminal 150 displayed on the display panels 2 and 4 is performed (514). In this case, a command is inputted through the use of the input units 200 and 400. The inputted command is transmitted to the central processing unit 203 of the terminal 150 in the same way as described in 512 and the central processing unit 203 performs the selected program function.

When the program function is performed, the central processing unit 203 transmits the result thereof to the central processing unit 105 of the display device via the third communication module. The central processing unit 105 in the invention controls the display driver to display the result of the program function on the display panels 2 and 4 (516).

The process flow goes to **520** when additional information for executing a program is inputted, and the process flow goes to **542** when additional information is not inputted (**518**).

Subsequently, according to the embodiment of FIG. 76, new information may be inputted at the time of inputting the additional information, but information stored in the memory unit of the terminal **150** is used (**520**). That is, when additional information stored in the memory unit **204** in the terminal **150** is inputted, a program is executed on the basis of the input information and the execution result is displayed on the display panels **2** and **4**. Then, the process flow goes to **542**.

On the other hand, when data stored in the memory unit **120** of the display panels **2** and **4** of the invention is selected (**526**), the central processing unit **105** transmits the information selected from the memory unit **120** to the terminal **150** via the third communication module **124c** (**528**), and the central processing unit **203** of the terminal **150** executes a program on the basis of the received data. When the execution result is transmitted to the display panels **2** and **4** of the invention, the execution result is displayed on the display panels **2** and **4** and the process flow goes to **542**.

According to the embodiment of FIG. 77, when the third terminal (terminal **180**) is designated in **526**, the process flow goes to **534**. That is, when the information stored in the terminal **180** (**534**), the central processing unit **105** transmits a signal (signal representing that the information stored in the terminal **180** is selected) to the terminal **180** via the third communication module.

Then, the central processing unit **181** of the terminal **180** transmits the selected information to the terminal **150** via the third communication module **185c** (**536**). The terminal **150** receives the information via the third communication module **243c**, executes a program on the basis of the received information, and transmits the result to the display panels **2** and **4** of the invention (**538**).

The central processing unit **105** displays the received result on the display panels **2** and **4** (**540**). Then, the process flow goes to **542**.

On the basis of a program end command is given (**542**), the program execution result is continuously displayed when the end command is not given (**544**), and the process flow goes to **498** when the end command is given.

In the flowcharts and block diagrams described above, a terminal (for example, **150** or **180**) capable of performing near-field communication is further provided to the display panels **2** and **4** of the invention, a program of the terminal capable of performing near-field communication can be controlled and information can be inputted through the use of the input units **200** and **400** disposed in the display device, the program execution result of the terminal capable of performing near-field communication can be displayed on the display panels **2** and **4** of the display device.

When a program of the terminal **150** is performed through the use of the input units **200** and **400** of the display device, information stored in the terminal **180** capable of performing near-field communication can be selected and utilized.

FIG. 78 is a flowchart illustrating one embodiment of a process flow of automatic reception of a near-field communication terminal. According to the embodiment of FIG. 78, when the display device is started up (**550**), near-field communication terminals can be mutually recognized (**552**). This function is enabled by the third communication module capable of performing near-field communication, a predetermined protocol, and an open systems interconnection (OSI) system. Possible OSI systems include but are not

limited to an TCP/IP, an HTTP, and an FTP. In addition, protocols determined to transmit and receive information through near-field communication may be used in the invention.

Then, the received data is checked (**554**). For example, when the terminal **150** receives a specific signal or executes a program (**556**), a signal is sent to the display panels **2** and **4** of the invention in accordance with a predetermined rule, and the central processing unit **105** displays the signal on the display panels **2** and **4** in accordance with a predetermined rule on the basis of the signal received via the third communication module **124c** (**558**). When data is not received, the checking operation is performed again.

For example, when the terminal **150** receives a call, the display panels **2** and **4** may be turned on.

When information is inputted through the input units **200** and **400** or the input unit **110** of the display panels **2** and **4** (**560**), the input information is transmitted to the terminal **150** (**562**). The terminal **150** executes a program on the basis of the transmitted information again, and this execution result is displayed on the display panels **2** and **4** of the display device as shown in the flowchart.

On the other hand, when information is not inputted, the process flow is ended (**566**) or is returned to the received data check operation depending on whether an end command is given or not.

That is, the display device can automatically display the program execution result of the terminal capable of performing near-field communication on the display panels **2** and **4** and can additionally control the program execution through the use of the input units **200** and **400** of the invention or the input units.

FIG. 79 is a flowchart illustrating an embodiment of a data telecommunication process. According to the embodiment of FIG. 79, when a power supply is turned on or a start button is pressed (**570**), terminals capable of performing near-field communication can be mutually recognized through the use of the near-field communication module **124c** (**572**). The terminals capable of performing near-field communication include the terminal **150**, the terminal **180**, and other terminals capable of performing near-field communication.

For example, the terminal **150** capable of performing near-field communication is displayed on the menu screen of the display panels **2** and **4** (**574**) and telecommunication with the terminal **150** is selected (**576**), detailed information of the terminal capable of performing near-field communication is displayed and the communication mode of the terminal capable of performing near-field communication is selected (**578**).

That is, when the second communication module **243b** of the communication modules of the terminal **150** is selected through the use of the input units **200** and **400** of the display device, the display device can perform telecommunication through the use of the second communication module **243b** of the terminal **150**.

Although not described herein, a program to be executed when the first communication module of the terminal **150** is selected and a program to be executed when the second communication module thereof is selected may be different from each other.

#### Embodiment 15

FIGS. 80 to 85 are diagrams illustrating embodiments where folded display panels are unfolded to be contiguous to each other.

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According to the embodiment of FIGS. 80 to 82, when the panel housings 20 and 40 having the display panels 2 and 4 mounted thereon are unfolded in the horizontal direction (A) from a folded state (FIG. 80), the panel housings are changed to the state shown in FIG. 81. When the panel housings further move vertically, the panel housings are changed to the state shown in FIG. 82. Then, the display panels 2 and 4 are contiguous to each other.

FIGS. 83 to 85 show another embodiment of the display device.

According to the embodiment of FIGS. 83 to 85, cover frames 34 and 54 having a rectangular shape are provided and have thin left and right sides and thick upper and lower sides. Similarly to the embodiment of FIGS. 80 to 82, this display device can be in a folded state (FIG. 83), a horizontally-moved state (FIG. 84), and a display-panel contiguous state (FIG. 85).

In the various above embodiments, the thickness of the frames on the side in which the display panels are contiguous to each other may be smaller than the thickness in the opposite side.

#### Embodiment 16

FIGS. 86 to 91 are diagrams illustrating an embodiment where display panels are directed to the outside when a display device is folded.

According to the embodiment of FIGS. 86 and 87, the panel housings 20 and 40 having the display panels 2 and 4 mounted thereon are folded so as to direct the surfaces of the display panels to the outside. This display device can be in a partially-folded state (FIG. 86) and a completely-folded state (FIG. 87).

Possible methods of causing the display panels to be contiguous to each other, of mounting the input units or the protection means, and of inputting information through the input units (or sensors) have been described previously.

According to the embodiment of FIGS. 88 and 89, cover frames 34 and 54 having a rectangular shape are provided and have thin left and right sides and thick upper and lower sides.

FIG. 88 shows a completely-folded state and FIG. 89 shows a partially-folded state. The thickness of the frame on the joint side may be smaller than that of the frame on the side opposite to the joint.

FIG. 90 is a cross-sectional view illustrating a folded state according to an embodiment of the invention. Since the corners come in contact with each other at the time of folding and unfolding, the corners of the panel housings are rounded (R).

According to the embodiment of FIG. 91, coupling units 21 and 41 may be disposed so as to fix the unfolded panel housings to each other. The coupling units 21 and 41 may include magnets having N and S poles or may include grooves and protrusions.

A connection shaft 60 has a structure surrounding a first shaft 60a and a second shaft 60b disposed in the panel housings 20 and 40. That is, the folding and unfolding operations are carried out using two shafts.

As already shown in the embodiment of FIGS. 1 and 2, the folding and unfolding operations may be carried out using a single shaft.

#### Embodiment 17

FIGS. 92 to 95 are diagrams illustrating a method where two panel housings are coupled according to an embodiment of the invention.

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FIG. 92 shows an embodiment where panel housings 20 and 40 having display panels 2 and 4 mounted thereon, and including coupling means 26 and 46 and covers 25 and 45, may be separated from each other. The covers 25 and 45 connected to the panel housings with shafts 23 may then be opened (FIG. 93), and then the panel housings 20 and 40 may be coupled to each other so as to cause the display panels 2 and 4 to be contiguous to each other (FIG. 94).

FIG. 95 shows an embodiment where cover frames 34 and 54 having a rectangular shape are provided and have thin left and right sides and thick upper and lower sides. The thickness of the frame on the joint side may be smaller than that of the frame on the side opposite to the joint. The two panel housings 20 and 40 may be coupled to each other with coupling means without using a cover as shown in the embodiment of FIG. 85.

#### Embodiment 18

FIGS. 96 to 99 are diagrams illustrating embodiments where a third display panel and panel housing are additionally provided.

FIGS. 96 and 97 show one embodiment in which three panel housings 20, 40, and 60 having display panels 2, 4, and 6, respectively, can be completely folded and completely unfolded.

According to the embodiment of FIGS. 96 and 97, when the first panel housing 20 and the second panel housing 40 are folded, the display panels 2 and 4 are folded to face the inside. When the second panel housing 40 and the third panel housing 60 are folded, the display panels thereof are folded to face the outside.

That is, a structure in which the display panels are folded to face the inside and a structure in which the display panels are folded to face the outside are alternately arranged. By employing this folding type, four or five or more display panels may be used. Previously disclosed embodiments of joints may be applied to the joint portion represented by J.

For an embodiment comprising three display panels, FIG. 96 shows a partially-folded state, and FIG. 97 shows a state where the display panels are unfolded to be contiguous to each other.

FIGS. 98 and 99 are diagrams illustrating an embodiment where a sliding structure and a folding structure are combined.

According to the embodiment of FIGS. 98 and 99, the first panel housing 20 and the second panel housing 40 are configured to fold and unfold the display panels 2 and 4, and the second panel housing 40 and the third panel housing 60 are configured to slide each other.

FIG. 98 shows a state where the panel housings are partially folded and partially move in a sliding manner. FIG. 99 shows a state where the folded display panels are unfolded to be contiguous to each other and the sliding movement further proceeds. When the second panel housing 40 and the third panel housing 60 are completely unfolded, a state resembling that of FIG. 97 is obtained. Previously disclosed embodiments of joints may be applied to the joint portion represented by J.

According to various embodiments of the invention, when panel housings are unfolded, two display panels can be joined to each other to form a single screen. Structures and designs of display panels and input units thereof are also provided.

Although a few embodiments of the present invention have been shown and described, it would be appreciated by those skilled in the art that changes may be made in this

embodiment without departing from the principles and spirit of the invention, the scope of which is defined in the claims and their equivalents.

What is claimed is:

1. A portable display device comprising:
  - a first case and a second case, which are foldable and unfoldable in a folding portion;
  - an external display, when the first case and the second case are folded, is provided outside one of the first case and the second case;
  - a main body provided in at least one of the two cases and having a central processing unit thereon;
  - a display provided on the first case, which is adapted to display information thereon and is a display that exists inside the first case when the first case and the second case are unfolded;
  - an input device provided on the display, which is adapted to input information;
  - a Flexible Printed Circuit Board (FPCB) or a connector, which connects the display and the input device to the main body; and
  - a side cover for protecting the side of the display and the side of the input device in the folding portion, wherein the FPCB or connector is connected to the display and the input device at an opposite side of the folding portion in the first case, and
  - an input device is also provided above the external display.
2. The portable display device according to claim 1, wherein a mask print is formed in the portion opposite to the folding portion and a portion perpendicular to the folding portion, respectively.
3. The portable display device according to claim 1, wherein an image is displayed on the display and the image is selected using a point, and
  - when the selected point moves, the image selected using the point also moves.
4. The portable display device according to claim 1, wherein a frame (side wall) is further provided in the first case, and the input device is mounted on the frame (side wall).
5. The portable display device according to claim 1, wherein the edge of the input device is equipped with a mask printing that makes it opaque.
6. A portable display device comprising:
  - a first case and a second case, which are foldable and unfoldable in a folding portion;
  - an external display, when the first case and the second case are folded, is provided outside one of the first case and the second case;
  - a main body provided in at least one of the two cases and having a central processing unit thereon;
  - a display provided on the second case, which is adapted to display information thereon and is a display that exists inside the first case when the first case and the second case are unfolded;
  - an input device provided on the display, which is adapted to input information;
  - a Flexible Printed Circuit Board (FPCB) or a connector, which connects the display and the input device to the main body; and
  - a side cover for protecting the side of the display and the side of the input device in the folding portion, wherein the FPCB or connector is connected to the display and the input device at an opposite side of the folding portion in the second case, and

an input device is also provided above the external display.

7. The portable display device according to claim 6, wherein a mask print is formed in the portion opposite to the folding portion and a portion perpendicular to the folding portion, respectively.

8. The portable display device according to claim 6, wherein an image is displayed on the display and the image is selected using a point, and
  - when the selected point moves, the image selected using the point also moves.

9. The portable display device according to claim 6, wherein a frame (side wall) is further provided in the first case, and the input device is mounted on the frame (side wall).

10. The portable display device according to claim 6, wherein the edge of the input device is equipped with a mask printing that makes it opaque.

11. A portable display device comprising:

- a first case and a second case, which are foldable and unfoldable in a folding portion;
- an external display, when the first case and the second case are folded, is provided outside one of the first case and the second case;
- a main body provided in at least one of the two cases and having a central processing unit thereon;
- a display provided on the first case, which is adapted to display information thereon and is a display that exists inside the first case when the first case and the second case are unfolded;
- an input device provided on the display, which is adapted to input information;
- a Flexible Printed Circuit Board (FPCB) or a connector, which connects the display and the input device to the main body; and
- a side cover for protecting the side of the display and the side of the input device in the folding portion, wherein the FPCB or connector is connected to the display and the input device at the perpendicular side of the folding portion in the first case, and
- an input device is also provided above the external display.

12. The portable display device according to claim 11, wherein a mask print is formed in the portion opposite to the folding portion and a portion perpendicular to the folding portion, respectively.

13. The portable display device according to claim 11, wherein an image is displayed on the display and the image is selected using a point, and
  - when the selected point moves, the image selected using the point also moves.

14. The portable display device according to claim 11, wherein a frame (side wall) is further provided in the first case, and the input device is mounted on the frame (side wall).

15. The portable display device according to claim 11, wherein the edge of the input device is equipped with a mask printing that makes it opaque.

16. A portable display device comprising:

- a first case and a second case, which are foldable and unfoldable in a folding portion;
- an external display, when the first case and the second case are folded, is provided outside one of the first case and the second case;
- a main body provided in at least one of the two cases and having a central processing unit thereon;

a display provided on the second case, which is adapted to display information thereon and is a display that exists inside the first case when the first case and the second case are unfolded;  
 an input device provided on the display, which is adapted to input information;  
 a Flexible Printed Circuit Board (FPCB) or a connector, which connects the display and the input device to the main body; and  
 a side cover for protecting the side of the display and the side of the input device in the folding portion, wherein the FPCB or connector is connected to the display and the input device at the perpendicular side of the folding portion in the second case, and  
 an input device is also provided above the external display.

**17.** The portable display device according to claim **16**, wherein a mask print is formed in the portion opposite to the folding portion and a portion perpendicular to the folding portion, respectively.

**18.** The portable display device according to claim **16**, wherein an image is displayed on the display and the image is selected using a point, and  
 when the selected point moves, the image selected using the point also moves.

**19.** The portable display device according to claim **16**, wherein a frame (side wall) is further provided in the first case, and the input device is mounted on the frame (side wall).

**20.** The portable display device according to claim **16**, wherein the edge of the input device is equipped with a mask printing that makes it opaque.

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